Order Artiodactyla

Even-toed ungulates, so-named because they usually have two or four toes on each foot, are a diverse group of relatively large, often gregarious, and usually herbivous, mammals of great economic importance. The reproductive physiology of the domesticated species (pigs, cattle, sheep, and goats) is extensive and outside the realm of this compendium. A review of reproductive cyclicity (9350) is available.

Order Artiodactyla, Family Suidae

Pigs (*Babyrousa* ?-100 kg; *Hylochoerus* 130-275 kg; *Phacochoerus* 40-150 kg; *Potamochoerus* 40-130 kg; *Sus* 3-350 kg) are terrestrial, often diurnal, omnivores from Eurasia and Africa. Males are often larger than females (937, 3571, 4645, 5762, 6752, 8466, 8501; cf 10941). The epitheliochorial, diffuse placenta implants into a bicornuate uterus and is not eaten after birth (*Phacochoerus* 1911). Neonates are furred and open-eyed (4152, 4638, 4643, 5762, 10028, 11253). Females usually have 4 (*Phacochoerus*), 6 (*Potamochoerus*), or 8-12 (*Sus*) mammae (331, 333, 1474, 3571, 5762, 9175, 9945, 10028, 10158). Males have seminal vesicles, a spiral penis (1911), and Sertoli cells with crystalloid inclusions (*Sus* 8140a, 10883a). Details of anatomy (1911, 6930), endocrinology (1760, 3221, 9641), and milk composition (3178, 9309) are available. The reproductive biology of domestic pigs is reviewed in supplements 33 and 40 of the Journal of Reproduction and Fertility published in 1985 and 1990 respectively.

NEONATAL MASS		
Babyrousa babyrussa	380-1050 g	9607a
Phacochoerus aethiopicus	400-550 g; term emb; via graph	1793
	f: 437 g; r330-490; n5	7184
	480-850 g; varies with litter size	937
	m: 553 g; r510-600; n3	7184
	est 720 g	9212
	860 g; injured; 910 g; intact	3443
Potamochoerus porcus	700-800 g	10158
Sus scrofa	500 g	2973
	557.43 g; r325-665; n7	35
	637.2 g; r603.1-753.5; n4	2588
	681 g; r630-715; n7	27
	720 g	8466
	740 g; sd170; n580	8220
	750 g; r600-1000	4643
	870 g; sd240; n28	10941
	1156 g; sd272; n>5000	2040
	f: 1432 g; se18.0; n296	4216
	f: 1440 g; se13.9; n676	4216
	m: 1470 g; se20.1; n257	4216
	m: 1534 g; se14.6; n631	4216
NEONATAL SIZE		
Babyrousa babyrussa	15-20 cm	7434
Sus scrofa	HB: 24 cm; sd2; n28	10941
	CR: 24.08 cm; r21.9-25.5; n2 lt, n5, 110 d emb	4638
WEANING MASS		
Sus scrofa	3.86 kg; sd1.32; n28, 4 wk	10941
	6.47 kg; sd2.10; 60 d	8220
	6.5-6.8 kg	2973
WEANING SIZE		
Sus scrofa	HB: 42 cm; sd6; n28, 4 wk	10941
LITTER SIZE		
Babyrousa babyrussa	1; n1 lt	12128
	1-2	7434
	mode 1-2; r1-3	1161
Hylochoerus meinertzhageni	2 yg; n1 lt	2206
	2-6	2449
	2-8 yg	4354a
	2-11	5762
	4-6 yg	2207

Phacochoerus aethiopicus	r1-4; n23	1230
	2-3; 4 being a normal full lt	9945
	mode 2-3; max 8	5762
	2-6	8514
	2 emb; n1 f	10154
	2.6 emb; r1-4; n21 f	1126
	2.79; r1-4	9212
	2.9 emb; mode 3; r1-4; n15 f	1793
	3-4	2439
	mode 3-4; r1-8	2449
	mode 3-4; r2-7	331, 333
	3-4 emb; n5 f	2275
	3-8 yg	10772
	3.0 emb; mode 2-4 82%; r1-8; n97 f	1793
	3.1 emb; mode 3-4; n119 f	1793
	3.12; r1-5; n16 lt	4142
	3.2 yg; n20 f	2275
	3.26; r1-5; n61	6929
	3.4 emb; r2-4; n7 f	3661
	3.5; r2-5; n2 lt	7184
	3.5; mode 3-4; r2-5	937
	3.6 cl	1911
	3.6 yg; r1-9	7467
	mode 4	5053
	4; n1 lt	345
	4; r3-5	11253
	6.0 yg; r5-7; n2 lt	3443
Potamochoerus porcus	1-4	12128
	1.5; r1-2; n4 lt	5575
	2-6 yg	10772
	2; n1 lt	2109
	mode 3-4; r2-6	331, 333, 334
	mode 3-4; max 8	10158
	3-6; primiparous: 3-4; adult: 5-6	11253
	3 emb; n1 f	8906
	3; mode 3-4; r1-6; n14 lt	6680, 10028
	4; n1 lt	647
	4 yg; max 10	8490
	mode 5-6	5771
	8-12	5304
Sus barbatus	4; n1 lt	7603
	8.2 emb; r4-11; n5 f	2313
Sus salvanius	2.25; r1-4; n4 lt	6752, 6756
	3.45; mode 3; r2-6; n11 lt	8095
	3.9; r3-6; n9 lt	6756, 6757
Sus scrofa	2.82 emb; r1-8; n56 f	2588
	3.2; r1-6	1230
	3.3	11254
	3.3; r1-5; n117 lt	5349
	3.5 emb; r3-4; n4 f	11255
	4 emb; n1 f	10079
	4 yg; n1 f	5136
	4.3 emb; se0.2; mode 4-5; r1-6; n29 f	9428
	4.6; r4.4-4.8; n2 samples	458
	4.5; mode 5; r1-8	2588
	4.62 emb; n21 f	6979
	4.70 emb; se0.33; r1-10; n34 f	1931
	5; r2-7	4152
	5.0 yg; r4-6; n2 lt	35
	5.00 emb; se0.36; 6.67 cl; se0.30; n27 f	473
	5.0; r1-9; n81 lt	1287
	5.3	10519
	5.4; r4-12	8466
	5.56 yg; r5.36-5.76; n2 morphs, 1st lt	2973

	6-12 yg	5146
	6.12; sd1.78	8220
	7.35 yg; r6.76-7.94; n2 morphs, 2nd lt	2973
	7.4 emb; r5-12; n8 f	10600
	8 yg; n1 lt	8720
	r9.0-10.6; n7 samples	4216
SEXUAL MATURITY		
Babyrousa babyrussa	f/m: 1-2 yr	9607a
Hylochoerus meinertzhageni	f: 1st mate: est 1 yr	5762
	f: 1st birth: 17-18 mo; n1	2206
	m: 18-19 mo	2206
	f/m: 18 mo	2207
Phacochoerus aethiopicus	f: 1st cl: 17-18 mo	1911
	f: 1st mate: 19 mo	1793
	f: 1st mate: 20 mo	1126
	f: 1st birth: 2 yr	937
	f: 18-19 mo	10158
	m: epididymis large: 18-19 mo	1911
	m: 1st sperm: 17-18 mo	6930
	m: elongation of spermatids: 17-18 mo	6930
	m: epididymal sperm: 26 mo	1793
	m: 18-20 mo	2165
	m: 24 mo	9212
Potamochoerus porcus	f: 1st conceive: 3 yr	10028
	f/m: 1st mate: ~86 wk	10239
	f/m: min 3 yr	5575
Sus scrofa	f: 1st ovulate: 15-18 mo	8360
	f: 1st mate: ~1 yr	4643
	f: 1st mate: 21 mo	4181
	f: 1st conceive: min 5-7 mo; mode >1 yr	473
	f: 1st conceive: 6 mo	2588
	f: 1st preg: 8-10 mo	2588
	f: 1st birth: 37 wk	2973
	f: 1st birth: 12 mo	9175
	f: 4-7 mo	8220
	f: 20 wk	2973
	f: 6 mo; r5-8	5349
	f: ~10 mo	10600
	f: 18-20 mo	4643
	m: 1st mate: 2 yr	4643
	m: 12-15 wk	2973
	m: 5-7 mo	10600
	m: 7.5-12 mo	5349
	m: 4-5 yr	4643
	m: 4-6 yr	8466
	f/m: 0.75-1.75 yr	3571
	f/m: 18-20 mo	8466
CYCLE LENGTH		
Babyrousa babyrussa	30 d; n1 cycle	1161
Phacochoerus aethiopicus	estrus: 48 h	1911
	6 wk; n2 cycles, 1 f	1911
Sus scrofa	20 d	8220
	~21 d	2936
	21.8 d; 64% r21-23; n11 cycles; isolated	4639
	23.8 d; 53% r21-23; n34 cycles; crowded	4639
GESTATION		
Babyrousa babyrussa	150-157 d	8242
	157-166 d	1161
Hylochoerus meinertzhageni	125 d	2449
	125 d	5762
	151.5 d; r149-154; n2	2206, 2207
Phacochoerus aethiopicus	140-170 d; field data	1126
	5 mo	11697
	~5 mo; r125-175 d	2439

	150-170 d	1793
	167-175 d	10158
	170-175 d	5762
	171 d	6929
	171-175 d	2449
	173 d; r171-175; n2 lt	1355
Potamochoerus porcus	~10 wk; field data, not many	5573
Sus salvanius	100-121 d	8095
	110-120 d	6756, 6757
Sus scrofa	15.5 wk	9175
	110 d	2588
	110-115 d	8220
	~112 d	4795
	113.38 d; r105-120; n2023 lt	2098
	16-17 wk	1474
	114 d	2040
	115 d	2973
	115.2 d; ±2.3; mode 115; 80% r114-118; r108-120; n41 lt	4639
	115.6 d; sd1.9	6885
	4 mo	5146
	~4 mo	8498, 8501
	~4 mo	8223
	~4 mo	2936
	~4 mo	4645
	~4 mo; field data	8466
	18 wk	4181
	18 wk; r16-20	3571
	r133-140 d; adults; r114-130 d; primiparous	4643
LACTATION		
Babyrousa babyrussa	wean: 6-8 mo	8242
Hylochoerus meinertzhageni	wean: 8-10 wk	2206
Phacochoerus aethiopicus	solid food: grass almost at once	5762
	solid food: ~1 wk	10158
	solid food: 17 d; n1 lt	2165
	solid food: <2 mo	1793
	den emergence: 2 wk	11253
	den emergence: 2 wk	5762
	wean: 9 wk	10158
	nurse: ~5 mo	10158
	wean: 3-4 mo	1793
Sus scrofa	nest emergence: 1 wk	4643
	solid food: 2-3 wk	4643
	wean: 2.5-3.5 mo	4643
	wean: ~4 mo	2588
INTERLITTER INTERVAL		
Hylochoerus meinertzhageni	12 mo	2207
Phacochoerus aethiopicus	10 mo	2439
	1 yr	9945
	1 yr	11253
	1.5 yr	8490
Potamochoerus porcus	1 yr	11253
Sus scrofa	230 d	2588
	230 d	2588
LITTERS/YEAR		
Phacochoerus aethiopicus	1-2	937
Sus scrofa	1; r1-2	5349
	1; r1-2	3571
	2	10348
	2	8501
SEASONALITY		
Hylochoerus meinertzhageni	birth: peak Mar, Sept; Zaire	2206
	birth: yr rd, peak July-Aug; Africa	5762
Phacochoerus aethiopicus	testes: large Mar-May, small Aug-Dec; Zululand, S Africa	6930
	estrus: May-June?; Angola	11697

	mate: May; Zimbabwe	1793
	mate: July-Dec, peak July-Oct; zoo	1230
	mate: fall; Zululand, S Africa	6929
	birth: May-July, Oct-Jan; n7; W Africa	937
	birth: peak May-July, Nov-Dec; Nigeria	4354a
	birth: June-Oct, peak July-Aug; nw plateau, Zambia	331, 333
	birth: Aug-Jan, peak Oct-Nov; Tanzania	9212
	birth: Sept-Nov; Tanzania	3661
	birth: Sept-Nov, peak Oct; Tanzania	1126
	birth: mid Sept-mid Nov, peak Oct; Rwanda	7467
	birth: Sept-Dec, peak Oct-early Nov; Zimbabwe	1793
	birth: Sept-Dec, peak Oct-Nov; Luangwa Valley, Zambia	331, 333
	birth: Oct-Nov; Namibia	9945
	birth: Nov-Dec, May-June; Zaire	11253
	birth: peak Nov-Dec, some Jan-Apr; S Africa	8514
	birth: peak Nov-Dec; S Africa	3145
	birth: peak Nov; e Africa	5053
	birth: Dec; Zimbabwe	2275
	birth: early summer; Zululand, S Africa	6929
	birth: yr rd; n Africa	2439
	birth: yr rd; Zimbabwe, Uganda	5762
	repro: Nov-Jan; S Africa	10772
Potamochoerus porcus	mate: June-Aug; S Africa	10772
	mate: dry period; Zaire	11253
	preg f: Apr-mid June; Madagascar	5573
	preg f: Aug; n1; Madagascar	8906
	preg f: Sept-Nov; n4; nw Madagascar	5575
	birth: Jan, June-Nov; n6; zoo	12128
	birth: May; n1; zoo	2109
	birth: July; s Cameroun	647
	birth: Oct-Jan; Tanzania, s central Africa	5762
	birth: peak Oct-Mar; Zambia	331, 333
	birth: Nov-Jan; S Africa	10772
	birth: end Nov-Feb; captive	10239
	birth: Dec-Jan; se Africa	5771
	birth: peak rainy season, none June, 1 lt July; n S Africa	10028
	birth: 1st half of rainy period; Zaire	11253
Sus salvanius	mate: Feb; nw Assam, India	8095
	mate: Dec-Feb; nw Assam, India	8095
	birth: Apr-May; n9; Assam, India	6756
	birth: May-June; n4; zoo	6756
	birth: Apr-May; captive	8095
	birth: Apr-May; Assam, India	6757
	birth: peak Apr-June, 1/16 Aug, 1/16 Sept; nw Assam, India	8095
Sus scrofa	sperm: yr rd; Smokey Mt, TE	5349
	anestrus: July-Oct; Chizé, France	6980
	estrus: Dec-Jan; Pyrenees, Spain	11255
	estrus: Dec-Feb; Chizé, France	6980
	mate: peak end Oct/Nov-Jan; Germany	3571
	mate: Oct-May, peak Nov-Dec; E Germany	1287
	mate: Nov-Jan; Europe, w USSR	4643, 4645
	mate: Nov-Feb; Chizé, France	6980
	mate: Nov-Dec; central Asia	8466
	birth: Jan-Aug, peak Mar-May; E Germany	1287
	birth: Jan-Sept; sw France	8360
	birth: peak mid Feb-end Apr; central Spain	9428
	birth: peak end Feb-May; Germany	3571
	birth: Mar, end July-Sept; Chizé, France	6979
	birth: Mar; central Asia	8466
	birth: Mar-Apr, July-Sept; zoo	10348
	birth: Mar-May; Europe, USSR	4643
	birth: mid Mar-mid Apr; Poland	4181
	birth: peaks Apr, Aug; Europe	1892
	birth: Apr-May, Oct; n6; zoo	44

birth: Apr-May; Chizé, France	6980
birth: Apr-May; Europe, USSR	4645
birth: peak Apr-July; Santa Catalina I, CA	473
birth: peak July-Oct; Pakistan	9175
birth: peak Oct-Feb 11/15, some July-Aug; s Pakistan	10079
birth: Feb-Mar; Sri Lanka	9487
birth: spring-summer, peak Sept; zoo	1230
birth: winter-spring; Morocco	8223
birth: yr rd, peak mid winter; SC	10600
birth: yr rd; Galapagos, Ecuador	1931
birth: yr rd; TN, NC	5349
birth: yr rd; WV	5129

Order Artiodactyla, Family Tayassuidae

Peccaries (*Catagonus* 30-40 kg; *Tayassu* 14-30 kg) are terrestrial, social omnivores that range from the southern United States through South America. A copulatory plug is reported (10238). The epitheliochorial, villous, diffuse or zonary placenta has central implantation into a bicornuate uterus (231, 6613, 7563, 11846) and is eaten after birth (8184). A postpartum estrus occurs 5-12 d (6613, 10238) after the birth of a furred, open-eyed neonate who can walk immediately (1474, 6989, 7624, 8798, 10136). Females have 2 (1474) or 8 (6613) teats. Details of anatomy (6613, 11846), endocrinology (4597, 7050, 10240), and milk composition (1376, 6529, 6532, 9309, 10241, 10524) are available.

NEONATAL MASS		
Catagonus wagneri	500 g; n2, ~1 wk	1516a
	678-763 g; n3, 1 wk	6989, 6990
Tayassu tajacu	454 g; r320-680; n6	7787
	561 g; n7; term emb	10136
	f: 608 g; n18 m & f	10136
	618 g; ±134; n7	8183a
	m: 639 g; n18 m & f	10136
	693 g; se18; n38, 21 lt	6528
	711 g; se24; n8	6530, 6531
	715 g; r681-766; n7	6613
	726 g; ±92; n9	8183a
	~900 g	3167
WEANING MASS		
Tayassu tajacu	3862 g; se163; n4, 6 wk	6530
	5400-6400 g	3167
LITTER SIZE		
Catagonus wagneri	mode 2	764
	2.46 yg; r1-4; n10 f	6989
	2.72 cl	6989
	2.73 emb; mode 2; r2-4; n11 f	6989
Tayassu pecari	1.9 yg; r1-2; n10 lt	6991
	mode 2; 4 rare; r1-4	8798
	2 emb; n1 f	7624
	2 ea; n3 lt	9268
	2.3 emb; r2-3; n3 f	5111
Tayassu tajacu	1.29 emb; mode 1; r1-4; n7 f	5298
	1.3 yg seen/f; mode 2; n82	957
	1.4 yg; mode 1; r1-3; n43 lt	5889
	1.5 yg; r1-3; n18 lt	9066
	1.5 yg; r1-3; n26 lt	7787
	1.53 yg; r1-2; n17 lt	3167
	1.6 yg; mode 2; r1-2; n11 lt	12128
	1.8 yg; se0.1; mode 2; r1-3; n32 lt	4598, 6528, 8184
	1.81 emb; r1-4	6613
	2 emb; n1 f	3024
	2 emb; n1 f	7040
	2 emb; n1 f	2341
	mode 2; r1-3; n71 lt	1230

	2.0; se0.3; n15 lt	4597
	2.13 emb; mode 2; r1-4; n23 f	10136
	2.17; r1-4; n29 lt	10238
	5 emb; n1 f	4248
	5.5 emb; r5-6; n2 f	5889
SEXUAL MATURITY		
Catagonus wagneri	f: 3 yr	7983
Tayassu pecari	f: 18 mo	9268
	f/m: ~2 yr	8798
Tayassu tajacu	f: 1st mate: 10.2 mo	6613
	f: 1st conceive: 11.3 mo	6613
	f: 9 mo	8798
	m: 1 yr	8798
	f/m: 1st mate: 33-67 wk	10238
CYCLE LENGTH		
Tayassu tajacu	estrus: 2 d	2656
	estrus: 2.60 d; r1-15; mate: 2.71 d; r1-6	6528
	estrus: 3.6 d	10238
	estrus: 5.8 d; r3-9; n4	6613
	23.46 d; r17-30; n64 cycles, 7 f	10238
	26 d; r8-42; n5 cycles, 1 f	6613
GESTATION		
Tayassu pecari	158.3 d; r156-162; n3	9268
	~158 d	8798
Tayassu tajacu	144.17 d; r142-148; n6	10237
	144.3 d; se0.6; r140-146; n13	6613
	144.88 d; r142-149; n9	10238
	145.5 d; se~0.5; n15	4597
	145.6 d; r144-149; n32	6528
LACTATION		
Tayassu pecari	solid food: end 1st wk	8798
Tayassu tajacu	wean: 6 wk	6530
	wean: 6-8 wk	10238
	wean: 6-8 wk	3167
	wean: ≤8 wk	5889
LITTERS/YEAR		
Tayassu tajacu	2	6613
SEASONALITY		
Catagonus wagneri	preg f: July-Sept; n11; Paraguay	6989
	birth: Sept-early Dec; Paraguay	6990
Tayassu pecari	birth: warm season; subtropics/temperate, New World	8798
	birth: yr rd; tropics, New World	8798
Tayassu tajacu	estrus: yr rd; zoo	2656
	mate: Apr-July; TX	6613
	mate: yr rd; zoo	1230
	mate: yr rd; sw US, n Mexico	1474
	mate: yr rd; TX	957
	conceive: Dec-Mar; TX	6613
	conceive: yr rd, except Feb; few data; TX	5298
	preg f: Aug; n1; Mexico	2341
	birth: Jan, May-Aug, Oct-Dec; n13; zoo	12128
	birth: Mar-Aug, Oct-Dec; n17; captive	3167
	birth: peak May-Oct; TX	957
	birth: peak July-Sept; zoo	2656
	birth: yr rd, peak May-July; TX	6613
	birth: yr rd, 74% June-Aug; AZ	7787
	birth: yr rd, peak June-Aug; n146; AZ	10238
	young found: yr rd; AZ	5889

Order Artiodactyla, Family Hippopotamidae

Hippopotamuses (*Choeropsis liberiensis* 160-275 kg; *Hippopotamus amphibius* 655-2100 kg) are semiaquatic, nocturnal herbivores in Africa south of the Sahara. The 9 kg (n1), epitheliochorial, villous, diffuse placenta has a central implantation into the horn of the bicornuate uterus that is ipsilateral to the side of ovulation (220, 231, 237, 4323, 6288, 7563, 10173; cf 238) and is not eaten after birth (220, 8798). A postpartum estrus occurs 4 (*Choeropsis* 9307) or 50 d (*Hippopotamus* 6288) after the birth of a mobile neonate (1353, 2109, 8798, 10158, 11253). Females have 2 inguinal mammae (331, 333, 9945, 10158). Males have seminal vesicles and a sigmoid penis (6288). Details of anatomy (6288, 9520), endocrinology (10035), and milk composition (4023) are available.

NEONATAL MASS		
Choeropsis liberiensis	4.5-6.5 kg	8798
	5-7 kg	8525
	5.73 kg; r3.6-7.0	6405
	6.56 kg; r5.0-7.0; n5	10365
Hippopotamus amphibius	24.4 kg; n1 stillborn	2109
	~25 kg; n1	220
	25.4-55 kg	5762
	30 kg	4023
	30-40 kg	11153
	30-50 kg	8798
	36.5-40 kg	5689
	44 kg; n1; heaviest fetus	6862
	50 kg	6288
NEONATAL SIZE		
Hippopotamus amphibius	748 mm	238
	750 mm; n1	220
	800 mm	2439
	m: 940 mm; n1	2595
	1270 mm	6288
LITTER SIZE		
Choeropsis liberiensis	1; n1 lt	6196
	1; n1 lt	2109
Hippopotamus amphibius	1	9064
	1; 2 very rare	5762
	1; 2 very rare	331, 333
	1 yg ea; n4 lt, 1 f	1353
	1 ea; n8 lt	1230
	1; r1-2; 0.7% twins; n274 lt	6288
SEXUAL MATURITY		
Choeropsis liberiensis	f: 3-4 yr	8798
	m: 4-5 yr	8798
Hippopotamus amphibius	f: 1st cl: 50% 9.5 yr; r3-17	6288
	f: 1st mate: 4 yr	11153
	f: 1st conceive: 3.5 yr; n1 pr	11332
	f: 1st preg: min 5 yr; mode 9-10	10173
	f: 1st birth: ~4 yr	10158
	f: ~5-6 yr; n1 pr	2109
	f: 9 yr; r7-15	5762
	f: mode 11-13 yr; r7-20	6862, 9520
	m: sperm: few 2-5 yr; many 6 yr	10173
	m: 1st sperm: 7-8 yr	10035
	m: testes large: 7-8 yr; r6-13	6288
	m: 1st mate: 3 yr	11153
	m: 1st sire: 3.5 yr; n1 pr	11332
	m: histology: 6-8 yr	6862
	m: r4-11 yr	5762
	m: ~6 yr; n1 pr	2109
	m: min 6 yr; mode by 8 yr	9520
	f/m: 3 yr; n1 pair	318
	f/m: mode 3-4 yr; r2.5-8	2609

CYCLE LENGTH		
Choeropsis liberiensis	estrus: ~3 d	8798
	estrus: 3 d	10365
	28-30 d	10500
	1 mo	10365
Hippopotamus amphibius	estrus: 3-4 d	8798
	estrus: 4-10 d	5689
	3-4 wk	5689
	~35 d; n1 pr	2109
GESTATION		
Choeropsis liberiensis	188 d; r187-204	6405
	r192-196 d; n4	10500
	199 d; r190-210	8798
	200 d; n1	2109
	205.3 d; r201-210; n3	10365
	7 mo; n2	9307
	~9 mo	1355
Hippopotamus amphibius	201-212 d	1230
	>7 mo	11332
	227-240 d	2449
	233.2 d; mode 235; r225-238; n15	5689
	236.75 d; r227-242; n4	11290
	~237 d	2439
	238.25 d; r234-243; n4 lt, 1 f	1353, 1355
	240 d; r225-257	8798
	~240 d	6288
	est 8 mo	11697
	est ~8 mo	10173
LACTATION		
Choeropsis liberiensis	solid food: 15 d	2109
	wean: 3 mo	10500
	wean: 6 mo	6196
	wean: 9-12 mo	8798
Hippopotamus amphibius	solid food: 1 mo, graze 5 mo	5762
	wean: 9 mo; n1	1353
	wean: min 10 mo; n1	6862
	wean: 10-12 mo	6288
	wean: ~1 yr	5762
	wean: 14 mo	6288
INTERLITTER INTERVAL		
Hippopotamus amphibius	17 mo; r13-26; n1 pr	5790
	21.8 mo	10173
	2 yr	6288
SEASONALITY		
Choeropsis liberiensis	mate: Aug; n1; zoo	2109
	birth: Jan, Mar, Aug-Sept, Nov; n41; zoo	5818
	birth: Mar, May-June; n3; zoo	2109
Hippopotamus amphibius	m fertile: yr rd; w Uganda	6288
	mate: peak Feb, Aug; w Uganda	6288
	mate: June, Sept-Oct; Zambia	333
	mate: July-Aug; Zaire	11253
	mate: yr rd; Namibia	9945
	mate: yr rd, peak May; n23; zoo, Netherlands	8339
	conceive: yr rd, peaks Feb, Apr, low Oct-Nov; w Uganda	6288
	birth: Feb-Mar; Zaire	11253
	birth: peak Mar-June; ne S Africa	3145
	birth: Apr-Dec; Zambia	333
	birth: Apr, Oct; w Uganda	6288
	birth: Oct-Mar 70%; S Africa	10173
	birth: yr rd, peak Feb; S Africa	10173
	birth: yr rd; Zambia	331

Order Artiodactyla, Family Camelidae

Camels (*Camelus* 300-1019 kg; *Lama* 40-155 kg; *Vicugna* 35-65 kg) are terrestrial herbivores from Asia, Africa, and South America. Ovulation is triggered by mating (1772, 1774, 1775, 2991, 3035, 3036, 6811, 6812, 7667, 7783, 7785, 8217, 9828) and sperm storage occurs (10366). The 21 kg (*Camelus*), epitheliochorial, villous, crescent-shaped placenta usually implants into the left horn of a bicornuate or bipartite uterus (1772, 2991, 3230, 3233, 5173, 7556, 7563, 7668, 7785, 9824, 9828, 9846, 10546, 10692) and is not usually eaten after birth (1420, 5915, 5994). Right horn implantations survive less than 50 d (389, 390). Estrus occurs 4-6 wk (*Camelus* 13, 11987) after the birth of a mobile, furred, and open-eyed neonate (2971, 3636, 5762, 5915, 5917). Males have a sigmoid penis and ampullary glands (151, 2432, 2456, 2456a, 2457, 2992, 5174, 7408, 10546).

Details of anatomy (389, 842, 2991, 3518, 3754, 9829, 10692), endocrinology (*Camelus* 11, 12, 77, 78, 713, 2146, 2968, 2969, 3882, 4947, 6811, 6812, 7982, 8268a, 9290, 9828, 11978, 11986, 11987; *Lama* 3036, 3234, 3235, 6383), spermatogenesis (10, 2457a, 8140, 8141), copulation (1751, 5709, 9998), birth (2971), and milk composition (23, 476, 2027, 2767, 3164, 3826, 4115, 6224, 8035, 10620, 11988) are available.

NEONATAL MASS		
Camelus bactrianus	35 kg; n1	3004
Camelus dromedarius	f: 24.5 kg; se2.5; n2	2971
	25.82 kg; sd2.14; n33	1450
	26-51 kg	896
	m: 31.3 kg; se1.69; n6	2971
	f: 32.5 kg; n3	2431
	f: 37 kg	896
	37.3 kg; r26.4-52.3	5926
	m: 38 kg	896
	f: 38.25 kg; se1.0	8870
	m: 41.84 kg; se1.2	8870
	42; ±5.6; n63	9846
Lama glama	8-16 kg	3475
Lama guanicoe	8-15 kg	3475
Lama pacos	6-7 kg	3475
	7.92 kg; r6.5-9.5; n4	9455, 10547
Vicugna vicugna	4-6 kg	3475
	6 kg; r5-6.8; n3	1420
	f: 6.2 kg; r5.0-7.9; n6	9607
	m: 6.5 kg; r5.9-7.8; n4	9607
LITTER SIZE		
Camelus bactrianus	1	8798
	1; 2 rare	8739
	2 mummified emb; n1 f	6682
Camelus dromedarius	1	896
	1; 2 rare	5762
	1; twins rare, 2 ovulations not rare	10692
	mode 1 cl; 12.45% 2-3 cl; r1-3	9831
	mode 1 cl; 15% 2-3 cl; n491 f	7668
	mode 1 emb; 0.26% 2-3; r1-3; n942 f	7784, 9828, 9831
	1.004 emb; r1-2; n497 f	7668
	14% twin conceptions	389
Lama glama	mode 1	2109
	mode 1	12128
	1 yg ea; n19 lt	9066
	1 ea; n40 lt	1230
Lama pacos	1.04 emb; r1-2; n22 f	3233
Vicugna vicugna	1; r1	3474
SEXUAL MATURITY		
Camelus bactrianus	f: 1st estrus: 4 yr	1772
	m: 3 yr	1772
	f/m: ~3 yr	8798
Camelus dromedarius	f: 1st ovulate: 3 yr	6942
	f: 1st mate: 4-5 yr	5926
	f: 1st birth: 3-4 yr	13

	f: 1st birth: 4 yr	389
	f: 1st birth: 54.3 mo; se6.3; r45.6-71.3; n37	11779a
	f: 1st repro: ~5 yr	5762
	f: 8-12 mo	13
	m: 1st rut: 3 yr	5926
	m: mature testes: 9-10 yr	5926
	m: 6 yr	6942
Lama guanicoe	m: testes anatomy: ~2 yr	2456
Lama pacos	f: ~1 yr	10546
Vicugna vicugna	f: 1st birth: 2-4.25 yr	9607
	f: 2 yr	3474
	m: 1st mate: 2.33-2.5 yr	9607
CYCLE LENGTH		
Camelus bactrianus	estrus:: 2-4 d	8798
	follicular maturation: 6.8 d; with ovulation	855
	follicular maturation: 12.4 d; no ovulation	855
	5-8 to 31-40 d	842
	19.0 d; ±4.25; follicular development to regression	1773
Camelus dromedarius	estrus: 3-4 d	6942
	estrus: 4.64 d; ±2.29; r1-15	7785
	estrus: 5.00 d; se0.26; r3-6; n11	5446
	14 d	6942
	17.2 d; n4	2969
	23.4 d; r22-24; se0.22; n9 cycles, 6 f	5446
	28 d; n35 cycles, 5 f	7667
GESTATION		
Camelus bactrianus	384 d; r370-395	2843
	385-406 d	8798
	13 mo	8739
	398 d; r375-412; bactrianus/dromedarius cross	2843
	402.22 d; se11.5; n23	1772
Camelus dromedarius	r345-360 d; n24	11987
	12-12.5 mo	1355
	12-13 mo	5762
	370-375 d	390
	370-390 d	6942
	377 d; r360-411	5926
	377.1 d; se12.08; r360-411; n142	11779a
	386.52 d; se1.75; n60	584
	389.87 d; se2.1; n33	7173
	391.1 d; ±16.7; n296	9847
	~13 mo	14
	398 d; r375-412; bactrianus/dromedarius cross	2843
	404.32 d; se4.82; n56	8870
	412 d; r380-439	2843
Lama glama	245-260 d	1046
	11 mo	1355
	330 d; se9.5	3035
	348-368 d	3475
	~13 mo	5994
Lama guanicoe	11 mo	1355
	345-360 d	3475
Lama pacos	342-345 d	3475
	345 d	10547
Vicugna vicugna	330-350 d	3474, 3475
	346-356 d	9607

LACTATION		
Camelus spp	wean: 6-10 mo	6342
Camelus bactrianus	solid food: ~14 d	8798
	wean: 10-11 mo	8798
Camelus dromedarius	solid food: <2 mo	5762
	wean: 47-67 wk	476
	wean: max 20 mo	5762
Lama glama	wean: 4-6 mo	6342
Vicugna vicugna	solid food: 1 wk	5915
	nurse: max 1 yr	5915
INTERLITTER INTERVAL		
Camelus bactrianus	2 yr	8798
Camelus dromedarius	380 d; r365-395	11987
	15 mo without lactation	2431
	20.2 mo; se5.3; r8-46; n460	11779a
	mode 2 yr	6942
	2 yr	896
	2 yr	3636
Vicugna vicugna	18 mo; n6+ lt ea, 8 f	9607
SEASONALITY		
Camelus bactrianus	mate: Feb; captive	8739
	mate: Mar; n17; zoo, Netherlands	8339
	mate: May-June; USSR	8798
	mate: Dec-mid Apr; captive	1772
	birth: Feb-Jan, Oct, n89; zoo	5818
	birth: Feb-May, occasionally June, July; USSR	8798
	birth: Mar-Apr; n6; zoo	12128
Camelus dromedarius	sperm: May-Oct; captive	1752
	spermatogenesis: peak Feb-Apr, low Aug-Oct; captive	11323
	spermatogenesis: peak Mar, decrease June, increase Sept; captive	10
	spermatogenesis: yr rd, peak Nov-Jan, low June; captive	10848
	spermatogenesis: yr rd; captive	3518
	testes: large Dec-Mar, small Apr-Nov; captive	9996, 9997
	testes: large Dec-Apr, small June-Aug; captive	1751
	rut: Mar-Apr; Egypt	5926
	rut: June-Sept; Australia	5926
	rut: Nov-Feb; India	5926
	rut: mid Dec-May; Morocco	5926
	ovarian activity: peak Mar-June; captive	9828
	follicles large: yr rd; captive	7783
	estrus: Jan-Feb; captive	5446
	estrus: Feb-?; captive	7667
	mate: Jan-Apr; Sinai desert	11986
	mate: mid Jan-May; captive	13
	mate: peak Feb-Mar; n Africa	5762
	mate: Apr-June, Sept-Nov; Somalia, n Kenya	5762
	mate: May-Feb; n10; captive	9
	mate: mid Oct-mid Mar; captive	5709
	mate: Nov-Mar; captive	6942
	mate: Dec-Mar; captive	713
	mate: Dec-Mar; captive	3518
	mate: Dec-Apr; captive	3636
	ovulate: peak Dec-May; captive	9831
	birth: Jan-Mar, rare Apr; captive	3636
	birth: Feb-mid May; captive	14
	birth: mid Dec-May; captive	896
	birth: yr rd, peak Dec-Feb; captive	389
	birth: yr rd; captive, Kenya	11779a
	birth: not seasonal; n9; zoo	1230
Lama glama	mate: peak Nov-Mar, low ?-Aug; captive	3035, 3036
	mate: sporadic; n40; zoo	1230
	ovulations after mating: rare June-Aug; captive	3035
	birth: Nov-Feb; captive	3035

	birth: yr rd; n38; zoo	12128
	birth: yr rd; n76; zoo	5818
Lama guanicoe	mate: Dec-Mar; s Chile	11779
	birth: yr rd; n141; zoo	5818
Vicugna vicugna	mate: Apr-June; Puna Zone, Peru	5917
	birth: Feb-Apr; Peru	3474
	birth: Feb-Apr; Peru	5915
	birth: Mar-May; Puna Zone, Peru	5917
	birth: Apr-early Dec; zoo	1420
	birth: July-Oct; zoos, n hemisphere	9607

Order Artiodactyla, Family Tragulidae

Chevrotains, *Hyemoschus* (10-15 kg) from central and west Africa and *Tragulus* (0.7-8 kg) from southeast Asia, are terrestrial, nocturnal herbivores of equatorial forests and savannas. Females are usually larger than males (5762, 8501). Copulatory bouts have multiple intromissions (1522, 2756). The villous, epitheliochorial placenta is transitional between cotyledonary and diffuse, and has a superficial implantation into a bicornuate uterus (3616, 7563, 9746, 11351). Placentophagia occurs (1522). A postpartum estrus (1522, 2109, 2332, 8868, 11614a) follows the birth of a furred and open-eyed neonate that can walk almost immediately (2332, 7161). Females have 4-6 mammae (5762, 7161, 8798). Details of birth (1522) and milk composition (9242) are available.

NEONATAL MASS		
Tragulus meminna	319 g; r288-382; n4	27
Tragulus napu	371 g; n1	2109
	375 g; r371-379; n2, 2 d	2332
NEONATAL SIZE		
Hyemoschus aquaticus	210 mm; n1	5762
LITTER SIZE		
Hyemoschus aquaticus	1	5762
	1-2	8798
Tragulus javanicus	1 yg	556, 557
	1 emb ea; n6 f	2313
	mode 1; r1-2	7161
	mode 1 yg; r1-2; n23 lt	613b
Tragulus meminna	1	8501
	1-2 yg	7382
	mode 2; r1-2	6653, 6654
Tragulus napu	1 yg	2332
	1 emb ea; n3 f	2313
	1 ea; n4 lt	2109
	1 yg; r1-2; n176 lt	11614a
SEXUAL MATURITY		
Hyemoschus aquaticus	f/m: 9 mo	5762
	f/m: 9-26 mo	2757
Tragulus napu	f: 1st conceive: 4.5 mo	2332
	f: 1st birth: 4.5 mo + 152 d	2332
	f: 1st birth: 9.5 mo	11614a
	f: 4.5 mo	11614a
	m: 1st mate: 135 d	11614a
CYCLE LENGTH		
Tragulus napu	14 d	8868
GESTATION		
Hyemoschus aquaticus	4-5? mo	8798
	5-6 mo	5762
	6-9 mo	2755, 2757
Tragulus javanicus	~5 mo; field data	2251
Tragulus meminna	150-160 d	7382
Tragulus napu	153.5 d; r152-155; n6; excludes 172 d outlier	2332

LACTATION		
Hyemoschus aquaticus	wean: 2-3 mo	2755
	wean: 8 mo	2756
	wean: max 9 mo	5762
	wean: 3-6 mo	2757
Tragulus javanicus	nurse: short time	7161
	wean: 21 d; n1; survived mother's death	9242
Tragulus napu	solid food: 2 wk	2332
	nurse: 3 mo	2332
INTERLITTER INTERVAL		
Tragulus napu	5 mo	2109
SEASONALITY		
Hyemoschus aquaticus	birth: Mar; n1; Lake Kivu, Zaire	8854
	birth: Apr, July; n2; captive, e Zaire	5762
	repro: yr rd; ne Gabon	2755
Tragulus javanicus	mate: Nov-Dec; Vietnam	2251
	preg f: end Apr-early Aug; n6; n Borneo, Indonesia	2313
	birth: end Apr-May; Vietnam	2251
	birth: Dec-Jan; Borneo, Indonesia	556
Tragulus meminna	mate: June-July; s India, Sri Lanka	6653, 6654
	mate: June-July; Nepal	7382
	preg f: latter part of yr; Sri Lanka	8501
	birth: end rainy season; India, Sri Lanka	6653, 6654
	birth: yr rd; Sri Lanka	8501
Tragulus napu	preg f: June-early Aug; n3; n Borneo, Indonesia	2313
	birth: Jan, Apr, June, Nov; n1 pr; zoo	2109

Order Artiodactyla, Family Cervidae

Deer (*Alces* 200-825 kg; *Blastocerus* 100-150 kg; *Capreolus* 15-50 kg; *Cervus* 18-475 kg; *Elaphodus* 17-50 kg; *Elaphurus* 150-200 kg; *Hippocamelus* 45-65 kg; *Hydropotes* 8-15 kg; *Mazama* 8-35 kg; *Moschus* 7-17 kg; *Muntiacus* 14-34 kg; *Odocoileus* 14-215 kg; *Ozotoceros* 30-41 kg; *Pudu* 6-11 kg; *Rangifer*; 44-318 kg) are terrestrial grazers and browsers from the New World, Eurasia, and Africa. Males are usually larger than females and have antlers (52, 806, 1474, 1523, 1924, 2109, 2288, 2618, 3571, 4390, 4643, 4645, 5331, 5637, 6387, 8235, 8501, 8711, 9019, 9175, 10730, 11084; cf 412, 6387). *Rangifer* females have antlers.

Implantation is preferentially in the right horn in *Muntiacus* (1700a, 1712, 1713) and is delayed in *Capreolus* (96, 97, 98, 99, 101, 2259a, 5616, 6459, 9061, 9939). The epitheliochorial, syndesmochorial, or endotheliochorial, villous, cotyledonary or diffuse placenta implants in a bicornuate uterus (297, 371, 1700a, 1708, 1712, 1713, 3127, 4262, 4263, 4414, 4425, 4426, 6116, 6817, 7055, 7563, 9158, 9707, 9713, 10005, 10477, 10510, 10789, 10964, 10965, 11057, 11157, 11630, 11860) and is eaten after birth (1708, 1924, 4505, 5897, 7046, 7174, 7320, 7323, 9545, 9809, 10881, 11157, 11500). Estrus may occur postpartum (2204, 3447a, 4836, 9175; cf 1924, 10730) or be synchronous (1924, 5124, 9978). Ovulation may occur without estrus (480, 804, 7058, 10776, 11267).

Neonates are precocial with fur and open eyes (322, 360, 361, 806, 1474, 1708, 1713, 1924, 3092, 3477, 3571, 4426, 4505, 4643, 4645, 4717, 4719, 5066, 5232, 5338, 6491, 7174, 7263, 7320, 7543, 7623, 7624, 9545, 9688, 9809, 9931, 10221, 11156, 11500, 11638). The sex of the offspring may vary with the social rank of the mother (*Cervus* 509, 609, 1921, 1923; *Odocoileus* 2378a). Females have 4 mammae (182, 1474, 3571, 4494, 6491, 7921). Males have ampullae and seminal vesicles (1730).

Details of anatomy (*Capreolus* 6738; *Cervus* 2703, 3367; *Odocoileus* 3877, 11856; *Rangifer* 6320), intersex (2144, 6804), endocrinology (1705a, 6596; *Alces* 10409; *Capreolus* 98, 620, 1385, 4880, 6738, 9552, 9764, 9766, 9767, 9768, 9940, 9953, 11341; *Cervus* 51, 371, 413, 414, 415, 416, 417, 418, 613, 619, 1385, 1924, 2175, 2178, 2921, 4141, 4207, 4208, 4214, 5229a, 5634, 5636, 6450, 6462, 6463, 6595, 6596, 7475, 7592, 10562a, 10563, 11182, 11509; *Elaphurus* 2175, 6594, 6596; *Odocoileus* 18, 1372, 1401, 1401a, 1402, 1403, 1404, 1405, 1697, 2965, 4367, 7102, 7359, 8596, 8601, 8603, 8605, 8606, 9661, 9662, 9663, 11884, 11909; *Rangifer* 2696, 7240, 9400a, 10449, 11644), spermatogenesis (*Capreolus* 7920; *Cervus* 4207, 4208, 6451, 6591; *Odocoileus* 6164; *Rangifer* 6319), birth (360, 6208, 10881), suckling (3635, 7322), milk composition (4643, 5179; *Alces* 2036, 3479, 9031; *Cervus* 362, 6078; *Odocoileus* 5841, 7588; *Pudu* 2204; *Rangifer* 406, 4489, 8744),

reproductive energetics (124, 126, 1920, 1921, 1922, 1923, 1926, 5932, 6595, 11266), and light (*Cervus* 50, 52, 53, 2178, 2682, 6458, 6850, 10564a, 11520, 11520a; *Odocoileus* 1373, 1400, 1406, 8597, 8598, 8599, 8600, 9664) are available.

NEONATAL MASS

Alces alces	f: 6-14 kg	4643
	m: 8-16 kg	4643
	f: 9.35 kg; n1 stillborn	6243
	11-16 kg	3477
	11.2 kg; n6	11262
	~12 kg	3571
	17 kg; n1 stillborn	9031
	17.7 kg; n1, 2 wk	2483
Capreolus capreolus	0.93 kg; r0.5-1.26	11442
	1.09 kg; r0.95-1.23; n2 term emb	4643
	~1 kg	3571
	1.34 kg; r1.28-1.39; n2	6459
Cervus axis	f: 2.77 kg; r1.53-3.45; n3 stillborn	6243
	2.991 kg; r2.2-4.0; n34	27
	m: 3.45 kg; n1	6243
	4.06 kg; r3.63-4.5; n2	9475
Cervus dama	m: 3.2 kg; n1	6243
	f: 4.4 kg; n93 m & f	1708
	4.5 kg; n1, 230 d emb	10380
	f: r4.55-6.4 kg; n5	1244
	m: 4.6 kg; n93 m & f	1708
	f: 4.7 kg; n7	6079
	f: 4.9 kg; sd0.8; n28, <3 d	1245
	m: 5.3 kg; n3	6079
	m: 5.7 kg; sd0.7; n29, <3 d	1245
Cervus elaphus	5.8-7.2 kg	1924
	f: 6.00 kg; se0.22; r3.4-7.6; n25 1st calves	361, 362
	f: 6.23 kg	1920
	6.35 kg; r3.63-9.99; n204	7364
	f: 6.44 kg	4138
	m: 6.59 kg; se0.21; r4.0-7.7; n21 1st calves	361, 362
	6.71 kg; r3.17-10.43; n244	7364
	m: 6.72 kg	1920
	m: 6.90 kg	4138
	7.5 kg; r5.5-9.1; n13	2200
	f: 7.6 kg; n1 not first calf	361, 362
	m: 8.12 kg; se0.41; r7.9-9.2; n4 not 1st calf	361, 362
	f: 8.3 kg; r7.77-9.1; n3	6243
	m: 8.95 kg; r8.7-9.2; n2	6243
	9 kg	7042
	11-22 kg; 1st wk	4643
	<11.4 kg; <50% survive	10811
	11.5 kg; se2.4; moribund	4840
	13 kg	1182
	mode 13-16 kg	9385
	15 kg; r8.6-20; n23	5338
	>16 kg; 90% survive	10811
	m: 17.4 kg; n1 term	7543
	18.5 kg; se1.8	4840
Cervus eldi	4.54-6.1 kg; n8	9475
	f: 4.7 kg; se1.1; n18	11590
	m: 4.8 kg; se1.4; n16	11590
Cervus nippon	2.25 kg; r1.5-3.0; n2	9475
	f: 2.4 kg; n1	6243
	m: 2.52 kg; n1	6243
	f: r4.2-6.2 kg; n9	4643
	4.5-7.0 kg	3198
	m: r4.75-7.32 kg; n12	4643

Cervus porcinus	2.365 kg; r2.0-2.74; n6	27
	m: 2.70 kg; n1 stillborn	6243
Cervus unicolor	10.270 kg; r7.0-12.3; n20	27
Elaphurus davidianus	3.1 kg; n1 emb, prob not near term	4262
	9.73 kg; n11 stillborn or dead within 24 h	11590a
	f: 11.49 kg; sd1.707; n49	11590a
	m: 11.66 kg; sd1.434; n43	11590a
Hydropotes inermis	m: 0.91 kg; n1	6243
	1.175 kg	2054
Mazama gouazoubira	f: 1.3 kg; r1.2, 1.4; n2	3444
	0.510-0.567 kg	10788a
Moschus berezovskii	0.350-0.558 kg; n15	4032a
	0.455-0.609 kg	4032a
Moschus chrysogaster	0.6 kg; n2	4032a
Moschus moschiferus	0.3-0.5 kg	4643
	0.400-0.635 kg	4032a
Moschus sifanicus	1 kg; 2 d	5232
Muntiacus muntjak	f: 1.32 kg; n1	6243
	m: 1.42 kg; n1	6243
	1.554 kg; r1.2-2.01; n22	27
Muntiacus reevesi	1.05 kg; n1	1713
Odocoileus hemionus	2.5 kg; n1	7320, 7321
	f: 2.5 kg; n53; 2.7 kg; n11 singletons; 2.5 kg; n42 twins	7590
	2.7 kg; sd0.4; r1.7-3.8; n101	7590
	f: 2.8 kg; n1, 174 d fetus	5066
	m: 2.8 kg; n48; 3.3 kg; n9 singletons; 2.7 kg; n39 twins	7590
	f: 2.89 kg; r2.8-2.95; n2	2618
	2.97 kg; r2.0-3.6; n11	2089
	~3 kg; n3	9376
	3.03 kg; n4	3876
	3.2 kg; r2.1-4.5; n11	3876
	3.3 kg; r2.3-4.5	9183
	m: 3.30 kg; r3.01-3.46; n3	2618
	3.4 kg; r3.2-3.7; n11	7878
Odocoileus virginianus	1.8-3.6 kg; n4 emb	10909
	2.1 kg; r1.9-2.5; n3	7878
	2.3-2.7 kg; r1.13-3.97; 1-2 d	11638
	f: 2.5 kg; n8	4500
	2.54 kg; r2.49-2.58; n2	7263
	f: 2.6 kg; r1.47-3.74; n12	4503
	m: 2.61 kg; n1	6243
	2.75 kg; r2.27-3.24; n2 samples	11263
	2.9 kg; sd0.3; r2.5-3.3; n8, <1 d	7845
	2.95 kg; r0.9-4.58; n201	11259
	m: ~3 kg; n10	4500
	3.2-3.6 kg	4250
	m: 3.37 kg; r2.04-6.58; n17	4503
	3.4 kg; ±0.59; n large	11261
	f: 3.5 kg; se0.1; n74 twin	11266
	m: 3.7 kg; se0.1; n78 twin	11266
	f: 3.9 kg; se0.1; n30 singleton	11266
	3.9 kg; n8?, 3 d	10790
	m: 4.1 kg; se0.1; n28 singleton	11266
Ozotoceros bezoarticus	2.1 kg; r1.2-2.5	3446
Pudu mephistophiles	0.4 kg; n1	3445
Pudu pudu	0.78 kg; n1	2204
	1.0 kg	4719
Rangifer tarandus	2.5-3.5 kg; n3	9022
	2.75 kg; r2.5-3.0; n2 twins	7057
	2.98 kg; se0.47; n7 dead	10047
	3.2-5.9 kg	8210
	3.72 kg; se0.27; n11	10047
	f: 4.5 kg	5637
	4.7 kg; se0.2; n14	10221

	4.8 kg; r2.7-6.16; n5	55
	5-9 kg	806
	5 kg ea; singletons	7057
	f: 5.0 kg; se0.1; n397	2987
	5.1 kg; se0.04; r1.8-8.5; n823	2987
	m: 5.3 kg; se0.1; n423	2987, 10837
	f: 5.52 kg; r4.63-6.36; n3	6243
	5.6 kg; se0.1; n37	10221
	f: 5.7 kg; twin	7985
	6-7 kg	4643
	m: 6.14 kg; r3.81-7.42; n5	6243
	m: 6.2 kg; twin	7985
	6.4 kg	2288
	6.76-9.07 kg; across n8 yr	801
	f: 6.8 kg; singleton	7985
	m: 7.2 kg; singleton	7985
	f: 7.2 kg; n153	804
	f: 7.3 kg	2288
	m: 7.8 kg; n121	804
NEONATAL SIZE		
Alces alces	80 cm	3571
Cervus dama	HB: 66.5-73.8 cm	3199
Cervus elaphus	92-101 cm	4643
	m: CR 92.5 cm; n1 term	7543
Cervus nippon	TL: 570 mm	3198
Hydropotes inermis	45 cm; n3 stillborn	2054
WEANING MASS		
Cervus elaphus	56.5 kg; se4.9; text; 59.9 kg; 92 d table	4840
Odocoileus hemionus	21.6 kg; ±1.80; n16, 105 d	9731
	21.9 kg; sd4.71; n7, 105 d	8240
	23.9 kg; ±1.44; n16, 105 d	2089, 8240
	~25 kg; r20-30; 111-120 d	7590
Odocoileus virginianus	f: 35 kg; 6 mo	9975
	m: 37 kg; 6 mo	9975
Rangifer tarandus	~20 kg	5637
LITTER SIZE		
Alces alces	1	9542
	1-2	9427
	1-2	2483
	1-2; twins not rare	11156
	1; r1-3	3477
	1 yg ea; n6 lt	11262
	mode 1 yg; 4.5% twins; r1-2	1871a
	mode 1; <5% twins; r1-2	7623
	5-33% twins; n13 studies	3478
	1.06; r1-2; n160 lt	5993
	1.1 emb; r1-2; n64 f	8862
	1.11-1.16/100 f; r1-2	3620
	1.12 cl; 1 emb; yg f; 1.25 cl; 1.16 emb; adult; n424 lt	9597
	1.12-1.50 cl; across regions, yr	6818
	1.135; 13.5% twins; 3 rare; r1-3; n1217 f	8411
	1.14 emb; r1-2; n330 f	1012
	1.15; 15% twins; r1-2; n1158	8547
	1.15; r1-2; n2047 lt	8411
	1.22; r1-2; n129 lt	2903
	1.28 cl; r1.06-1.65; n8 regions; 11-29% twins 5 regions	9597
	1.33 emb; se0.05; mode 1 55%; r1-3	5990
	1.38 yg; 38% twins; n238 f	543a
	1.47 cl; r1-3; n121 f	9979
	1.543 yg; r1-2; n151 lt	3480
	1.59 emb; se0.06; mode 2 53%; r1-3	5990
	1.89; r1-2; n19 lt, 2 f	10056
	3 yg; n1 lt; anecdote	5004

Blastocerus dichotomus	mode 1	10809
	1 emb; n1 f	9551
Capreolus capreolus	1-2; r1-3	3571
	mode 1 73%; r1-2	8759
	1+; mode 2	3388
	1.25; r1-2; n4 lt	953
	1.3; r1-2; n3 lt	6243
	1.54-1.86	6118
	1.76 emb; mode 2; r1-3; n17 f	1699
	1.8	266
	1.83; r1-3; n12 lt	3092
	1.85 emb; mode 2; r1-3; n149 f	10510
	1.91 emb; mode 2; r1-4; n108 f	1707
	1.92; mode 2; r1-3; n38 lt	8759
	2 emb; mode 2; r1-4; n73 f	11443
	2 yg each of 2 lt	11500
Cervus axis	1	9486
	1	2936
	1; n1 lt	6654
	1 emb ea; n4 f	7382
	1 ea; n9 lt	6243
	1; 2 sets twins	9545
	1 yg; 1 set twins; n225 lt	2109
	1.01; mode 1; 1.2% twins; r1-2; n80 lt	12128
	1.03 yg; r1-2; n38 lt	9066
	1.25 yg; r1-2; n4 lt	5062
Cervus dama	1-2	7652
	1 ea; n17 lt	6243
	1 emb ea; n52 f	10380
	1 ea; n93 lt	1230
	1; 0.4% twins; r1-2; n250 lt	4426
	1 yg; 0.5% twins; r1-2; n200 lt	2109
	mode 1; 11% twins	11023
Cervus duvauceli	1	9545
	mode 1	12128
Cervus elaphus	1; mode 1	4214
	1	5635
	1	1924
	1	1230
	1	7652
	1; twins rare	1182
	1; twins extremely rare	5636
	1; r1-3	5146
	1 ea; n40 lt	6243
	1 emb ea; n211 f	10884
	1 emb; 0.08% twins; r1-2; n1186 uteri	3336
	1 emb; 0.1% twins; r1-2; n896 f	5840
	1 conceptions; 0.2% twins; r1-2; n600 lt	7365
	1 yg; 0.26% twins; r1-2; n380 lt	2109
	1 emb; 0.7% twins; r1-2; n149 f	12002
	1; 0.8% twins; r1-2; n117	12128
	1.09; n11 lt	2200
	1.4 cl; n10 lt	7647
	1.85; r1-2; n26 lt	4263
Cervus eldi	1	6654
	1 ea; n3 lt	12128
	1 yg ea; n8 lt	9475
	1; 97.5% single; n158 lt; twins died	11590
	1.04 yg; 4% twins; r1-2; n24 lt	2109
Cervus nippon	1 ea; n9 lt	6243
	1 emb ea; n9 f	7594
	1 yg ea; n187 lt	2109
	1.05; 4.6% twins; r1-2; n108 lt	12128

Cervus porcinus	1 yg ea; n32 lt	2109
	1.01; r1-2; n124 lt	1230
Cervus timorensis	1	11180
	1 ea; n21 lt	12128
Cervus unicolor	1	1230
	mode 1	12128
	1 ea; n4 lt	12128
	1 emb ea; n5 f	5710
	1 ea; n6 lt	6243
	1.02 yg; 1.5% twins; r1-2; n64 lt	2109
Elaphodus cephalophus	1; n>1	4592
Elaphurus davidianus	mode 1	11592
	1-2	4425
	1 yg; n1 lt	10536
	1 ea; n6 lt	1230
	1 ea; n6 lt	6243
	1 yg ea; n16 lt	2109
	1 yg ea; n143 lt	11590a
Hydropotes inermis	mode 2; max 3	2109
	mode 2-3 yg; 1,4 rare	2653
	2.3 yg; mode 2; r2-4; n12 lt	7289a
	3; 1 dead; n1 lt	2054
	3 yg; 1 lt	11999
	3.5; r3 yg, 4 emb; n2 lt	3127
	4 cl	3127
	5-6	182
Mazama americana	1-2	3633
	1-2	11642
	1	11387
	1 emb; n1 f	7624
	2; mode 2	4263
Mazama gouazoubira	1 emb ea; n12 f	10302
	1 yg; n17 lt	3444
Mazama rufina	1	7652
Moschus chrysogaster	mode 1 yg; r1-2; n7	4032a
Moschus moschiferus	mode 1; 2 rare	9836
	mode 1; 2 occasional	6654
	mode 1; r1-2	9175
	1-2	4263
	1-2	7652
	1-2	11642
	1.82; n119	613b
	mode 2	3034
	2	4643
Muntiacus muntjak	1	12128
	mode 1; 2 occasional	8498, 8501
	1-2	6653, 6654
	1-2	7652
	1-2	4263
	1 emb; n1 f	7382
	1 emb ea; n2 f	2313
	1 ea; n3 lt	6243
	1 ea; n38 lt	1230
Muntiacus reevesi	1	12128
	1 yg ea; n5 lt	9066
	1 emb ea; n11 f	1700a
	1 emb ea; n21 f	1712, 1713
	1 yg ea; n100 lt	2109
Odocoileus hemionus	1-2	2086, 10775, 10776
	1-3; mode 1 yearling, mode 2 adult	5146
	1.2 emb; r1-2; n9 yearling	7156
	1.375; r1-2; n8 lt	3876
	1.38 emb; n26 f	10613
	1.4; mode 1; r1-2	9183

	1.44; r1-2; n9 lt	11369
	1.44; n152 lt	6232
	1.45 yg; mode 1; 3 rare; r1-3; n64 lt	6491
	1.47; r1-3; n127 lt	6244
	1.5; r1-2; n2 lt	6243
	1.5; r1-2; n1 f, 2 lt	4185
	1.5 emb; r1-2; n16 f	10595
	1.54 emb; r1-4; n50 f	9713
	1.58 emb; r1-3; n246 f	9182
	1.6 yg; r1-2; n9 lt	9066
	1.60; n47 lt; r1.38-1.91 by age	10773
	1.63; r1-2; n73 lt	6244
	1.67 emb; r1-2; n3 f	9000
	1.67; mode 2; r1-2; n61 lt	7590
	1.67; mode 2; r1-3; n113 lt; r1.27-1.91; n6 herds	952
	1.67; r1-2; mode 2; n162 lt, 5 herds	952
	1.67 emb; mode 2; r1-3; n562 f	9184
	1.7 emb; mode 2 58%; r1-3; n50 f	5065
	1.75; mode 2; r1-3	1758
	1.80 emb	1466
	1.8 emb; r1-3; n41 adult	7156
	1.81 cl; r1-2; n79 f; 1.67 emb; r1-2; n85 f	7825
	1.83 yg; r1-2; n6 lt	7878
	1.9 emb; mode 2; r1-3; n8 f	10595
	1.92 emb; mode 2 67%; r1-3; n24 f	5303
	1.98 cl; n40 lt	7156, 7157
	2 repeatedly; n1 f	9303
	2.06 cl; 1.73 emb; r1-3; n59 f	12072
	5 emb; n1 f	7827
Odocoileus virginianus	mode 1; 2 occasional	3024
	mode 1; 2 occasional	9385
	mode 1; r1-2	7145
	1-1.6 cl	11637
	1-3 cl	10730
	1-3 yg	8182
	1-4	1770, 9809
	r1.0-2.267 emb; n~100+?	2263
	1 yg; n1 lt	6142
	1.0 emb ea; n6 fawns	5874
	1.09 cl; fawn; 2.27 cl; adult	4506
	1.1 cl; r1-4	5464
	1.2 emb; n5 f	8605
	1.2 yg; r1-2 n6 lt	1013a
	1.21 cl; n19 f	1309
	r1.22-2.18 cl; r1.00-1.92 emb; n8 samples	8917
	1.25 cl; fawns; 2.04 cl yearling; 2.22 cl adult	7942
	1.25; r1-2; n16 lt	12128
	1.25 emb; se0.07; n91 f	8180
	1.28 emb; n32 f	5134
	1.28 emb; mode 1; r1-2; n186 f; incl 4 fawns	9112
	1.3 fawns; minimal production 1974-1977	3620, 7837
	1.3; r1-2; n7 lt	6243
	1.3; r1-2; n10 lt	4503
	1.4 emb; n12-15 f	5861
	1.45; mode 1; r1-2; n42 lt	10005
	1.5 yg; r1-2; n2 lt	7878
	1.5 emb; n100 f	7614
	~1.5 cl	6783
	1.52 yg; r1-3; n132 lt	11259
	1.59 emb; 1.85 cl; n331 f	615
	1.61 emb; 1.75 cl; mode 2; r1-3; n130 f	5208
	1.61 emb; n318 f	5329
	1.62 emb; n441 f	5723
	1.63 emb; n109 f	4502

	1.63 emb; n1103 f	9088
	1.66 yg; 1.77 cl; r1-3	11780
	1.66 emb; n108 f	11780
	1.67; r1-2; n12 lt	4500
	1.67 emb; n936 f	4706
	1.7 emb; n12 yearling	5874
	1.74; r1-3; 3 rare; n27 lt	11260
	1.79 emb; se0.09; n389 f	8180
	1.8 emb	5464
	1.8 emb; n65 f	548
	1.86 emb; mode 2 70%; r1-3; n361 f	9282
	1.86-2.25	7941
	2	7652
	2 emb	4504
	mode 2	9164
	mode 2; r1-2	7235
	2; r1-3	4250
	2 cl; 3 emb	7043
	2 emb; n1 f	10965
	2 cl; n1 f	4705
	twins n10 birth	10881
	2 yg; n10 lt, 1 f	11258
	2.0 emb; n11 adult f	5874
	2.03 cl; n460 f	9282
	2.27; r1-3; n44 lt	6244
	2.33 yg; r1-3; n12 lt, 1 f	8211
	3 emb; n1 f	4705
	4 emb; n1 f	10909
Ozotoceros bezoarticus	1 ea; n20 lt	3446
Pudu mephistophiles	mode 1; r1-2	10809
Pudu pudu	1 yg; n1 lt	4719
	1-2	11642
Rangifer tarandus	1	7652
	1 yg ea; n9 lt	2109
	1 ea; n15 lt	6243
	1; 2 rare	806
	1 set of twins, died w/in 12 h	7057
	2 rare; 1/10,000 lt	7055
	2; n1 lt	7046
	2.9 cl; 11% 5-7; r1-7; n73	2288
SEXUAL MATURITY		
Alces alces	f: 1st cl: 24/68 yearling; 84/99 adult	9978
	f: 1st cl: 50% yearling	6816
	f: 1st estrus: 2nd autumn	4643
	f: 1st mate: 1.5 yr	1871a
	f: 1st mate: 2-3 yr	1474
	f: 1st mate: 3-4 yr	4643
	f: 1st preg: yearling	1012
	f: 1st preg: 80% yearling; n87	8862
	f: 1st preg: 3rd yr	4643
	f/m: 1st mate: yearling	8547
	f/m: 1st mate: 16-17 mo	11156
	f/m: 1st repro: 1 yr; n1 ea; high nutritional plane	9686
	f/m: 16-18 mo	3477
	f/m: 16.5-30 mo	3571
Capreolus capreolus	f: 1st cl: 4-12 mo	11443
	f: 1st mate: 14 mo	8759
	f: 1st mate: 5-6 mo	10925
	f: 1st birth: 2 yr	3388
	m: 1st mate: 3 yr	3388
	m: >1 yr	10925
	f/m: ~14 mo	3571
	f/m: 2nd yr	4645

Cervus axis	f: 14-17 mo	9545
	f/m: 1st mate: 3-4 mo; n1	3989
	f/m: 1st mate: 12-24 mo	2936
Cervus dama	f: 1st birth: 3 yr	11023
	f: 15-18 mo	4426
	f: ~16 mo	371
	f: 16 mo	1708
	f: 16-17 mo	10380
	m: testes: large 13 mo	3267
	m: spermatogenesis: 7 mo	3199
	m: spermatogenesis: yearling	1701
	m: sperm: 14 mo	1706, 1708
	m: sperm: 16 mo	1709
	m: epididymal sperm: 13 mo	3267
	m: eversion of prepuce transition zone: stops 9 mo, recommences 17 mo	1711
	m: 2nd yr	10380
	f/m: 1st mate: 17-20 mo	3266
	f/m: 1st mate: active 4-5 yr	4643
	f/m: 2.25 yr	3571
Cervus duvauceli	f: 1st birth: ~3 yr	9545
Cervus elaphus	f: 1st mate: 1-3 yr	7365, 7367
	f: 1st mate: 2 yr 4 mo	644
	f: 1st mate: 2-4 yr	7366
	f: 1st mate: 3.5-3.9 yr	1921
	f: 1st conceive: 1.5 yr	11532
	f: 1st conceive: 20 mo; n1	643
	f: 1st conceive; 2-3 yr	6463
	f: 1st preg: 1 yr	2377
	f: 1st preg: 18 mo; n1	9510
	f: 1st preg: 20 mo; n1	1957
	f: 1st preg: 0-78% yearling	3336
	f: 1st preg: 8-17% yearling	3368
	f: 1st preg: yearling; but not many	5840
	f: 1st birth: 14-16 mo; 65 kg	5636
	f: 1st birth: 2 yr	88
	f: 1st birth: 2 yr	5146
	f: 1st birth: 3.25-3.64 yr	1923
	f: 1st birth: 3-4 yr	4139
	f: 1st birth: 3-4 yr	6213
	f: 1st lact: 2 yr; n4	11531
	f: 1 yr	1871a
	m: mature sperm: yearling	2000
	m: motile sperm: 14-16 mo	5636
	m: epididymal sperm: 2nd yr	7365
	m: secondary sex characters: 9-15 mo	6450
	m: testes large: ~4 yr	5634
	m: 1st sire: yearling	7505
	f/m: mature gonads: 18 mo	12002
	f/m: 1st repro: ~3 yr	12002
	f/m: active 5-6 yr	4643
Cervus eldi	f: 1st conceive: 468.9 d; r340-583; n16 f	11590
	f: 1st birth: 31.5 mo; r21-42; n2	8640
Cervus nippon	f: 1st preg: 8 mo; n1	1714
	f: 1st preg: yearling	2306
	f: 1st birth: 3 yr	4643
	f: min 6 mo	7594
	f/m: 1st mate: 3-4 yr	4643
	f/m: 1.25-1.5 yr	3571
	f/m: 16-18 mo	3198
	f/m: 1.5 yr	4643
Cervus timorensis	f: 1st mate: 11-12 mo	11060
	f: 1st conceive: 2 yr 3 mo	11084
	f: 18-20 mo	11180

	f: 54-59 kg	6473
Cervus unicolor	f: 1st birth: 2 yr 1 mo 28 d	30
	f: 1st birth: ~3 yr	9545
Elaphodus cephalophus	f: 18 mo	4592
Hydropotes inermis	f: 1st preg: 6 mo	7289a
Mazama gouazoubira	f: 20.5 mo	3444
Moschus moschiferus	f: 1st mate: < 1 yr	6653, 6654
	f: 18 mo	9175
	m: 1st mate: 21 mo	3034
	m: 1st sire: 1 yr	3444
	m: 3 yr	9175
	f/m: 1st mate: 15-17 mo	4643
Moschus sifanicus	f: 1st birth: 2 yr; n1	5232
	f/m: 1st mate: 1.5 yr; n1	5232
Muntiacus crinifrons	f/m: 1st mate: subadult	4836
Muntiacus muntjac	f: 8 mo	613a
	m: 12 mo	613a
Muntiacus reevesi	f: 1st ovulate: <12 mo	9423a
	f: ~6 mo	1709a
Odocoileus hemionus	f: 1st cl: 6 mo	5444
	f: 1st conceive; 6-8 mo; but few do	9182
	f: 1st mate: 16 mo	2086
	f: 1st mate: 17 mo	10614
	f: 1st mate: 18 mo	10615
	f: 1st mate: 1.5 yr	12072
	f: 1st mate: 2.5 yr; some yearling	10595
	f: 1st mate: 85% yearling	10776
	f: 1st birth: 50% 1 yr; 22.5-33.5 kg	7587, 7589
	f: 1st birth: 2 yr	7156
	f: 1st lact: 1.5 yr; n1	10777
	f: ~1 yr; n1	2110
	f: 2nd yr	9184
	f/m: 1st mate: 18 mo	7878
	f/m: 1st mate: few as fawns, yearling	10595
	f/m: 1st repro: 1 yr	10775
	f/m: 195-212 d	5146
	f/m: some 16 mo	2618
	f/m: some 1.5 yr	952
Odocoileus virginianus	f: 1st estrus: 11 mo	1227a
	f: 1st ovulate: 1/25 yearling; est 2 yr	5464
	f: 1st cl: 4-7 mo	4506
	f: 1st cl: 6-8 mo 5/21; 8-10 mo 17/17	1308, 1309
	f: 1st conceive: 0.5 yr; n3/73; mode 1.5 yr	5723
	f: 1st conceive: 6 mo; 5/160	10730
	f: 1st conceive; 6 mo 24%; 18 mo 21%	548
	f: 1st conceive: 214 d; n1	9976
	f: 1st mate: 5-8 mo	7942
	f: 1st mate: 0.5 yr	5329
	f: 1st mate: 1.5 yr	1474
	f: 1st preg: 6 mo	9088
	f: 1st preg: 9 mo; mode 10-15	1308
	f: 1st birth: 1 yr	7145
	f: 1st birth: 463 d; n1	9976
	f: 6-9 mo	11265
	m: epididymal sperm: 48% fawns	3369
	m: 1st sire: 202 d; n1	9976
	m: 1st sire: <1.5 yr; n5	9667a
	m: 16 mo	1871a
	f/m: 1st mate: 4-7 mo; 83.6% of fawns	4506
	f/m: 1st mate: 6-18 mo	1769, 9809
	f/m: 1st mate: no fawns	1651
	f/m: 1st birth: 1.5 yr	4250
Ozotoceros bezoarticus	f: 1 yr	3446

Pudu pudu	f: 1st mate: 0.5 yr	11221
	m: 1st sire: 9 mo	2204
Rangifer tarandus	f: 1st mate: 18 mo; some 4-5 mo	8744
	f: 1st ovulate: 1.5 yr; 2/100 yearling, 46/100 2 yr	2288
	f: 1st conceive: 6-18 mo	9021
	f: 1st conceive: 16-17 mo	6320
	f: 1st conceive: 3-4 yr	7055
	f: 1st preg: 8 mo; occasionally 5 mo	7921
	f: 1st preg: 17 mo	7055
	f: 1st preg: 21 mo if >46 kg	10774
	f: 1st preg: 22 mo; 9/21	8235
	f: 1st preg: 29-30 mo	801
	f: 1st preg: 2.5 yr	2987
	f: 29-41 mo	806
	m: sperm: 17-18 mo	7055
	m: testes large: 7 yr	6319
	m: 1st mate: 18 mo	7921
	m: 16 mo; n1	2109
	f/m: 1st mate: 3rd yr	4643
	f/m: 2 yr	4643
CYCLE LENGTH		
Alces alces	estrus: <24 yr	1871a
	estrus: mode 4-5 d; r2-5	4643
	estrus: several d	11156
	receptivity: 7-12 d	1871a
	20-22 d	1871a
	25 d	6817
Capreolus capreolus	estrus: 3-4 d	3388
	estrus: 4-5 d	4643
Cervus dama	22.4 d; r20-27; n142 cycles	413
	24-26 d	3199
Cervus elaphus	18.2 d; se0.4; n17 cycles	5636
	18.3 d; se1.7; r5-22; n78 cycles, 11 f	2200, 6463
	21.2 d; r19-25; n9 cycles	7539
Cervus eldi	mode 17 d; n3 f	11590
Cervus nippon	estrus: 1 wk	4643
Cervus timorensis	estrus: 2 d	11060
	~13 d	11060
Elaphurus davidianus	18.464 d; sd2.017; r17-25; n33 cycles, 4 f	11590a
	19.5 d; se0.6; n5 f	2175
	~31 d	11592
Muntiacus muntjac	estrus: <48 h	613a
	14-21 d	613a
Odocoileus hemionus	estrus: 24-36 h	10773
	estrus: 24-36 h	11598
	8+ d; r6.5-9.0	10773, 10776
	3-4 wk	6491
	21-23 d	7878
	22-28 d	11598
	23.7 d; sd2.9; r18-30; n20 cycles, 4 f	11882
Odocoileus virginianus	estrus: ~24 h	4250
	estrus: 24-36 h	1769, 9809
	26.2 d; sd2.1; r21-30; n8 f	5899
	28 d	4250
	28-29 d	1769, 9809
Rangifer tarandus	estrus: 1-3 d	804
	10-12 d	2561
	10-12 d	806
	~2 wk	4643
	2-3 wk	7921
	17 d; r11-26; n3	7058
	r18-25; n12 cycles, 2 f	804
	23.5 d; n14 f	2696

GESTATION

Alces alces	217 d; r216-218; n2	10401
	7 mo; field data	6817
	225 d; n2	10056
	227-235 d	11156
	234 d; r226-244	3478
	~236 d	480
	~240 d	2483
	240-246 d	3478
	~8 mo	7623
	~8 mo; field data	5993
	~8 mo	7652
	242.3 d; n3	11262
	35-38 wk	3571
Blastocerus dichotomus	271 d; n1	3447a
Capreolus capreolus	implant: 2-5 mo	4643
	implant: 3 mo	12096
	implant: 4-5 mo	10417, 10418
	implant: 4-5 mo	3571
	implant: 150 d	8759
	implant: 5.5 mo	11443
	implant: 5.5 mo	9939
	postimplantation: 4 mo	9939
	postimplantation: 4.5 mo	11443
	postimplantation: 144 d	8759
	postimplantation: 5 mo	10417, 10418
	postimplantation: 5-6 mo	3571
	postimplantation: 28 wk	12096
	4 mo	3388
	~22 wk; at 2nd mating	3571
	~9 mo	4645
	~9 mo; r6-10	4643
	9-9.5 mo	7652
	9-10 mo	10417, 10418
	9-10 mo	3571
	40 wk	953
	40 wk	12096
	9.5 mo	9939
	294 d	8759
	10 mo	11443
Cervus albirostris	10 ? mo; field data	3034
Cervus axis	210-238 d	4263
	7 mo 8 d	1355
	~229 d	3989
	8-8.5 mo	9545
	~8 mo; some observers say ~6 mo	8498, 8501
	8.25 mo	7652
Cervus dama	7 mo	1523
	~7 mo; field data	4426
	7 mo	6243
	229 d; r225-238; n8	7425
	230 d; r225-234; n11	8742
	230-240 d	4263
	33-35 wk	3199
	7.5 mo	11023
	7.5 mo; field data	4414
	8 mo	1355
	8 mo; field data	8344
	8 mo; field data	2682
Cervus duvauceli	8 mo	7652
	256 d; n1	7425
Cervus elaphus	implant: 6 wk	10789
	~7 mo; field data	9222
	~7 mo; field data	644

	7.5 mo; ~8.5 mo	7652
	225-246 d	4263
	~226 d	1230
	228 d; r227-229; n2	6463
	231-238 d	3571
	230.2 d; sd4.4; n11 1st preg; 234.3; sd0.95; n6 later preg	6460
	231.0 d; sd4.5; n11	2200
	231.491 d; captive; 236.587 d; r220-251; wild	4139
	~233 d	7367
	233.1 d; se0.6; n38	5636
	234 d; r230-239; n9	8742
	234 d; se0.5; n32	51
	234.2 d; ±5.04; f emb; 236.1 d; ±4.75; m emb; n70	1920, 1924
	234.25 d; r231-238; n4	5635
	235 d	2151
	237 d; field data	3873
	237 d; r230-244; n4	7425
	34 wk	7256
	8 mo	2682
	8-8.5 mo	1355
	249-262 d	9385
	~250 d	7042
	255 d	5146
	~8.5 mo; r249-262 d	6213
	265 d; n1	4390
	8.5-9 mo; field data	7366
	9 mo; field data	11642
	~9 mo	7652
	10-11 mo	4263
Cervus eldi	> 190 d	8640
	~7 mo	6654
	240.4 d; r236-244; n5	9475
	241.7 d; r240-244; n3	985
Cervus nippon	~30 wk	3198
	7.25-7.5 mo	7652
	222 d; r218-229; n11	8742
	223 d; r218-230; n8	7425
	225-246 d	4263
	7.5; field data	11642
	8 mo	1355
Cervus porcinus	est 7 mo	1013a
	7.5 mo	7652
	est max 8 mo	9175
	8 mo	1355
	~8 mo	8498, 8501
	~8 mo	6654
Cervus timorensis	est 7.5 mo	6698
	240-264 d	11060
	248.6 d; sd13.4; r217-277; n19	11180
	267-284 d	11084
Cervus unicolor	~240 d	1230
	8 mo; field data	2682
	8 mo	8498, 11642
	~8 mo	8501
	246 d; n1	7425
	8-9 mo	7652
Elaphodus cephalophus	~6 mo	3034
Elaphurus davidianus	8.5 mo; field data	4262
	283.38 d; sd6.11; mode 280; n21	11590a
	9.5 mo	4582
	285-300 d; field data	9548
	288 d	11592
	11 mo	9939
Hippocamelus antisensis	240-270 d	7233a

Mazama gouazoubira	206 d; courtship to birth	3444
	~225 d	10788a
Mazama rufina	7.25 mo	7652
Moschus berezovskii	178-192 d	4032a
Moschus chrysogaster	170-198 d; n4	4032a
Moschus moschiferus	3-4 mo; field data	3034
	~5 mo	6653
	~160 d	9175
	5.5 mo	7652
	185-195 d	4643
	7.5 mo; field data	9836
Muntiacus muntjak	5-6 mo	6653, 6654
	~180 d	9175
	~6 mo	8501
	~6 mo	7652
	210 d; r205-215; n40	3447a
Muntiacus reevesi	6 mo	7652
	209-220 d	1700
Odocoileus hemionus	199.7 d; r194-205; n9 lt, 4 f	7878
	r200-203 d; n207	10773
	202-212 d	6387
	203 d; r199-207; n5	3876
	~205 d	6491
	207 d; r205-209; n4	2618
	7 mo; field data	6386
	7 mo; r183-212 d; n10	2086
	~7 mo; r205-212 d; field data	2618
Odocoileus virginianus	implant: 5-6 wk	9809
	6 mo; field data	11637
	193.8 d; r187-198; n12	4500
	196 d	1769
	mode 196 d; r189-210; 7 lunar mo	372
	199.2 d; se0.4; n119	8605, 11266
	199.4 d; n21	9809
	~200 d; n2 lt, 2 f	7878
	200.8 d; se0.4; n174	11263
	202 d	5213
	202 d; r195-212	4250
	202.1 d; ±4.1; r196-213; n55	11261
	204 d; 90% 197-208; r197-222; n10	4503
	7 mo; field data	11642
	~7 mo	627
	~7 mo	1355
	~7 mo	7652
	~7 mo; r205-212 d	6213
	270 d	10529
Ozotoceros bezoarticus	est 7 mo	3446
	7+ mo	5212
Pudu pudu	5.5 mo	4717
	est 202-203 d	2204
	202-223 d	6293
	207 d; n1	11221
Rangifer tarandus	implant: 27-29 d postovulation	7055
	~6 mo	4389
	200-210 d; n1	3714a
	207 d; r204-213; n3	7425
	208.0 d; se2.9; n5	7058
	7 mo; field data	6319
	216.4; se1.7; n15	7058
	217 d; n2; artificial insemination; 209, 214 d; n2 f	2696

	220-225 d	7921
	~223 d; gestation of m emb 2.7-5.4 d > for f emb	4643
	~227 d	7055
	229 d; r227-230; n4	800, 804, 806
	240 d	8210
	8 mo	7652
LACTATION		
Alces alces	solid food: 14 d	11156
	solid food: 2 wk	4643
	solid food: 2 wk	3477
	solid food: 1.5 mo	4645
	solid food: 1.5-2 mo	3571
	wean: 3-3.5 mo	4645
	wean: max 92 d	4840
	wean: 3.5-4 mo	4643
	wean: ~4 mo	3571
Capreolus capreolus	solid food: 6-8 d	3092
	solid food: 1 mo	4643
	nurse: max > 15 mo	8759
	wean: ~2 mo	4645
	wean: 2-3 mo	4643
	wean: 2-3.5 mo	6342
	wean: 4.5 mo	3092
Cervus axis	wean: 4 mo	3989
Cervus dama	solid food: 4 wk	4643
	wean: min 20 d	3635
	wean: 3.5-4 mo	3571
	nurse: 4 mo	1708
	nurse: min 6 mo	1701
	milk present: ~7 mo; field data	5211
Cervus elaphus	solid food: 3-4 wk	5338
	solid food: 3-4 wk	9688
	solid food: 1 mo	4643
	solid food: 6 wk	9385
	wean: 4 mo	9688
	wean: 4-5 mo	7042
	wean: 5-7 mo	1924
Cervus eldi	wean: 35 wk	11590
Cervus nippon	solid food: 10-12 d	4643
	wean: 3-4 mo	3571
	wean: 4-5 mo	4643
	wean: 8-10 mo	3198
Cervus timorensis	wean: 7-8 mo	11180
Elaphurus davidianus	nurse: 4-7 mo	11592
Odocoileus hemionus	solid food: few d after birth	6491
	solid food: 2-3 wk	7878
	solid food: 2-3 wk	2926
	solid food: 1 mo	9376
	wean: 5-16 wk; some 32-56	263
	1st rumination: 50.2 d; r36-58; n≤7	8240
	wean: 60 d	7878
	wean: 60-75 d	2618
	nurse: 3 mo	9376
	wean: 3-4 mo	9183
Odocoileus virginianus	solid food: 2 wk	11259
	solid food: 2-3 wk	9809
	wean: min 20 d	3635
	wean: 4 mo	9809
	wean: 10 wk	10150
	wean: 5 mo	1308
	wean: 5-7 mo	9975
	wean: min 4 mo; max 6-7	9525

Pudu pudu	solid food: 12 d	4717
	lact: 60 d; n1	4719
Rangifer tarandus	wean: 2-2.5 mo	5637
	wean: 20 wk	11640
	wean: 5+ mo	5932
INTERLITTER INTERVAL		
Alces alces	1 yr	2483
Capreolus capreolus	1 yr	8759
Cervus axis	~14 mo	3989
Cervus dama	1 yr	11023
	1 yr	1708
Cervus eldi	11 mo; r9-13; n5	8640
	380.7 d; se105.0; n60 yg live	11590
	384.7 d; se93.3; n28 yg die	11590
	387 d; r298-635; n13 intervals, 4 f	985
Cervus timorensis	366 d; sd6.8; n34	11180
Elaphurus davidianus	359.5; sd17.7; n102	11590a
Muntiacus muntjac	241 d; n60	3447a
Muntiacus reevesi	209-220 d	1700
	216 d; r214-219; n4	613a
Odocoileus hemionus	1 yr	7156
	1 yr; n1	4185
Odocoileus virginianus	~10 mo	1309
Ozotoceros bezoarticus	252-408 d	3446
Rangifer tarandus	1 yr	7055
SEASONALITY		
Alces alces	spermatogenesis: Aug-Oct; Europe, w USSR	4645
	estrus: Aug-Jan; zoo	6243
	estrus: mid Sept-Oct; zoo	11156
	estrus: end Sept-Oct; n Sweden	6817
	mate: ?-Sept 15; Mongolia	9542
	mate: mid Aug-Oct; USSR	4643
	mate: end Aug-Oct; Europe, w USSR	4645
	mate: end Aug-Oct; n Poland	3571
	mate: end Aug-Oct; Trondelag, Norway	7652
	mate: early Sept-?; n Sweden	6817
	mate: Sept; BC, Canada	3666
	mate: Sept; MI	11262
	mate: Sept-mid Oct; MI	7623
	mate: Sept-Oct; Canada, N US	1474
	mate: mid Sept-mid Oct; n USSR	4643
	mate: mid Sept-mid Oct; NEW, ONT, Canada, AK	8411
	mate: mid Sept-Oct; Sweden	480
	mate: mid Sept-early Nov; WY	2483
	mate: end Sept-early Oct; Canada	6379
	mate: end Sept-1st 3 wk Oct; se Norway	5993
	ovulate: Sept-early Oct; Sweden	480
	ovulate: Oct; Sweden	6817
	ovulate: Oct; Trondelag, Norway	4179
	ovulate: Oct-Nov; no data other mo; Sweden	6816
	conceive: yearling before Oct 16; Finland	8862
	conceive: end Sept-early Oct; nw ONT, Canada	9978
	conceive: est mid-Dec; Fairbanks, AK	1929
	birth: mid Mar-Oct; Trondelag, Norway	4179
	birth: Apr-May; WY	2483
	birth: Apr-June; n33; zoo	5818
	birth: end Apr-early May; n Poland	3571
	birth: early May; Europe, w USSR	4645
	birth: May; n1; zoo	6243
	birth: May-June; Canada, n US	1474
	birth: peak mid-end May 90%; AK	3620
	birth: mid-end May; Sweden	480
	birth: mid May-early June; NEW, ONT, Canada, AK	8411
	birth: mid May-mid June; MI	7623

	birth: mid May-mid June; se Norway	5993
	birth: mid May-mid June; USSR	4643
	birth: end May-early June; WY	1871a
	birth: end May-early June; Sweden	6817
	birth: end May-June; N Am	3477
	birth: Aug-Sept; Mongolia	4645, 9542
	birth: mid Aug, rare; QUE, Canada	7437
Blastocerus dichotomus	birth: May-Sept; Pantanal, Brazil	10809
Capreolus capreolus	sperm: epididymis end May-Dec; Germany	10418
	sperm: epididymis ?-Dec; Bern, Switzerland	11443
	spermatogenesis: Apr-?; Germany	12096
	spermatogenesis: May-Sept; USSR	10925
	spermatogenesis: mid May-mid Aug; Germany	10418
	spermatogenesis: June-Aug; Germany	10417
	spermatogenesis: peak June-Aug; Hungary	7920
	spermatogenesis: end July/early Aug-Sept; Europe, w USSR	4645
	testes: large Apr-July, small Sept-Mar; Bern, Switzerland	11443
	testes: large June-Aug; Bavaria, Germany	1384
	testes: peak June-Aug; Germany	10417
	testes: peak mid July-mid Aug; England, UK	9940
	testes: increase Aug; Germany	12096
	estrus: June-July; zoo	6243
	estrus: July-Aug, Oct-Nov; Germany	10418
	mate: July; w USSR; a few wk later; e USSR	4643
	mate: July-Aug; UK	9939
	mate: July-Aug; Bavaria, Germany	1384
	mate: July-Aug; central Germany	10510
	mate: July-Aug; Switzerland	11443
	mate: July-Aug; captive	9553
	mate: mid July-mid Aug; Europe	7652
	mate: mid July-mid Aug; Dorset, England, UK	1226, 8759
	mate: mid July-mid Aug; England, UK	9940
	mate: mid July-mid Aug; UK	3388
	mate: mid July-mid Aug; captive	620, 9554
	mate: mid July-Aug, some Oct-Nov; Germany	10417, 10418
	mate: peak mid July-early Sept, minor peak end Nov-early Dec; Germany	3571
	mate: end July-Aug; Germany	953
	mate: end July-Aug; Dorset, England, UK	8759
	mate: end July-Sept; Europe, w USSR	4645
	mate: Aug; Germany	12096
	mate: mid Aug-Sept; USSR	10925
	ovulate: end Aug; Germany	953
	cl: well formed Nov, regressed Dec; Germany	12096
	preg f: Nov-Jan; n4; Germany	953
	implant: Dec; UK	3388
	implant: Dec; Germany	10417, 10418
	implant: ~Dec; Germany	953
	implant: Dec-Jan; central Germany	10510
	implant: end Dec; Dorset, England, UK	8759
	implant: end Dec-mid Jan; USSR	4643
	implant: Jan-Feb; Bern, Switzerland	11443
	birth: Apr-?, peak May; Europe, w USSR	4645
	birth: Apr-May; Germany	10417, 10418
	birth: end Apr-June, peak May; England, UK	8759
	birth: end Apr-early July; Germany	953
	birth: May, sometimes Apr; Europe, w USSR	4645
	birth: May; UK	9939
	birth: peak May; w USSR; 2-3 wk later; e USSR	4643
	birth: May-mid June; Germany	4880
	birth: May-June; Dorset, England, UK	1226
	birth: May-June; Germany	3571
	birth: May-June; Switzerland	9430
	birth: May-early July; zoo	6243

	birth: end May-early June; UK	3388
	birth: end May-early July; captive, Sweden	3092
	birth: peak end May-early June; Switzerland	6118
	birth: June; Germany	12096
Cervus albirostris	mate: Sept-Oct; native accounts; e, central Tibet	3034
	birth: July; native accounts; e, central Tibet	3034
Cervus axis	spermatogenesis: increase Mar-June, peak May; India	9545
	mate: Apr-May; Nepal	7382
	mate: Apr-Aug; HI	3989
	mate: June-Aug, Jan-Feb; Sri Lanka	2936
	mate: yr rd, peak May-July; India	9545
	mate: yr rd, except Jan-Feb; zoo	6243
	conceive: peak May; India	9545
	birth: Jan-Feb; Nepal	7382
	birth: yr rd, peak early in year; Sri Lanka	8498, 8501
	birth: yr rd, peak Jan-early Mar; India	9545
	birth: yr rd, peak Feb-Mar; n110; zoo	44
	birth: yr rd, peak May-July; captive	2109
	birth: yr rd, peak Sept-Oct, Dec-Jan; Sri Lanka	2936
	birth: yr rd, peak Nov-Apr; HI	3989
	birth: yr rd, peak cold season; India	6654
	birth: yr rd, low May-Aug; n576; Chitwan Nat Park, Nepal	7360a
	birth: yr rd; n215; zoo	5818
	birth: yr rd; n80; zoo	12128
Cervus dama	ovarian activity: ceased Aug-Sept; New Zealand	414
	sperm: Mar, June, Sept, none Nov-Jan; captive, New Zealand	415
	sperm: Aug-Jan; England, UK	1706
	testes: peak Oct; s England, s Scotland, UK	1701
	testes: peak Oct-Nov, small Jan-Feb; Essex, England, UK	1706, 1708
	testosterone: high Apr; captive, New Zealand	415
	estrus: Apr, Oct-Dec; zoo	6243
	estrus: end Apr-Aug; captive	413
	estrus: Sept-Mar, peak Oct; zoo	4005
	mate: Feb-Apr; TAS, Australia	11642
	mate: Apr-May; New Zealand	2682
	mate: Apr-May; captive, New Zealand	415, 418
	mate: mid Apr-mid May; New Zealand	11642
	mate: end Aug-early Sept; captive	4486
	mate: end Aug-Sept; Essex, England, UK	1708
	mate: end Sept-Oct; USSR	4643
	mate: Sept/Oct-?; captive	2109
	mate: Oct; se Europe, Asia	7652
	mate: peak Oct; Spain	1243
	mate: peak Oct; Alsace, France	9528
	mate: Oct-early Nov; England, UK	1706
	mate: Oct-early Nov; UK	1523
	mate: Oct-early Nov; UK	4263, 4414, 4426
	mate: Oct-Nov; captive	8344
	mate: Oct-Dec; zoo	6243
	mate: mid-end Oct; W Germany	11023
	mate: mid Oct-Nov; 2 wk; Germany	3571
	mate: Nov-Dec; zoo	1230
	conceive: end Sept-early Dec; Essex, England, UK	1708
	conceive: Oct-Feb; captive	3266
	conceive: mid-end Oct; e Europe	10380
	birth: May-July; captive, s England, UK	4263, 4414, 4426
	birth: May-Sept; n8; zoo	7425
	birth: mid May-early June; W Germany	11023
	birth: end May-June; UK	1523
	birth: peak end May-early June; Spain	1244, 1245
	birth: June; Essex, England, UK	1708
	birth: June; s England, s Scotland, UK	1701
	birth: June; USSR	4643
	birth: June; captive	8344

	birth: June; zoo	7174
	birth: June, Sept; n3; W Germany	3267
	birth: June, Oct; captive	3266
	birth: mid-end June; Hampshire, England, UK	5211
	birth: June-early July; Germany	3571
	birth: June-Aug; zoo	6243
	birth: June-Dec, peak June; n202; captive	2109
	birth: June-Dec, peak June; n51; zoo	5818
	birth: Dec; Australia, TAS, Australia	11642
	birth: Dec-Jan; New Zealand	2682
	birth: Dec-Jan; New Zealand	11642
Cervus duvauceli	mate: peak Sept; Assam, India	11033
	mate: Oct; India	7652
	mate: mid-end Oct; India	6653
	mate: mid-end Oct; captive	4488
	mate: mid Dec-mid Jan; India, Nepal	11642
	mate: Aug-early Oct; England, UK	11642
	birth: May-Nov; n54; captive	5818
	birth: Apr-Dec, peak Apr-July; zoo, Germany	3447a
	birth: June-July; England, UK	9545
	birth: June-Aug; n8; captive	12128
	birth: July-Aug; n India	9545
	birth: Sept-Oct; central India	9545
Cervus elaphus	sperm: Sept-Mar; captive	4208
	spermatogenesis: mid Aug-Oct; Europe, w USSR	4645
	testes: peak Apr-May, low Aug; New Zealand	613
	testes: peak Aug; captive	4208
	testes: peak Sept; Canada	3336
	testes: large Sept-Oct; Bavaria, Germany	1384
	testosterone: peak Aug; captive	4208
	testosterone: peak Sept; captive	10563
	estrus: Apr; Invernary, New Zealand	5634, 5636
	estrus: end Aug-Oct, peak mid Sept-mid Oct; Germany	10789
	estrus: Oct-May/June, some Oct-Nov; Rhum, Inner Hebrides, UK	2200
	estrus: yr rd; zoo	4535
	mate: Mar-Apr; New Zealand	2682
	mate: Mar-Apr; QLD, Australia	9222
	mate: Mar-July; CA	4390
	mate: mid Mar-mid Apr; New Zealand	11642
	mate: mid Apr-mid May; New Zealand	5633, 5635
	mate: July; location not specified	4263
	mate: Aug-Sept; MT	5888
	mate: Aug-Nov, peak mid Sept-mid Oct; USSR	4643
	mate: mid Aug-Oct; Europe, w USSR	4645
	mate: end Aug-Sept; s Sweden	88
	mate: end Aug-Sept; Algeria, Tunisia	7256
	mate: end Aug-early Oct; N America	7652
	mate: Sept-Oct; Rhum, Inner Hebrides, UK	6463
	mate: Sept-Oct; Bavaria, Germany	1384
	mate: Sept-Nov; Rhum, Inner Hebrides, UK	4139, 5124
	mate: mid Sept-Oct; OR, WA	1182
	mate: mid Sept-Oct; Rhum, Inner Hebrides, UK	6450
	mate: mid Sept-mid Oct, & 1 Mar; Apache Co, AZ	11420
	mate: mid Sept-mid Oct; ID	9385
	mate: mid Sept-mid Nov; Amu-Daria R, USSR	6765
	mate: mid Sept-mid Nov; captive	3873
	mate: mid Sept-Nov; Rhum, Inner Hebrides, UK	7365
	mate: end Sept-early Oct; WI	5210
	mate: end Sept-mid Oct; Afognak I, AK	644
	mate: end Sept-Oct; Asia	6654
	mate: end Sept-mid Nov, peak Oct; Scotland, UK	7366
	mate: early Oct-Feb; Rhum, Inner Hebrides, UK	2200
	mate: Oct; according to natives; w Tatsienlic, Tibet, China	3034

mate: Oct; Rhum, Inner Hebrides, UK	1920
mate: Oct; Rhum, Inner Hebrides, UK	2151
mate: Oct-Feb; semiwild, central Bohemia, Czechoslovakia	631
mate: Oct-Nov; captive	50
mate: Oct-Nov; zoo	6243
mate: Nov; Germany	12096
ovulate: mid Mar-May; New Zealand	5633
ovulate: Sept-Oct; Rhum, Inner Hebrides, UK	6460
ovulate: Sept-Nov, peak Oct; WA, MT, OK	4214
ovulate: peak mid Oct; captive	52
ovulate: Oct-Jan; Rhum, Inner Hebrides, UK	6463
ovulate: Nov; Germany	12096
cl: mid Mar-June; New Zealand	5633
conceive: Sept-Oct; Germany	10789
conceive: Sept-Oct; Canada	3336
conceive: Sept-Nov, peak Oct; Rhum, Glen Feskie, Inner Hebrides, UK	7367
conceive: mid Sept-Oct, +1 mid Dec; Rhum, Inner Hebrides, UK	6463
conceive: mid Sept-Oct; MT	7543
conceive: Oct; Rhum, Inner Hebrides, UK	1924, 2151
preg f: Oct-May; Crimea, Russia	12002
preg f: end Nov-Mar; Germany	10789
birth: Jan; New Zealand	5635
birth: Apr; Asia	6654
birth: Apr-May; Owens Valley, CA	7042
birth: Apr-Nov; n168; zoo	2109
birth: peak end Apr; Algeria, Tunisia	7256
birth: May; Crimea, USSR	12002
birth: May-early June; WY	1871a
birth: May-June; location not specified	4263
birth: May-June; w US	6213
birth: May-Aug; Rhum, Glen Feskie, Inner Hebrides, UK	7366, 7367
birth: May-Aug; Rhum, Inner Hebrides, UK	4139
birth: May-Aug, peak June; n64; zoo	5818
birth: peak May-June, some Oct-Nov; Rhum, Inner Hebrides, UK	1920, 1924, 2151, 2200, 6460, 6463
birth: May-Oct; n218; zoo	2109
birth: May-Sept, peak June-July; n50; zoo	12128
birth: mid May-early July; Europe, w USSR	4645
birth: mid May-June; Germany	3571
birth: mid May-mid June; ID	9385
birth: mid May-mid June; OR, WA	1182
birth: mid May-mid June, & 1 Nov; Apache Co, AZ	11420
birth: mid May-mid June; USSR	4643
birth: end May; Afognak I, AK	644
birth: end May-early Nov; n67; zoo	12128
birth: peak early June; WA	9688
birth: June; WY	2105
birth: June-July; n5; zoo	2109
birth: June-Oct; n4; zoo	7425
birth: Aug-Sept; CA	4390
birth: Nov-Dec; QLD, Australia	9222
birth: Nov-Dec, late births Jan, Mar; zoo	1230
birth: mid Nov-Dec; New Zealand	2682
birth: end Nov-early Dec; Invemary, New Zealand	5636
birth: Dec; n3; New Zealand	11520
birth: end Dec; New Zealand	11642
wean: Jan; Canada	3336
wean: Sept; Crimea, USSR	12002
wean: Sept; CA	7042
wean: Dec-Feb; Rhum, Inner Hebrides, UK	2151

Cervus eldi	mate: mid Mar-mid May; Burma	6654
	mate: Nov-June; zoo	985
	mate: winter; zoo	4488
	birth: May-Nov, peak Oct; zoo Germany	3447a
	birth: Aug-Jan; zoo	985
	birth: Sept-Mar; n25; zoo	2109
	birth: 80% mid Sept-Nov; captive	11590
	birth: Oct-Nov; Burma	6654
Cervus nippon	mate: May; New Zealand	11642
	mate: Sept-?; captive	3964
	mate: Sept-Jan, peak Oct-Nov; Japan	7388
	mate: Sept-mid Nov; Europe, w USSR	4645
	mate: mid Sept-mid Nov; Dnjepr, USSR	3126
	mate: end Sept-Nov; 25-35 d; USSR	4643
	mate: Oct-Nov; Germany	3571
	mate: Oct-Nov; s Ukraine, USSR	10368
	birth: Jan; New Zealand	11642
	birth: end Mar-mid Oct, Nov-Dec, peak May-June; n115; zoo	12128
	birth: Apr-Oct; n8; zoo	7425
	birth: May-June; Dnjepr, USSR	3126
	birth: May-June; s Ukraine, USSR	10368
	birth: May-June; USSR	4643
	birth: May-Oct; n187; captive	2109
	birth: May-Dec, peak July; n95; zoo	5818
	birth: end May; Europe, w USSR	4645
	birth: end May-June; Dnjepr, USSR	3126
	birth: June; Germany	3571
	birth: peak mid Dec; New Zealand	2306
Cervus porcinus	antlers cast: Nov-Feb; Orissa, India	45
	estrus: Aug, Nov-Dec; zoo	6243
	mate: Apr-May; Sri Lanka	8501
	mate: Sept-Oct; Asia	6654
	mate: Sept-Dec; zoo	6243
	mate: Nov-Dec; Sri Lanka, India	11642
	birth: Feb-May, peak May; Chitwan Nat Park, Nepal	7360a
	birth: Feb-June; Baewan I, Indonesia	1013a
	birth: Mar-July; Pakistan	9175
	birth: Mar-Aug, Nov-Dec; zoo	6243
	birth: May-June; Asia	6654
	birth: Nov-Dec; Sri Lanka	8498, 8501
	birth: yr rd; captive	2109
	birth: yr rd; zoo	1230
	birth: yr rd; zoo	5818
	lact f: Mar; n1; Nepal	7382
Cervus timorensis	mate: May, Aug, Oct, Dec; captive	11060
	mate: July-Aug; New Zealand, New Caledonia	11642
	birth: Jan, May, July-Sept; captive	11060
	birth: Feb, Apr-June, Sept-Dec; n21; zoo	12128
	birth: Mar-July; VIC, Australia	11180
	birth: Mar-Sept; Java, Indonesia	11642
	birth: Apr-May; New Caledonia	11642
	birth: Oct-Nov; Flores, Indonesia	11642
	birth: peak Apr-June; NSW, Australia	4255
	birth: Apr-July; Pulan Rusa, Java, Indonesia	11084
	birth: Aug-Sept; Indo-Australian archipelago	11084
	birth: almost yr rd; Celebes, Indonesia	11642
	birth: yr rd; n18; zoo	5818
	lact f: 67% summer, 44% winter; captive	6698
	wean: Nov; VIC, Australia	11180
Cervus unicolor	mate: Feb-June, Aug, Nov-Dec; zoo	6243
	mate: Mar-Apr; New Zealand	2682
	mate: Mar-Apr; North I, New Zealand	11642
	mate: July-Nov; Borneo, Indonesia	11642
	mate: Oct-Nov; Nepal	7382

	mate: Oct-Nov; India, China	7652
	mate: Oct-Nov; plains India	6654
	mate: Oct-Nov; Sri Lanka	8498
	mate: Nov-Apr; s India	5331
	mate: peak Nov-Dec; India	9545
	mate: yr rd, peak May; n21; zoo	8339
	birth: May; Nepal	7382
	birth: May-Sept; Borneo, Indonesia	557
	birth: June-July; India	8498
	birth: June-July; India	6653, 6654
	birth: Nov-Dec; New Zealand	2682
	birth: Dec; North I, New Zealand	11642
	birth: yr rd, peak May-Aug; n89; Chitwan Nat Park, Nepal	7360a
	birth: yr rd, low Apr-May; India	9545
	birth: yr rd, except June-July; zoo	6243
	birth: yr rd; England, UK	6653
	birth: yr rd; n35; zoo	44
	birth: yr rd; n58; zoo	12128
	birth: yr rd; zoo	1230
	birth: yr rd; captive	2109
Elaphodus cephalophus	birth: end spring-early summer; China	4592
Elaphurus davidianus	estrus: almost yr rd; zoo	6243
	mate: mid May-Aug; zoo	9548
	mate: June-July; captive	4425
	mate: June-Oct; zoo	6243
	mate: end June-early Sept; Bedford, England, UK	9939
	mate: July; captive, England, UK	4262
	mate: mid Dec-early Jan; captive, Bedford, England, UK	2175
	birth: Mar-May; zoo	6243
	birth: Mar-July; n16; captive	2109
	birth: Mar-Aug, peak end Mar-May; n144	11590a
	birth: Mar-Sept, peak Apr; n17; zoo	5818
	birth: Mar-Nov; zoo	11592
	birth: Apr-May; zoo, Netherlands	11154
	birth: Apr-June, peak end Apr-mid May; zoo	9548
	birth: peak mid Apr-May; captive, England, UK	4262, 4425
	birth: May-Aug; Bedford, England, UK	9939
Hippocamelus antisensis	birth: Jan-Mar; s Peru	7233a
Hippocamelus bisulcus	courtship: mid Feb-mid Apr; 15 d; few data; Rio Claro, Chile	8711
Hydropotes inermis	estrus: Oct-early Dec; zoo	6243
	birth: May, July; zoo	6243
	birth: May; Hunan, Hupeh, China	182
	birth: end May-June; captive	2109
Mazama americana	birth: June; Yucatan, Mexico	3633
	birth: Sept-Apr, peak Nov-Feb; Suriname	1227a
	birth: yr rd; zoo, NY	6709a
Mazama gouazoubira	repro: not seasonal; zoo	3444, 3447a
Mazama rufina	mate: June; Guyana to Paraguay	7652
Moschus chrysogaster	birth: June-mid July; Gharwal, Himalaya, India	4032a
Moschus moschiferus	mate: Jan; Himalayas, India	6653, 6654
	mate: Jan; s Asia	7652
	mate: Jan; China	3034
	mate: est Jan; China	11642
	mate: Sept-Feb, peak Nov-Dec; USSR	4643
	mate: Nov-Dec; Pakistan	9175
	mate: end Nov; Altai, USSR	9836
	birth: Apr-May; China	3034
	birth: May-July, peak June; USSR	4643
	birth: end May-early June; Pakistan	9175
	birth: June; Nepal	7382
	birth: June; Himalayas, India	6653, 6654
	birth: est June; China	11642
	birth: June-early July; Altai, USSR	9836
Muntiacus crinifrons	preg f: yr rd; e China	4836

Muntiacus muntjak	antlers cast: Apr-May; zoo, Orissa, India	25
	estrus: Jan, Mar, May, Aug, Nov-Dec; zoo	6243
	mate: Jan-Feb; India	7652
	mate: Jan-Feb; n India	6653, 6654
	mate: Jan, Mar, May; zoo	6243
	mate: Aug-Sept; Sri Lanka	8501
	mate: yr rd; s India	6653, 6654
	preg f: Mar; n1; Nepal	7382
	birth: Feb-Mar; Pakistan	9175
	birth: Feb-Mar, July, Nov-Dec; zoo	6243
	birth: end Feb-Nov; zoo	12128
	birth: May-May, Aug-Sept, Nov-Dec; Chitwan Nat Park, Nepal	7360a
	birth: July-Aug; n India; yr rd; s India	6653, 6654
	birth: nearly yr rd; zoo	1230
	birth: yr rd; Java, Indonesia	11085
	repro: yr rd; zoo	44
Muntiacus reevesi	mate: end Jan-Feb; Kwantung, China	182
	birth: Apr-Sept, Nov-Dec; n17; zoo	12128
	birth: early June; Kwantung, China	182
	birth: yr rd, peak Apr-July; captive	2109
	birth: yr rd, peak Apr-July; captive, Japan	12030
	birth: yr rd; England, UK	1710
Odocoileus hemionus	antlers shed: Apr; few data; zoo	6243
	spermatogenesis: June-Dec; CA	6491
	estrus: Mar; few data; zoo	6243
	mate: Sept; BC, Canada	2086
	mate: end Sept-mid Nov; 10-17 d; CA	6491
	mate: Oct, fawns; captive	7587
	mate: Oct-Dec; CA	10615
	mate: mid Oct-Jan; w coast N America	2926
	mate: mid Oct-Nov; ID	9385
	mate: end Oct-Dec peak mid Nov; CA	10613
	mate: Nov usually; WY	9376
	mate: Nov-Dec; captive, Vancouver I, Canada	7589
	mate: Nov-Dec; CO	7157
	mate: Nov-Dec, rarely to mid Jan; CO	7156
	mate: Nov-Dec; WA	12072
	mate: Nov-mid Dec; CA	6386
	mate: peak mid-end Nov; WI	5210
	mate: mid Nov-Dec; CA, NV	6232
	mate: mid Nov-mid Dec; central UT	9182, 9183
	mate: peak mid Nov-early Dec, some to Jan; OR, CA	1758
	mate: early Dec; captive, AZ	7878
	mate: Dec; UT	9183
	mate: Dec-Feb; AZ	10595
	mate: mid Dec-Jan; CA	2618
	mate: mid Dec-Feb, peak mid-Dec; captive, AZ	7878
	mate: end fall-early winter; Baja CA, Mexico	6387
	ovulate: Oct-Nov; OR	5444
	ovulate: Nov; Vancouver I, Canada	10773, 10776
	conceive: Jan-Feb; AZ	10595
	conceive: end Sept-Oct; CA	6491
	conceive: mid Nov-early Feb, peak Nov-Dec; CO	7156, 7157
	birth: Mar-Nov; BC, Canada	2086
	birth: Apr; Baja CA, Mexico	10809
	birth: mid Apr-July, peak early-mid May; CA	6491
	birth: May-Aug; n11; captive	2109
	birth: last 3 wk May-early June; CA	10614, 10615
	birth: peak end May-early June; WA	10334
	birth: end May-early June; WI	5210
	birth: end May-mid July; WA	12072
	birth: early June-Aug, peak July; CA	6386
	birth: ?-June; w US	5146
	birth: June; n207; Vancouver I, Canada	10773

	birth: June; captive, Vancouver I, Canada	7587, 7589
	birth: June; WY	1871a
	birth: June; few data; zoo	6243
	birth: June; zoo	11369
	birth: peak June; ID	9385
	birth: June-mid July; central UT	9182, 9183
	birth: June-Aug; Baja CA, Mexico	6387
	birth: June-Aug; w coast N America	2926
	birth: June-early Oct, peak June-July; n22; zoo	12128
	birth: end June-early July; WY	9376
	birth: July; CA	2618
	birth: mid July; captive, AZ	7878
	birth: mid July-Sept, peak end July-early Aug; AZ	10595
	birth: end July; n1; NM	3761
	birth: end spring-early summer; AZ	7145
	wean: Aug-Sept; CA	6491
	wean: end Aug; CA	10615
	wean: Oct; UT	9183
Odocoileus virginianus	antlers shed: Feb-Apr; zoo	6243
	epididymal sperm: mid Aug-Feb, SC	8282
	sperm: peak Nov; PA	6164
	sperm: peak Nov-Dec; TX	9198
	spermatogenesis: July-Dec-Jan; MA	11856
	testes: large Aug-Jan; AR	11637
	testes: peak Oct-Nov; captive	8599
	testes: large Sept-Jan; TX	5134
	testosterone: peak Aug-Nov; captive	7102
	testosterone: peak Nov-Dec; captive	1373
	estrus: Oct-early Mar; zoo	6243
	estrus: mid-end Nov; Finland	5464
	mate: mid Jan-Mar; captive, AZ	7878
	mate: Apr-May; New Zealand	11642
	mate: Apr-July; Virgin I	11507
	mate: Apr-Oct; Suriname	1227a
	mate: June-Nov; Honduras	5861
	mate: Oct-Nov; ID	9385
	mate: Sept-Oct; TX	10730
	mate: Sept-Nov; central N America	4250
	mate: Sept-Dec; e N America	7652
	mate: end Sept-Feb, peak mid Nov; NY	9809
	mate: Oct; PA	4368
	mate: Oct-Dec; Canada	8916
	mate: Oct-Dec; NY	7235
	mate: Oct-Feb, peak early Nov; OH	7942
	mate: mid Oct-mid Jan; s IL	9282
	mate: end Oct-mid Dec; MI	8181
	mate: end Oct-Dec; ME	548
	mate: end Oct-Dec; NY	5213
	mate: Nov; zoo, Germany	627
	mate: Nov-Jan; zoo	6243
	mate: mid Nov-mid Dec; SC	8282
	mate: mid Nov-mid Mar; n387; MS	5208
	mate: peak end Nov-early Dec; NY	5214
	mate: end Dec-Feb; fawns; mid Sept-Nov; adult; SC	5329
	mate: yr rd; Venezuela	1309
	ovulate: Nov; OH	4367, 11909
	cl: none Aug-Nov; MS	5208
	conceive: end Sept-mid Dec; AR	11637
	conceive: Sept-mid Feb, peak mid Nov-early Dec; AR	11780
	conceive: end Oct-mid Feb; TX	5723
	conceive: Nov-Dec; Finland	5464
	conceive: Nov-mid Mar; n AL	6636
	birth: Mar-July; Guarica, Venezuela	10529
	birth: Mar-Sept, peak June; NY	9809

	birth: Apr-June; AR	11637
	birth: Apr-June; NY	7235
	birth: May-June; PA	4368
	birth: May-Aug, peak May-June; n16; zoo	12128
	birth: May-Sept; n177; captive	2109
	birth: mid May-early June; n10; ONT, Canada	10881
	birth: mid May-mid June; ID	9385
	birth: mid May-mid June; NY	5213
	birth: mid May-Aug, peak end May-early June; OH	7942
	birth: peak end May-early June; PA	9164
	birth: peak end May-early June; VA	7061
	birth: end May-June; central N America	4250
	birth: end May-mid Oct, peak June; TX	1506
	birth: June; ME	548
	birth: peak June; AR	11780
	birth: June; zoo, Germany	627
	birth: June-Apr; n13; captive	2109
	birth: June-mid July; e MT	2821a
	birth: June-Aug; zoo	6243
	birth: peak mid June; NY	5214
	birth: July; sw US, Sonora, Mexico	7145
	birth: end Aug; captive, AZ	7878
	birth: Sept; n1; MI	3945a
	birth: Sept; n1; MI	11257
	birth: Sept-Mar; Colombia	1013a
	birth: Sept-Apr; Suriname	1227a
	birth: mid Oct-mid Nov; New Zealand	11642
	birth: yr rd; Venezuela	1309
	preg f: end Nov-mid May, SC	8282
	lact: none Apr-June; MS	5208
	wean: July; ID	9385
Ozotoceros bezoarticus	territorial m: Apr-Oct; zoo, Germany	3446
	birth: July-Dec, peak Oct-Nov; central Brazil	8999a
	birth: Oct-Nov; Buenos Aires District, Argentina	10809
	birth: yr rd, peak Sept-Nov; Buenos Aires District, Argentina	5212a
	birth: yr rd; zoo, Germany	3447a
Pudu pudu	estrus: May; Chile	11221
	mate: Apr, July; n3; zoo	4717
	birth: Jan; n1; zoo	4717
	birth: end Apr-July; zoo	3445
	birth: Oct-Feb; zoo, NY	6709a
	birth: est Nov-Jan; w S America	11642
	birth: Dec; Chile	11221
Rangifer tarandus	tubluar sperm: Sept-early Dec; NWT, Canada	7055
	spermatogenesis: mid Sept-mid Nov; Sweden	7240
	testosterone: high Sept-Nov; captive	11644
	estrus: early Oct; Finland	5934
	mate: Mar-Apr; S Georgia, Antarctica	6319, 6320
	mate: mid Mar; S Georgia, Antarctica	1082
	mate: end Aug-Sept; captive	2109
	mate: end Aug-mid Oct; AK	8210
	mate: end Aug-Oct; AK	4184
	mate: Sept-Oct; captive, USSR	8744
	mate: Sept-Oct; AK	2561, 11640
	mate: Sept-Oct; Canada	1474
	mate: Sept-Oct; zoo	6243
	mate: Sept-Nov; USSR	4643
	mate: mid Sept-Oct; AK	7625
	mate: mid Sept-Oct; captive	8210
	mate: end Sept; Swedish Lapland	3091
	mate: end Sept-mid Oct; QUE, Canada	802
	mate: end Sept-Oct; n N America, n Europe, n Asia	7652
	mate: end Sept-Oct; WI	5210
	mate: Oct; Canada	2288, 2289

mate: Oct; NEW, Canada	803, 805
mate: Oct; nw AK	6377
mate: Oct-mid Nov; captive	7240
mate: Oct-Nov; Mackenzie, BC, Canada	5637
mate: Oct-Nov; NWT, Canada	7055
mate: Oct-Nov; captive	804, 806
mate: mid Oct-Nov?; Canada	4389
conceive: Oct; Canada	2288
birth: Apr-June; AK	8210
birth: Apr-June; zoo	6243
birth: Apr-June; n24; zoo	12128
birth: Apr-Aug; n3; zoo	7425
birth: mid Apr; AK	11640
birth: mid Apr-mid May; captive	2109
birth: May; n1; Scotland, UK	2696
birth: May; s Norway	9021
birth: May-June; Canada	4389
birth: May-June; Canada	1474
birth: May-June; n N America	804, 805, 806
birth: May-June; USSR	4643
birth: mid May; AK	7625
birth: end May-early June; captive	2499
birth: end May-June; Mackenzie, BC, Canada	5637
birth: end May-mid June; nw AK	6377, 6380
birth: end May-early July; NWT, Canada	7055
birth: June; AK	1568
birth: June-July; Canada	2288
birth: Nov; S Georgia, Antarctica	6319, 6320
wean: Aug-Sept; Mackenzie, BC, Canada	5637
wean: Sept; Canada	4389

Order Artiodactyla, Family Giraffidae

Giraffes (*Giraffa camelopardalis* 500-1930 kg) and okapis (*Okapia johnstoni* 200-250 kg) are terrestrial herbivores in Africa south of the Sahara. Male giraffe are larger than females, while female okapis are reported slightly larger than males (2109, 5762). The epitheliochorial, cotyledonary or diffuse placenta implants into a bipartite or bicornuate uterus (297, 4241, 6634, 7563, 7711, 8160, 8161) and is occasionally eaten (2109, 3441, 6034, 8290, 8798, 8823). Neonates are mobile, furred, and have open eyes (2109, 5153, 5762, 6034, 8290, 8798, 8823, 9191). Fetal ovaries have corpora lutea with progesterone concentrations similar to those of preimplantation gestation (3881, 5600). These fetal corpora lutea do not disappear until puberty (5600).

One captive giraffe returned to estrus 49 d after birth (5935). Giraffes have 2-4 mammae (331, 8798). Males have seminal vesicles and a sigmoid penis (11243). Details of anatomy (3823, 4240, 4242, 5600, 5627, 8160), endocrinology (5549, 6582; *Giraffa* 3881, 5627, 11704; *Okapia* 6580, 6581), birth (8823), and milk composition (408, 4243, 9770) are available.

NEONATAL MASS

Giraffa camelopardalis	f: 30 kg; n1	6178
	f: 34-45 kg; n1	6034
	36-45 kg	3441
	m: 39 kg; n1	2109
	r48.5-63.5 kg; n4	9514
	m: 50-55 kg; n1	6178
	~54 kg	5153
	54.5 kg	3247
	54.5 kg; n16	2212
	67.5 kg; n1 stillborn	9068
	77 kg; n1	11785
	f: 89.0 kg	4239
	m: 95.1 kg	4239
	102.0 kg	10030

Okapia johnstoni	f: r14-22.5 kg	6191
	17 kg; r14.5-19.5; n4	8823
	18 kg; n1	8798
	18.5 kg	6191
NEONATAL SIZE		
Giraffa camelopardalis	164 no units; n1; stretched out, tip nose to base of tail	11785
LITTER SIZE		
Giraffa camelopardalis	1	9068
	1	9064
	1; 2 occasional	5153
	1-2	4241
	1 yg ea; n4 lt	319
	1; n40 lt	3431
	1 ea; n77 lt	8699
	1.12 yg; r1-2; n8 lt	9066
	2 stillborn; n1 lt	2212
Okapia johnstoni	1; n1	6184
	1; n1 lt	2109
	1 emb; n1 f	7711
SEXUAL MATURITY		
Giraffa camelopardalis	f: 1st mate: 56 mo	4239
	f: 1st conceive: 3.83 yr; ± 3 mo; captive	4241
	f: 1st conceive: 4.67 yr, ± 3 mo; wild	4241
	f: 1st preg: 2nd half 4th yr	8798
	f: 1st preg: 6 yr; ±3 mo	4241
	f: < 4 yr	5762
	f: 4.67 yr	10158
	m: spermatogenesis, testes large: 3-4 yr	4242
	m: min 2.67 yr; 3-4 yr	10158
	m: 7 yr	5762
	f/m: 3 yr; n1 pr	318
	f/m: 3-4 yr	8798
Okapia johnstoni	f: 1st mate: 2.83 yr; n1	8798
	f: 1st conceive: 2 yr	6191
CYCLE LENGTH		
Giraffa camelopardalis	estrus: max 24 h	2109
	r14-19 d; n6 cycles, 1 f	2109
	15 d	2439
	15.5 d; n12 cycles, 4 f	6582
Okapia johnstoni	estrus: 24 h	8798
	estrus: several d	6179
	~14.5 d	6581
	15.2 d; n12 cycles, 4 f	6582
	r20-21 d; n3 cycles, 1 f	4860
	3 wk	8798
GESTATION		
Giraffa camelopardalis	400-480 d	2449
	13.75 mo; r13 mo 4 d, 14.5 mo; n2	6178
	~14 mo	4582
	~14 mo; n1	10612
	427-488 d	5762
	434 d; n1	11785
	14.5 mo	3441
	~14.5 mo	9068
	444 d	8161
	r445-446 d; n10	2439
	450 d	8421
	450 d	3247
	15 mo; n5	3441
	15 mo; n35	2212
	455.25 d; r450-461; n4	12088
	457 d	2109
	457 d	4239, 4240
	462 d; r459-465; n3	9514
	468 d; n1	11704

	488 d; n1	9191
Okapia johnstoni	435-449 d; n1; 421 d; n1	8290
	441 d; n1	2109
	427-457 d	5762
	439.6 d; r427-452; n6	6191
	444 d; r435-453; n5	8823
	446 d; n1	6184
	450 d	8421
	15 mo	6581
LACTATION		
Giraffa camelopardalis	solid food: 13 d; n1	2109
	solid food: 2-3 wk	8798
	solid food: 2 wk	5762
	wean: min 1 mo	3431
	wean: min 1 mo; wild; to 1 yr; captive	5762
	ruminate: 6-8 wk	8823
	wean: 6 mo	8823
	wean: 6 mo	2439
	wean: 6-8 mo	6206
	wean: 7-11 mo	6342
Okapia johnstoni	solid food: 6 wk	2109
	ruminate: 6-7 wk	8823
	wean: 6 mo	8823
	wean: 6-9 mo	8798
INTERLITTER INTERVAL		
Giraffa camelopardalis	14 mo	10158
	19.9 mo; ±0.6; wild	4241
	21-22 mo; n4; yg live	3431
	21.5 mo; ±0.7; captive	4241
	645 d	4239
	2 yr; yg live	5762
SEASONALITY		
Giraffa camelopardalis	mate: yr rd; Africa, south of Sahara	8798
	mate: yr rd; S Africa	4239
	conceive: yr rd, 60% Dec-Mar; n143; Transvaal, S Africa	4244
	mate: est June-Sept; Okawango, Angola	11697
	birth: Jan-Mar; ne Uganda	5762
	birth: May-Aug; Serengeti, Tanzania	5762
	birth: not after Apr; n Uganda	5762
	birth: Feb-Mar, Aug-Oct; S Africa	3145
	birth: Mar-Sept; n32; Africa, south of Sahara	8798
	birth: Mar, Oct; n2; zoo	2109
	birth: May-Sept; Zimbabwe	2275
	birth: Mar-?; Okawango south bank, Angola	11697
	birth: yr rd, except Dec; captive	2109
	birth: yr rd, peaks Jan-Apr, Sept-Oct; S Africa	8514
	birth: yr rd, peaks Feb-Mar, Aug-Oct; S Africa	3145
	birth: yr rd, peak near end of rains; Luangwa valley, Zambia	843
	birth: yr rd; Kenya, Uganda, Tanzania	5762
	birth: yr rd, may peak dur dry season; Kenya	3431
	birth: yr rd; Transvaal, S Africa	5153
	birth: yr rd; Zambia	331
Okapia johnstoni	birth: Oct; n1; zoo	2109
	birth: yr rd; n125; zoo	8823

Order Artiodactyla, Family Bovidae

Bovids (*Addax* 60-125 kg; *Aepyceros* 34-83 kg; *Alcelaphus* 100-218 kg; *Ammodorcas* 23-34 kg; *Ammotragus* 40-145 kg; *Antidorcas* 26-48 kg; *Antilocapra* 28-70 kg; *Antilope* 19-57 kg; *Bison* 350-1020 kg; *Bos* 272-1000 kg; *Boselaphus* 108-270 kg; *Bubalus* 228-1200 kg; *Budorcas* 196-350 kg; *Capra* 16-150 kg; *Capricornis* 30-140 kg; *Cephalophus* 4-80 kg; *Connochaetes* 118-295 kg; *Damaliscus* 56-172 kg; *Dorcatragus* 9-27 kg; *Gazella* 2.5-98 kg; *Hemitragus* 39-100 kg; *Hippotragus* 150-300 kg; *Kobus* 50-309 kg; *Litocranius* 28-52 kg; *Madoqua* 3-7 kg; *Nemorhaedus* 22-35 kg; *Neotragus* 2-9 kg; *Oreamnos* 45-154 kg; *Oreotragus* 8-16 kg; *Oryx* 91-238 kg; *Ourebia* 8-21 kg; *Ovibos* 143-650 kg; *Ovis* 19-200 kg; *Pantholops* 25-55 kg; *Pelea* 20-30 kg; *Procapra* 20-40 kg; *Pseudois* 25-80 kg; *Raphicerus* 6-16 kg; *Redunca* 14-95 kg; *Rupicapra* 16-50 kg; *Saiga* 21-69 kg; *Sigmoceros* 125-204 kg; *Sylvicapra* 12-25 kg; *Syncerus* 265-900 kg; *Tetracerus* 17-21 kg; *Tragelaphus* 24-1000 kg) are large, often diurnal, terrestrial herbivores from North America, Eurasia, and Africa. Males are usually larger than females (269, 491, 546b, 1237, 1432, 1649, 1890, 2079, 2439, 3571, 4643, 4645, 4656, 4832, 4950, 5260, 5300, 5763, 5764, 6189, 6192, 6241, 6387, 6407, 6409, 6525, 6584, 6922, 7144, 7855, 7884, 8477, 8593, 8740, 8972, 9121, 9122, 9175, 9540, 9541, 9545, 9547, 9636, 10027, 10637, 10740, 11697, 11784, 11787, 12019) except in *Capricornis* (7389; cf 5947), *Madoqua* (5625, 9944, 9945), *Neotragus* (2716), *Ourebia* (5763, 10631), *Raphicerus* (9945; cf 11783), and *Sylvicapra* (4926, 5763, 11789) where females are reported larger. For *Nemorhaedus* (9547) the sexes are reported equal in size.

Female *Tragelaphus* have an extra chromosome (3492). The 1-3 kg, epitheliochorial or syndesmochorial, villous, cotyledonary placenta implants into a bicornuate or duplex (*Addax* 2487; *Connochaetes* 10472) uterus (22, 239, 269, 271, 297, 1424, 1435, 1649, 2296, 2308, 2746, 4190, 4907, 5597, 5625, 5626, 5699, 6043, 6044, 6525, 7563, 7616, 7617, 8008, 9330, 9415, 9593, 9715, 10271, 10310, 11790, 11797, 11854, 11860). Only the right uterine horn is used in *Aepyceros* (4907, 5597, 6352, 7562, 10158), *Antidorcas* (10038, 11211), *Antilope* (22), *Gazella* (5122, 9180; cf 6044), *Kobus* (1434, 6044, 9181, 10158, 10450, 11731a; cf 1431), *Madoqua* (5625), *Neotragus* (6581a), *Oreotragus* (10158), *Oryx* (10154), and *Sylvicapra* (1791, 6352, 10610). The right ovary is preferentially used in *Kobus* (11731a).

In utero siblicide occurs (*Antilocapra* 8008). Placentophagia is common (208, 269, 785, 1047, 2081, 3099, 3100, 3667, 3871, 3955, 4086, 4196, 5018, 5115, 5229, 5248, 5249, 5258, 5763, 5789, 5828, 5876, 6801, 9311, 9593, 9617, 9691, 10158, 10269, 10638, 10939, 11044, 11506, 11758, 11789, 11854; cf 2296, 2746, 3105, 6657, 8641, 12107). Estrus may return within a month of birth (269, 270, 493, 922, 1434, 2608, 2815, 4972, 5229, 5763, 5764, 5896, 5999, 7541, 7616, 9691, 10158, 10269, 10628, 11697, 11758; cf 1432, 2605, 11253), except in *Bos* (10905, 11088). *Connochaetes* births are synchronous (3099, 3102, 3105, 9993; *Ovis* 8844).

Neonates are mobile, furred and have open eyes (207, 334, 459, 560, 561, 1047, 1474, 1649, 1684, 2079, 2296, 2598, 2602, 2746, 3099, 3102, 3105, 3146, 3377, 3571, 3667, 3871, 4254, 4643, 4645, 4950, 4972, 5229, 5249, 5258, 5763, 5764, 5828, 5876, 5996, 6586, 6657, 6801, 7051, 7395, 7625, 7910, 8458, 8739, 8798, 9121, 9545, 9945, 9994, 10043, 10158, 10216, 10535, 10638, 10946, 11021, 11044, 11253, 11433, 11578, 11789, 12019; cf 7910).

Females have 2 mammae (*Alcelaphus* 9945; *Capra* 2081, 3571; *Connochaetes* 324, 331, 9945, 10158; *Damaliscus* 324, 331, 9945, 10158; *Ovis* 1474, 3571; *Sigmoceros* 331, 333, 9030), 2-4 mammae (*Antidorcas* 324, 921, 9945, 10158; *Ovibos* 1474, 2364), or 4 mammae (*Aepyceros* 324, 331, 333, 10158; *Antilocapra* 1474, 8009; *Bison* 1474, 3571; *Budorcas* 3034, 8798; *Capricornis* 3034; *Cephalophus* 324, 331, 10158; *Hippotragus* 324, 10158; *Kobus* 331, 333, 374, 9945, 10158; *Madoqua* 324, 5625, 10158; *Nemorhaedus* 9175; *Neotragus* 324, 10158; *Oreamnos* 1474, 9122; *Oreotragus* 324, 331, 10158; *Oryx* 324; *Ourebia* 324, 331, 10158; *Pelea* 324, 10158; *Raphicerus* 324, 9945, 10158, 10392; *Redunca* 324, 331, 9945, 10158; *Rupicapra* 2079, 3571; *Sylvicapra* 331, 333, 10158; *Syncerus* 331; cf 8515; *Tragelaphus* 324, 331, 10158).

Males have seminal vesicles (1424, 10602). *Damaliscus* and *Kobus* may mate on leks (546a, 546b).

Details of anatomy (3715, 3823; *Aepyceros* 3048, 5596; *Antilocapra* 8010; *Capricornis* 5825; *Gazella* 6922, 10271; *Madoqua* 5606, 5625; *Ovibos* 9330; *Kobus* 7540; *Rupicapra* 3715; *Tragelaphus* 1381), intersex (10031), endocrinology (9428a, *Addax* 2486; *Aepyceros* 1227, 3148, 5132; *Alcelaphus* 5608; *Ammotragus* 3332; *Antilope* 4860, 4938; *Bubalus* 380, 381, 467, 649, 650, 4191, 4615, 5496, 5509, 8206, 8926, 8927; *Damaliscus* 10029; *Oryx* 2818; *Ovibos* 2498, 9331; *Ovis* 8897, 11645; *Redunca* 10025; *Rupicapra* 3663), spermatogenesis (938, 10817, 10818), birth (2296, 3105, 3955, 5018, 6586, 7068, 9971), milk composition (408a, 561, 1776, 1927, 2035, 2079, 2581, 3417, 4116, 4118, 4643, 4950, 4972, 5595, 6058, 6246, 6522, 8690, 8892, 9594, 10299,

10300, 10620, 10739, 10740, 11214, 11291, 11350), reproductive energetics (789a, 4047, 4048), and light (270, 9318) are available. Supplements 30 and 43 of the Journal of Reproduction and Fertility, published in 1981 and 1991, review the reproduction of domestic ruminants (cattle, sheep, and goats).

NEONATAL MASS

Addax nasomaculatus	f: 4.8 kg	2603
	m: 5.6 kg; r4.5-6.4; n4	6802
	f: 5.9 kg; r4.5-7.3; n4	6802
	6.1 kg; n1	6819
Aepyceros melampus	~5 kg	3146
	6.1 kg; n1	633
Alcelaphus buselaphus	m: ~5-6 kg; n1	9741
	12.5 kg	3247
Ammotragus lervia	4.5 kg	4020
Antidorcas marsupialis	3.82 kg; se0.12; n13	11213
	m: 4 kg	10040
Antilocapra americana	1.04-5.78 kg	11208
	2.3-3.2 kg	12019
	3.0 kg; r1.5-4.16	2607
	~3.2 kg	4647
	3.5 kg	8009
	f: 3.96 kg; n46, 1-6 d	1515a
	m: 4.13 kg; n44, 1-6 d	1515a
Antilope cervicapra	3-4 kg	7616
	3.276 kg; r2.6-4.2; n33	27
	4.5-5.0 kg	10628
Bison bison	est 15-25 kg	9386
Bison bonasus	22-23 kg	4643
	24.25 kg; r24.0-24.5; n2	9505
Bos frontalis	m: 13 kg; n1	2109
	f: 23.1 kg; r19.5-26.5; n4	9593
	m: 33.0 kg; n1	9593
Boselaphus tragocamelus	5.875 kg; r5.5-6.25; n2	9475
Capra caucasica	3.5-4.2 kg	4643
Capra hircus	m: 1.55 kg; n1 stillborn	7051
	f: 1.7 kg; n15	9223
	m: 1.9 kg; n11	9223
	f: 2.5 kg; se0.23; r2.0-3.0; n4	11054
	m: 2.9 kg; se0.18; r2.5-3.3; n4	11054
	<3 kg	11389
Capra ibex	m: 2.21 kg; n1 term emb	2081
	m: 2.55 kg; se2.43; r1.7-3.15; n5	8594
	f: 2.75 kg; se3.21; r1.97-3.45; n5	8594
	2.89 kg; n20	8594
	2.95 kg; r2.9-3; n2	8178
	3.5-4 kg	4643
Capricomis crispus	3.313-3.708 kg; extrapolated	10535
	3.5 kg; n1	5177
Cephalophus dorsalis	m: 1.60 kg; n1	2605
	f: 1.62 kg; n1	2605
Cephalophus maxwelli	710-954 g	8866
Cephalophus monticola	10% of mother's weight	5763
Cephalophus rufilatus	m: 0.79 kg; n1	2605
	f: 1.14 kg; n1	2605
Cephalophus sylvicultor	6 kg; n1, 4 d	6642
Cephalophus zebra	1.615 kg; n1	9691
	~1.8 kg est n1	11021
Connochaetes gnou	~8 kg	5901
Connochaetes taurinus	f: ~15 kg	10638
	16.5 kg; n1 emb	10272
	m: ~19 kg	10638
	~22 kg	10158

Damaliscus dorcas	6.36 kg	5702
	6.4-7.3 kg	2746
	7 kg	10023, 10027
Damaliscus hunteri	8.35 kg; n1	5249
Damaliscus lunatus	10-12 kg; term emb	1792
Gazella dorcas	1.65 kg; sd0.23; n33	6764
Gazella granti	5.2 kg	3247
	m: 5.4 kg; n1	2603
	5.63 kg; r5.2-6.2; n3 near term	10271, 10272
Gazella leptoceros	f: 4.64 kg; n1	2603
Gazella spekei	f: 1.150 kg; r0.7-1.7	8991
	1.25 kg; n26	8991
	m: 1.36 kg; r0.7-1.8	8991
Gazella subgutturosa	2.4 kg; r2-3.2; n10	4643
	r2.6-3.1 kg; n5	4643
Gazella thomsoni	2.315 kg; r2.0-3.5; n64	1196
	~2.5 kg	9180
	2.7 kg	3247
Hippotragus equinus	f: 11 kg; n1	334
	f: 16.5 kg; n1	2602, 2603
	m: 18 kg; n1	2603
Hippotragus niger	m: 13.6 kg; n1	4085
	m: 13.75 kg; r12.3-14.5; n4	6802
	f: 14.0 kg; r11.4-16.1; n4	6802
	15-20 kg	5972
	f: 16.5 kg; n1	2603
	m: 18.95 kg; se0.76; r13.50-19.80; n26	4086
	f: 20.33 kg; se1.05; r13.50-22.50; n22	4086
Kobus kob	f: 5.405 kg	1434
Kobus leche	5.1 kg; n5	9181
Madoqua kirki	f: 0.6 kg; n1	2603
	0.605-0.795 kg; n5	2598
Madoqua saltiana	f: r0.56-0.68 kg	5763
	m: r0.725-0.795 kg	5763
Nemorhaedus goral	~2 kg; ~10-15 d before birth	4643
Neotragus batesi	max 0.5 kg; term emb	2716
Oreamnos americanus	f: 2.95 kg; n1	1237
	m: 3.19 kg; sd0.43; n8	5112
	3.32 kg; sd0.50; n17	5112
	f: 3.45 kg; sd0.50; n9	5112
Oreotragus oreotragus	1.130 kg; n1 term emb	11788
Oryx dammah	m: 10.05 kg; n1	2603
Oryx gazella	m: 11.14 kg; r9.5-12.3; n7	6802
	f: 12.12 kg; r11.4-13.6; n3	6802
	12.8 kg	11212
Ourebia ourebi	m: 2.2 kg; n1	2603
	2.27 kg; n1	6315
Ovibos moschatus	7.3 kg; n1	4985
	r9-ll kg; n14	11639
	f: 10.5 kg; n7; 2 stillborn twins: 7, 8 kg	11705
	f: 14.0 kg; r9.1-18.1; n29	6241
	m: 15.5 kg; r10.4-20.9; n32	6241
Ovis ammon	2.25-2.5 kg	3571
	2.8-6 kg; to 6 d; includes subspecies variation	4643
Ovis aries	m: 1.8 kg; r1.4-2.4; singletons; 1.5 kg; 1.1-1.8; twins	5307
	f: 1.8 kg; r1.4-2.5; singletons; 1.5 kg; 1.2-1.8; twins	5307
	1.9 or 2.0 kg; r0.85-2.9	1449
	2.8 kg; se0.21; r2.0-4.3; n10	11054
	3.5 kg; se0.18; r2.5-4.3; n11	11054
Ovis canadensis	3.3 kg; r2.8-3.7; n9	3667
	~3.6 kg	4333
	4.2 kg; se0.27; n16	4482
	4.4 kg; se0.37; n4	4482
	4.5 kg; r4-5; n4	11910

	5.4 kg; near term	3417
	f: 5.4 kg; r5.3-5.5; n2	1021
Ovis dalli	3.2-4.1 kg	7884
	3.3 kg; sd0.4	1446, 9732
	3.3 kg; r2.9-3.6; n5	3667
	4.025 kg; n1 fetus	7883
Ovis musimon	2 kg; r1.6-2.6	8467
	2.25 kg; r2.0-3.4	1449
	f: 2.523 kg; r2.3-3.07; n5	9415
	2.7 kg; n1	10946
	m: 2.852 kg; r2.1-3.39; n5	9415
Procapra gutturosa	2.8-3 kg	4643
Raphicerus campestris	0.84 kg; term emb; 0.936 kg newbom	11790
	1 kg; n1	5763
Redunca arundinum	4.5 kg	5763
Redunca fulvorufula	2.5 kg? term emb, from graph	5165
	>3 kg	5763
Rupicapra rupicapra	f: 1.89 kg; n1 near term	2079
	2-2.5 kg	2079
	2-3 kg	8477
	m: 2.38 kg; n1 near term	2079
	2.8-3.2 kg; 3-5 d	4643
Saiga tatarica	m: 3.135 kg; n4	8641
	3.5 kg; r2-4.4	560
	3.5 kg; m ~200 g heavier; r2-4.4 kg; n45 f, n45 m	4643
	f: 3.657 kg; n6	8641
Sigmoceros lichtensteini	~15 kg	10158
Sylvicapra grimmia	f: 1.35-1.8 kg; n3	11789
	1.61 kg	334
	m: 1.65 kg; n1	11789
Syncerus caffer	f: 34 kg; r32-36; n2	8515
	m: 37.2 kg; r26-42; n6	8515
	40 kg; r38-45	9994
	47 kg; r37-54; n3	11292
	f: 50.0 kg; n1	11292
	m: 53.6 kg; n1	11292
Tetracerus quadricornis	0.942 kg; r0.740-1.065; n7	27
	1.15 kg; r1.0-1.3; n2	9956, 9957
	1.04 kg; r0.75-1.2; 0.75 kg yg died 13 d	37
Tragelaphus angasi	4.15-5.5 kg	6525
	5.6 kg; n10	270
	m: 5.68 kg; n12	269
Tragelaphus eurycerus	16-18 kg	11022
	19.5 kg; n3	8867
	20.4 kg; n1	2720
	22.3 kg; n1	11973
Tragelaphus imberbis	m: 5.2 kg; n1	6411
	m: 6.1 kg; r5.3-7.6; n8	6189
	f: 6.3 kg; n1 term emb	6411
Tragelaphus oryx	22-36 kg	5763
	f: 28.4 kg; n21	10895
	f: 28.6 kg; r24.5-32.7; n4	8690
	f: 29.1 kg	11784
	m: 31.3 kg; n25	10895
	31.5 kg	10894
	f: 32 kg	10158
	m: 36 kg	10158
	m: 36.4 kg; r32.7-40.9; n3	8690
	m: 36.8 kg	11784
Tragelaphus scriptus	m: 2.948 kg; r2.778-3.118; n2, 1-7 d	11787
	f: 3.044 kg; r2.664-3.203; n4, 1-7 d	11787
	f: 3.375 kg; n1	201
	3.5-4.5 kg	10158
	m: 4.175 kg; n1	201

Tragelaphus spekei	~4 kg; n4	6192
Tragelaphus strepsiceros	~15 kg; n1	11291
	m: 16.3 kg; n1, 3 d	11781
NEONATAL SIZE		
Antilocapra americana	58.4 cm	11208
Aepyceros melampus	~50 cm	5597
Bison bonasus	88.5 cm; r87-90; n2	9505
Capra hircus	m: HB: 37 cm; n1 stillborn	7051
Capricomis crispus	48-50 cm; extrapolated	10535
Cephalophus monticola	m: 31 cm; n1	185
Gazella granti	f: max 62.5 cm	10271
	m: max 71.0 cm	10271
Gazella subgutturosa	54.3 cm; r49.0-59.0; n10	4643
	r55-58 cm; n5	4643
Hippotragus equinus	33 cm	5764
Kobus leche	70.3 cm; n1 term emb	9945
Oreamnos americanus	TL: 55.9 cm; n1	1237
Oreotragus oreotragus	f: 39 cm; emb	9945
Ovis dalli	63.7 cm; n1 fetus	7883
Procapra gutturosa	51.35 cm; r51-56; n2	4643
Redunca fulvorufula	HB: 57.5 cm; term fetus; from graph	5165
Saiga tatarica	60 cm; r53-67	560
Tragelaphus angasi	81-83 cm	6525
Tragelaphus oryx	body: 52.0 cm	10894
Tragelaphus scriptus	m: 56.1 cm; n1	11787
	f: 58.2 cm; r54-61; n4	11787
WEANING MASS		
Antilocapra americana	8.9 kg; 2 mo	11596
Bison bison	135-180 kg; 8-9 mo	7143
Ovis canadensis	f: ~27 kg; 6 mo	4333
	m: ~29 kg; 6 mo	4333
Ovis musimon	17.50 kg; 180 d	8467
LITTER SIZE		
Addax nasomaculatus	1 yg ea; n5 lt	2109
	1.002 yg; 0.2% twins; n1012 lt	2487
Aepyceros melampus	1; twins 1 ignored & died	10158
	1 yg	5259
	1 emb ea; 2 f	6612
	1 emb; n7 f	4907
	1 yg ea; n58 f	2274, 2275, 7562
	1 yg; n144 lt	1230
	1; sometimes 2	331, 333
Alcelaphus buselaphus	1	3956
	1	5764
	1 yg; n1 lt	2603
	1 yg; n1	9741
	1 yg; n8 lt	1230
Ammodorcas clarkei	1	7167
Ammotragus lervia	1.01 yg; 1% twins; r1-2; n96 lt	1230
	1.07; n190 lt	12128
	1.2 yg; r1-2; n162 lt	2109
	1.3 emb; mode 1; r1-3; n44 f	4020
	1.35 emb; n20 f	4020
	1.5; r1-2; n4 lt, 1 f	9734
	1.54 yg; r1-2; n13 lt	9066
Antidorcas marsupialis	1	10038
	1 yg; 1 lt	2109
	mode 1; 2 rare; r1-2	8351
	1.01 yg; 1.4% twins; r1-2; n146 lt	1230
Antilocapra americana	1-2	10809
	1.5; r1-2; n4	11370
	1.78; r1-2; n14 lt	2607
	1.80; mode 2; r1-3	11208
	1.94 emb; r1-2; n18 f	1759

	2	5018
	mode 2	8644
	mode 2; 1,3 rare	4647
	2; r1-2; at first birth 1 fawn, thereafter 2	12019
	2; n1 lt	7376
	2 yg; n1 lt	2109
	2 emb; n5 f	7145
	2 emb; n6 f	1424
	2 emb; n6 f	10924a
	2 yg; n17 lt	5828
	2.1 emb; mode 2 92%; r1-3; n37 f	6217
	4-6 cl	11860
	5.1 cl; r2-7; n66	8008
Antilope cervicapra	1-2	6653, 6654
	1 ea; n4 lt	6541
	1 yg ea; n38 lt	9066
	1 yg ea; n46 lt	1230
	1 yg ea; n97 lt	2109
	1 yg ea; n338 lt	7616
Bison bison	1	7144
	mode 1 yg; r1-2	7068
	mode 1; 2 occasional	10216
	1; 2 occasional	7068
	1 yg ea; n9 lt	1230
	1 yg ea; n19 lt	9066
	1 yg ea; n91 lt	2109
	1 yg ea; n111 lt	4047
	1 emb ea; n481 f	3559, 3560
	2 very rare	4501
Bison bonasus	1	3571
	1	4643
	1 yg	3822a
	1 yg ea; n2 lt	2109
	1 yg ea; n2	9505
	1 ea; min n132 lt	5999
	2 rare	5229
Bos frontalis	1	9545
	mode 1	7161
	1-2	2802
	1 yg; n16 lt	2109
	1 ea; n16 lt	12128
Bos grunniens	mode 1	12128
	1 yg ea; n2 lt	9066
Bos javanicus	1	6652
Boselaphus tragocamelus	1-2	6654
	mode 1; r1-2	12128
	1 yg ea; n8 lt	2109
	1.3 yg; r1-2; n17 lt	9066
	1.3 yg; r1-2; n~60 lt	1230
	1.5; r1-2	9175
	2 yg; n1 lt	3871
	2 yg; n1 lt	6145
	2 yg; n1 lt	30
Bubalus bubalis	1-2	6654
	1 yg; r1-2	8501
	1 yg ea; n27 lt	1230
	2 yg	7382
Bubalus depressicornis	1 ea; n7 lt	12128
	1 yg ea; n8 lt	1230
Budorcas taxicolor	1	11419
	1 yg	9547
	1; seldom 2	8798
	maybe 1-2/lt; anecdotal	154
	1 yg; n1	459

Capra caucasica	1	4643
	mode 1	8458
	1.08; r1-2; n13 lt	12128
Capra cylindricornis	1.2; r1-2; n9 lt	8328
Capra falconeri	2 common; r1-2	9175
	2 yg; n1 lt	2109
	3 nursing; 1 lt	8694
Capra hircus	1-2	4643
	1-2 yg	9223
	1-2 yg	9175
	mode 1; 2 rare; r1-2	2080
	1-3 yg	11389
	1 emb; n1 f	6293
	1.22 yg; r1-2; n9 lt	2109
	1.25; r1-2; n8 lt	12128
	1.25 yg; r1-2; n60 f	9357
	1.26 emb; r1-2; n27 f	9357
	1.5 emb; r1-2	12020
	1.63 cl; n27 f	9357
	1.93 yg; n15 lt	4934
	2.05 yg; n18 lt	4934
Capra ibex	1; seldom 2	3571
	1; 2 rare	2081
	1-2	6652
	mode 1; r1-2	4643
	1 stillborn	8707
	1.1; mode 1; r1-2	6764
	1.11; r1-2; n213 lt	7910, 7911
	1.20 yg; mode 1; r1-2; n365 lt	10530
Capricornis crispus	1 yg; 1 lt	5177
	1 yg; 1 lt	5948
	1.01; r1-3; n209 f; 1 set each twins, triplets	7390, 10535
Capricornis sumatraensis	1 yg; n1 lt	496
Cephalophus dorsalis	1 yg ea; n2 lt	2603, 2605
Cephalophus leucogaster	1 yg ea; n3 lt, 1 pair	2109
Cephalophus maxwelli	1	74
	1 yg	4354a
Cephalophus monticola	1 emb; n1 f	11342
	1 emb; n1 f	331
	1 yg ea; n12 lt	1230
	1 yg ea; n16 lt	2109
	1 ea; n39 lt	11349
Cephalophus natalensis	1 yg ea; n7 lt	1230
Cephalophus ogilbyi	1	5277
Cephalophus rufilatus	1 yg ea; n3 lt	2603, 2605
Cephalophus sylvicultor	1	5277
	1	254
	1	5575
	1 yg	5996
Cephalophus zebra	1 yg	11021
Connochaetes gnou	1 yg	5901
	1 yg ea; n5 lt	2109
	1 yg ea; n66 lt	1230
	1.02; r1-2; n50 lt	4557
Connochaetes taurinus	1	10158
	1	10638
	1 emb	2275
	1 yg; n1 lt	2109
	1 yg ea; n3 lt	9066
	1 yg ea; n43 lt	1230
	mode 1; r1-2	2449
	1; 2 rare	331
	2 rare	8514

Damaliscus dorcas	1 yg	2746
	1 yg; n1 lt	2603
	1 yg ea; n74 lt	1230
	2 yg; n1 lt	7774
Damaliscus hunteri	1; n4 lt	5249
Damaliscus lunatus	1	331
	1 emb; n1 f	1792
	1? yg ea; n2 lt	1230
Gazella cuvieri	1.41 yg; r1-2; n195 lt	8100
	1.5 yg; n18 lt	3567
Gazella dama	1	5575
	1	254
	1 yg	3567
Gazella dorcas	1	493
	1	8223
	1; 2 rare	5997
	1-2	5575
	1-2	254
	1 yg; 1 lt	2109
	1 yg ea; n4 lt	2600, 2603
	1; n7	12128
	1; mode 1; n33 lt	6764
Gazella gazella	1-2	9175
	1-3 yg	2109
	1 yg; 2 rare	492, 493
	1 emb; n1 f	6293
Gazella granti	1 yg; 1 lt	9294
	1 yg ea; n3 lt	2600, 2603
Gazella leptoceros	1 yg	3567
	1 yg ea; n2 lt	2600, 2603
Gazella rufifrons	1 yg	3567
	1 yg ea; n2 lt	2600, 2603
Gazella spekei	1 yg	3567
Gazella subgutturosa	1-2	4643
	1-2; one third of births are twins	9175
	est 1.5	3567
	1.3; r1-2	12128
	1.3; r1-2; n3 lt	5876
	2 yg; n1	2273
Gazella thomsoni	1	6922
	1	4972
	mode 1	2449
	1; n1 lt	9180
	1 emb ea; n2 f	6612
	1.02 yg; r1-2; n42 f	5122, 5123
Hemitragus hylocrius	1 yg ea; n7 lt, 2 f	8693
	1 yg ea; n22 lt	11750
	2	6654
Hemitragus jemlahicus	mode 1	6652, 6654
	1 yg ea; n105 lt	1230
	1.01 emb; n157 births	1669
	1.01 yg; r1-2; n180 lt	2109
	1.09; r1-2; n115 lt	12128
Hippotragus equinus	1	5764
	1	11253
	1	11758
	1	2449
	1 emb; n1 f	6612
	1 yg; n1 lt	5248
	1 emb ea; n2 f	5827
	1 yg ea; n2 lt	2602, 2603
Hippotragus niger	1	11758
	1 ea; n2 lt	5972
	1 yg ea; n3 lt	2603

	1 ea; n4 lt	1437
	1 yg *ea*; n18 lt	1230
Kobus ellipsiprymnus	1	8350
	1	11253
	1	5763
	1 yg	2109
	1; r1-2	333
	1; perhaps 2	331
	mode 1; r1-2	2449
	mode 1; r1-3	10158
	1 emb; n1 f	186
	1 emb; n1 f	6612
	1 *emb ea*; n3 f	5052
	1 yg ea; n62 lt	1230
	2; n1 lt	9248
Kobus kob	1	1432
	1 yg ea; n11 lt	2109
	1 ea; n17 f	3543
Kobus leche	1	10158
	1	12128
	1	9945
	1	331, 333
	mode 1	2449
	1; no twins observed	7368
	1 *ea*; n78	9181
	1 yg ea; n149 lt	1230
Kobus vardoni	1	331, 333
	1	5763
	1	11253
Litocranius walleri	1	6410
	1 yg; n1 lt	2109
	1	5789
Madoqua guentheri	1 emb ea; n2 f	5606
Madoqua kirki	1	10158
	1	5763
	1 emb; n1 f	6612
	1 emb; n1 f	185
	1 yg ea; n3 lt	2603
Madoqua saltiana	1	5763
Nemorhaedus goral	1	6654
	1	4643
	mode 1; 2 rare; r1-2	9175
Neotragus moschatus	1	10158
	1	5763
	est 1 yg; n8 lt	5191
Oreamnos americanus	1; 2 reported	9121
	mode 1; r1-2	4933
	mode 1; r1-2	1237
	1-2; 3 occasional	1474
	most 1-2 yg; r1-3	6381a
	1.4; r1-2; n15 lt	5112
Oreotragus oreotragus	1	10158
	1	9945
	1	5763
	1	11253
	1 emb; n1 f	331
	1 yg ea; n2 lt	1230
	1 yg ea; n3 lt	2109
Oryx dammah	1 yg; n1 lt	2603
	1 yg *ea*; n2 lt	2109
	1 ea; n12 lt	12128
Oryx gazella	1	5764
	1 yg; n1 lt	2603
	1 yg; n1 lt	10787

Oryx leucoryx	1 yg ea; n33 lt	1230
	1 yg	6619
	1 yg	10395
Ourebia ourebi	1 yg ea; n4 lt	2109
	1	10158
	1	2439
	mode 1	2449
	1; seldom 2	11253
Ovibos moschatus	1 emb ea; n2 f	331
	1 yg	4221
	1; <1% twins	10259
	1; 2 rare	10740
	mode 1; 2 rare	10259
	1 yg; n8 lt	8033
	1.01; 1.3% twins; r1-2; n75 lt; twins stillborn	11705
Ovis ammon	1.04; 3.9% twins; n125 f	6381
	mode 1 especially first birth; r1-2	4643
	mode 1; 2 occasional; r1-2	9175
	mode 1; 2 rare	3571
Ovis aries	1-2	3034
	1-2 yg	1181
	1.1; r1-2; n76 lt	9547
	1.3; n120 lt	11053
	1.5 yg; r1-2; n2 lt	2109
Ovis canadensis	2.5/lt; r1-7; highest ovulation rate 4.2 r1-10	939
	1	4643
	1	3667
	1; no 2	11578
	1; no 2	10257
	1; occasionally 2	1474
	1+; few 2	4950
	1.36; r1-2; n11 f	10243
	11 records of twinning	2890
Ovis dalli	1	1447
	1	3667
	mode 1; 2 occasional	7884
	1-2; one set twins	7625
	1 emb ea; n21 f	7883
	1 yg ea; n73 lt	9980
	1 record of twins	2890
	2 rare	4878
Ovis musimon	1 yg	8467
	mode 1; r1-2	10946
	1.07 yg; r1-3; n159 lt	1230
	1.09 yg; 9.3% twins; r1-2; n86 lt	2109
	1.10; r1-2; n115 lt	12128
	mode 2	10945
Ovis nivicola	1	10722
Ovis vignei	1-2	6652, 6654
	1-2	12128
Pantholops hodgsoni	1	6654
	1 yg	9547
Pelea capreolus	1	10158
	1 yg ea; n14 lt	1230
Procapra gutturosa	1	4643
	1 yg	7395
	2	11419
Pseudois nayaur	1-2	9547
	1-2 yg	11449
Raphicerus campestris	1	11697
	1	5763
	1; 2 rare	10158
	1 yg ea; n2 lt	1230
	1 yg ea; n6 lt	1684

	1 emb ea; n7 f	4907
Redunca arundinum	1	10158
	1	5763
	1 emb	11697
	mode 1; r1-2	2449
	1; 2 occasional	11253
	1; 2 rare	331
	1 emb; n1 f	186
	1 yg ea; n31 lt	1230
Redunca fulvorufula	1	10158
	1	5763
	1 emb; n1 f	4926
	1 yg; n1 lt	2603
	1 yg ea; n6 lt	2109
	1 yg ea; n30 lt	1230
Redunca redunca	1	5763
	1 yg	4354a
Rupicapra rupicapra	1; yg animals; 2; older animals; r1-3	3571
	1; <1% twins; r1-2	2079, 2080
	1; 2 rare	10888
	1+; 2 rare; r1-2	8477
	1 emb; n1 f	11255
	1 emb ea; n38 f	11056
Saiga tatarica	1 ea; n4 lt	11370
	1.3; r1-2; n18 lt	8641
	1.5 yg; r1-2; n2 lt	9066
	1.75 yg; r1-3; n4 lt	3377
	2; 1,3 rare	4645
	mode 2	2215
	mode 2 >70%; r1-2	560, 561
	mode 2 70-90%	4643
Sigmoceros lichtensteini	1	10158
	1	2439
	1	7369
	mode 1	333
	mode 1; r1-2	2449
	mode 1; 2 rare	331
	mode 1; 2 sometimes	11253
	1 emb; n1 f	5052
Sylvicapra grimmia	1	11697
	1	5763
	1	11253
	1	333
	1; 2 rare	10158
	mode 1; r1-2	2449
	mode 1; 2 occasional	331
	1 emb; n1 f	185
	1 emb ea; n48 f	11789
	1 yg ea; n75 lt	1230
Syncerus caffer	1	9994
	1	331
	1	11253
	1	2449
	1 yg ea; n8 lt	1230
	1.12 emb; r1-2; n17 f	8515
	2 stillborn; n1 lt	7411
Tetracerus quadricornis	1-2	6654
	1 yg ea; n2 lt	9957
	1.83 yg; n6 lt	37
	2 yg	7382
Tragelaphus angasi	1	10158
	1 ea; n4 lt	11506
	1 yg ea; n43 lt	1230
	1.005; r1-2; n217 lt	269, 271

Tragelaphus eurycerus	1 yg ea; n11 lt	8867
Tragelaphus imberbis	1 yg; n1 lt	2603
Tragelaphus oryx	1	5277
	1	5763
	1	11253
	1; 2 rare	10158
	1; 2 sometimes	331
	1 emb ea; n3 f	5052
	1 yg ea; n4 lt	9066
	1 yg ea; n63 lt	1230
	1.02; r1-2; n94 lt	12128
	1 yg; 1 case stillborn twins n272	10895
Tragelaphus scriptus	1	10158
	1	2439
	1	5763
	1	331
	1	11253
	1 yg	201
	mode 1	2449
	1 emb; n1 f	185
	1 emb ea; n2 f	334
	1 yg ea; n3 lt	2603
	1 ea; n6 lt	8712
	1 yg ea; n32 lt	1230
	2 emb; n1 f	2603
Tragelaphus spekei	1	5763
	1	331
	1; n1 lt	5688
	1.0 yg; 0.2% twins; r1-2; n1085 lt	2485
Tragelaphus strepsiceros	1	2275
	1	331
	1	11781
	1 yg; n1 lt	11291
	1 yg ea; n2 lt	2109
	1 yg ea; n11 lt	1230

SEXUAL MATURITY

Addax nasomaculatus	f: 1st birth: 36.15 mo; r33.333-39; n2	2603
	f: 1st birth: 34.0 mo; sd6.6	2487
Aepyceros melampus	f: 1st mate: ~12-18 mo	5597
	f: 1st conceive: ~18 mo	5764
	f: 1st birth: 2 yr	3144
	m: 1st sire: ~13 mo	5764
	m: testis wt, spermatogenesis: 13 mo	5697
	m: 1st spermatogenesis: 18 mo	1334
	m: 1st territorial: 5+ yr	1334
Alcelaphus buselaphus	f: 1st mate: <2 yr	3247
	f: 1st birth: 25 mo; well fed; n3 of 5 f	7774
	f: 1st birth: ~3 yr	10158
Ammotragus lervia	f: 1st birth: 13.4 mo; n1	4020
	f: sperm: 11 mo; n1	4020
Antidorcas marsupialis	f: 1st ovulation: 28 wk	10023, 10038
	f: 1st conceive: 43-44 wk; n1	6510
	m: 1st sperm: 48 wk; n2	10040
	m: 1st mate: 52 wk	10023
	m: 1st mate: 24 mo; n1	6510
Antilocapra americana	f: 1st mate: 16 mo	11208
	f: 1st preg: ~8 mo; n1	7376
	f: 1st birth: 1 yr; n1	11954
	f: 5 mo min	8009
	f: 1.33 yr	2607
	f: 18 mo	1871a
	m: 1st mate: 16-17 mo	11208
	m: 2.33 yr	2607
	f/m: 1st mate: 15-16 mo	12019

	f/m: 1st mate: 1.5 yr	1474
Antilope cervicapra	f: 1st estrus: 7 mo	10628
	f: 1st birth: 14 mo	10628
	f: 1st birth: 22 mo	7617
	f: 1st birth: 26.5 mo; r24-29; n2	32
Bison bison	f: 1st mate: 1 yr rare	7068
	f: 1st mate: 15 mo 5%; 27 mo 40%; 39 mo 50%	3559
	f: 1st mate: 2-3 yr	6584
	f: 1st conceive: 1 yr rare; 2 yr 38%; 3 yr 52%	3560
	f: 1st conceive: 2 yr	4501
	f: 1st preg: 3.5 yr	1871a
	f: 1st birth: 3 yr	7068
	f: 1st birth: 3 yr	7805
	f: 1st birth: 3 yr; r2-4; n84	4047
	f: 2 yr	7068
	m: testicular sperm: yearling	4501
	m: epididymal sperm: 2-3 yr; 1 yr rare	3560
	m: 1st mate: 1 yr 16%, 2 yr 33%, 3 yr 100%	3559
	m: 1st breed: 6 yr	6584
	f/m: 1st mate: 2-3 yr	1474
Bison bonasus	f: 1st birth: 32-36.5 mo	5229
	f: 1st birth: 36+ mo	3822a
	f: 1st birth: 47.8 mo; mode 35-38; min 33 mo; max 6th yr; n52	5999
	f: 2 yr; active 3 yr	4643
	m: fertile insemination: 15-29 mo	5229
	m: active 6-7 yr	4643
	f/m: 2-3 yr	3571
Bos frontalis	f: 1st estrus: 451 d; n1	9593
	m: 1st mate: 1 yr; n1	9593
	f/m: 1st mate: ~2 yr	9545
Bos javanicus	f: 1st birth: 2.5 yr	6524
	f: 2 yr	11088
Boselaphus tragocamelus	f: 1st birth: 2 yr 9 mo 19 d; n1	30
	f: 1st birth: 3 yr 20 d; n1	32
Bubalus bubalis	f: 1st conceive: 28.5 mo; puberty 14-19 mo (cl present)	10940
Budorcas taxicolor	f/m: ~2.5 yr	8798
Capra caucasica	f: 1st estrus: 2 yr; often do not mate then	4643
	m: 1st mate: 4-5 yr	4643
Capra cylindricornis	m: < 3 yr	8328
Capra falconeri	f: 1st estrus: 30 mo	9175
	f: 1st birth: 2 yr	9547
	f: during 2nd yr	4643
	m: after 3rd yr	4643
Capra hircus	f: 1st mate: 6 mo	9357
	f: 1st mate: 1.5 yr; captive; 2.5 yr; wild	2080
	f: 1st conceive: 3 mo	9223
	f: 1st preg: 5 mo	12020
	f: 1st birth: 8 mo	9223
	f: 1st birth: 3 yr	2080
	f: 1st birth: 3 yr	9175
	f: 2nd yr; 1st birth: 3rd yr	4643
	m: 1st sire: 6 mo	9357
	m: 1st mate: 1.5 yr	2080
	m: 1st mate: 3-4 yr	4643
	f/m: 2 yr	9547
Capra ibex	f: prob repro: 212-242 d	8707
	f: 1st birth: 2-3 yr	4643
	f: 1 1/2 yr	2081
	m: 1st mate: active 5-6 yr	4643
	m: 1.5 yr	4643
	f/m: 1st mate: 3-4 yr; rarely 2 yr	8594
	f/m: 4th yr	3571

Capricomis crispus	f: 1st repro: 2-5 yr	7390
	f: ovarian histology: 2.5 yr	5825, 10535
	m: spermatogenesis: 6 mo	10817
Cephalophus dorsalis	f: 1st conceive: 14 mo	2605
	f: 1st birth: 24 mo	2603
Cephalophus maxwelli	f: 1st birth: 3 yr	74
Cephalophus monticola	f: 1st mate: 6-17 mo	1047
	m: 1st mate: 9 mo	1047
	f/m: ~3 yr	5763
Cephalophus rufilatus	f: 1st birth: 26.25 mo; r16.75-35.75; n2	2603
Connochaetes gnou	f: 1st conceive: 28 mo	11354
	f: 1st birth: most 3 yr; min 2 yr	11353, 11356
	f: 16-18 mo	11356
	m: 1st sire: 16 mo; n1	11356
	f/m: 1st mate: 3 yr	10042
Connochaetes taurinus	f: 1st mate: 1 yr	10637
	f: 1st mate: 15 mo	3099
	f: 1st conceive: 22% of yearlings, all by 2-3 yr	441
	f: 1st birth: 2 yr	3105
	f: 1st birth: 2 yr	10638
	f: 1.5-2.5 yr	5764
	m: 1st mate: 28 mo; mode 40 mo	3099
	m: over 5 yr	5764
Damaliscus dorcas	f: 1st mate: 18 mo	5702
	f: 1st mate: mode 28 mo	2746
	f: 1st birth: 2-3 yr	10158
	f: 1st birth: 36 mo	9328
	m: 1st mate: 18 mo some; most later	5702
	f/m: 1st mate: 18 mo	10023, 10027
	f/m: 28 mo	2746
Damaliscus lunatus	f: 1st mate: yearling	5308
	f: 1st birth: 2 yr 7/10, 3 yr 6/6	5101
	f: 1st birth: ~36 mo	1790, 1792
	m: testis wt: 32-34 mo	1792
	m: spermatogenesis: 40-42 mo	4084, 4085
	m: epididymal sperm: 18, 30 mo; n2	5308
	m: 1st mate: 2 yr 1/8, 3 yr 3/5, 4 yr 5/5	5101
	m: 1st repro: 40-42 mo	1792
Gazella cuvieri	f: 1st birth: min 344 d	8100
	m: 1st sire: 180 d	3567
Gazella dama	m: 1st sire: 120 d	3567
Gazella dorcas	f: 1st conceive: 18 mo	493
	f: 1st conceive: 1 yr 9 mo 23 d	2600
	f: 1st conceive: 716 d; n14	5997
	f: 1st birth: 27.5 mo	2603
	m: 1st sire: 589 d	3567
Gazella gazella	f: 1st conceive: 6-18 mo	493
	f: 1st birth: 12 mo	491
	f: 1st birth: 2 yr	492
	f: 18 mo	492
Gazella granti	f: 1st mate: 7-14 mo	3247
	f: 1st conceive: 152 d	2600
Gazella leptoceros	f: 1st birth: 10.5 mo	2603
Gazella soemmerringi	f: 1st conceive: 1 yr 7 mo	2600
	f: 1st birth: 26.375 mo; r25-27.75; n1	2603
Gazella spekei	f: 1st conceive: 16.8 mo; n6	8991
Gazella subgutturosa	f: 1st mate: 18-19 mo	4643
	m: 1st rut: 19 mo	9175
	m: 1st mate: active 2.5 yr	4643
Gazella thomsoni	f: 1st mate: 10-12 mo	1196
	f: 1st mate: ~13 mo	5122
	f: 1st mate: 1 yr	4972
	f: 1st mate: 1st yr	1328
	f: 1st conceive: 9-11 mo est	9180

	f: 1st conceive: 1 yr 4 d	2600
	f: 1st birth: ~18 mo	5122
	f: 1st birth: 18.33 mo	2603
	m: spermatogenesis: ~12 mo	5122
	m: 1st fertile: 480 d	3567
	m: 1st mate: 2 yr	4972
	m: 1st breed: 3.5 yr	5122
Hemitragus hylocrius	f: 1st birth: earliest 22 mo	11750
Hemitragus jemlahicus	f: 1st birth: 2 yr	9547
	m: 5-6 yr	9540
Hippotragus equinus	f: 1st conceive: 23-25 mo	10158
	f: 1st birth: 32-34 mo	10158
	f: 1st birth: 3 yr	11758
	f: 1st birth: 3 yr; n1	5101
	f: ~2 yr	5764
	m: electroejaculation: 16-19 mo	11758
	m: 1st mate: 3 yr; n1	5101
Hippotragus niger	f: 1st estrus: 27 mo; n2	4085, 4086
	f: 1st birth: 28.66 mo	2603
	f: 1st birth: 3 yr	11758
	f: 1st birth: 3 yr, 1/3; 4+ yr, 18/25	5101
	m: electroejaculation: 16-19 mo	4086
	m: sexual activity: 3 yr	4086
	m: 1st mate: 5 yr 1/2	5101
	m: 3 yr	11758
Kobus ellipsiprymnus	m: own territory: 6 yr	10269
Kobus kob	f: 1st ovulation: 13-14 mo	1434
	f: 1st birth: 13 mo; n1	1432
	m: fertile: 1 yr	1434
Kobus leche	f: 1st mate: 2-3 yr	9181
	f: 1st preg: 3rd yr	11731a
	f: 1st birth: yearling	10158
	f: 1st birth: 2-3 yr	7368
	m: testes size: 4-5 yr	9181
	m: ~15 mo	10158
Litocranius walleri	f: 1st birth: 18-19 mo; n2	6410
	f: end second yr	5789
Madoqua kirki	f: 1st birth: 15-18 mo	4622
	f/m: ≤8 mo	5625
Madoqua saltiana	f: ~10 mo	5763
	m: ~9 mo	5763
Nemorhaedus goral	f/m: 1st mate: 3 yr	1310a
	f/m: some 2nd yr	4643
	f/m: 2-3 yr	7125
Neotragus batesi	f: 1st birth: 8-16 mo	3191, 3192
	m: 8-18 mo	3192
Neotragus moschatus	f/m: 6 mo	5191
	f: ~6 mo	5763
Oreamnos americanus	f: ovarian histology: 2.5 yr	1237
	f: 1st mate: 18 mo; captive; later in wild	4997
	f: 1st mate: 2.5 yr	1871a
	f: 1st cl: 27.5 mo	6381a
	f: 1st birth: 3 yr; n2	5112
	f: preg: 34 mo; n1	6381a
	f: lact: 38 mo; n1	6381a
	m: sperm present: 1-2 yr	4620
	m: 1st mate: yearling	4997
	m: rut: 18 mo	4339
	m: large testes: 39.5 mo; n1	6381a
	f/m: 1st mate: 2.5 yr	1474
	f/m: >2 yr	5146
Oreotragus oreotragus	f/m: ~1 yr	5763
Oryx dammah	f: 1st birth: 32.92 mo; r30.333-35.5; n2	2603

Oryx gazella	f: 1st birth: 26 mo	11212
	f: 1st birth: 29-33 mo	10158
	f: ~24 mo	10158
Ourebia ourebi	f: 1st birth: 22.17mo; r19.333- 25; n2	2603
	f: ~10 mo	5763
	m: 14 mo; n1	1520
Ovibos moschatus	f: 1st estrus: 14-15 mo	8798
	f: 1st mate: 3 yr	5300
	f: 1st birth: min 3 yr; mode ~4	10259
	f: 1st birth: 4 yr; few data	10740
	f: 3 yr	8090
	m: 6 yr	5300
	m: 6 yr; few data	10740
	f/m: 3-4 yr	1474
Ovis ammon	f: 8/9 mo-1.5 yr	3571
	m: horn growth: ~4 mo	3571
	m: 1.5-2.5 yr	3571
	m: 2.5 yr	4643
Ovis aries	f: 1st mate: 18 mo	9175
	f: 1st birth: 2 yr	9547
	m: 1st mate: 2.5 yr; in herds only 4.5+ yr mate	9175
	m: spermatogenesis: ~6 mo	4109
Ovis canadensis	f: 1st mate: 1.5 yr few; mode 2.5 yr	10105
	f: 1st mate: 2 yr	4950
	f: 1st mate: 2.5 yr	10257
	f: 1st mate: ~2.5 yr	4643
	f: 1st birth: 290-313 d	7414
	f: 1-2 yr	1871a
	m: 1st mate: >3 yr	4643
	m: 1.5-2.5 yr	3667
	m: 18-36 mo	1871a
	f/m: 1-2 yr	788
	f/m: 18 mo	9815
Ovis dalli	f: 1st mate: 18-30 mo	7883
	f: 1st birth: some 2 yr; mode 3 yr	4878
	f: 1st repro: 2 yr captive; 50% 4 yr; adult wt 5 yr	1447
	m: 1.5-2.5 yr	3667
	f/m: ~18 mo	7884
Ovis musimon	f: 1st mate: 18 mo	10946
	m: spermatogenesis: ~6 mo	7579
	m: 1st mate: 30 mo	10946
Ovis nivicola	f: 2 yr	10722
Pseudois nayaur	f: 1st birth: 2 yr	9547
	f: yearling	11449
Raphicerus campestris	f: 1st conceive: 9.5 mo; n1	1684
	f: 1st preg: when the second molars erupt	11790
	f: 6-7 mo	10158
	m: 1st sire: 8.5 mo; n1	1684
	f/m: 6-9 mo	5763
Raphicerus sharpei	f: 1st cl; when the second molars erupt	5699
	m: spermatogenesis: when the third molars erupt	5699
Redunca fulvorufula	f: 9-24 mo	5165
	f: min 9 mo; mode 12-14 mo	10158
	m: 1st sperm: 8-12 mo	5165
	m: 1 yr	10158
Rupicapra rupicapra	f: corpora albicantia: 2.5 yr; no data <2.5 yr	3663
	f: 1st mate: 1.5 yr	2080
	f: 1st mate: 96% 3 yr	8477
	f: 1st preg: 7 mo; n1	658
	f: 1st preg: 2 yr, n1 of 3; 3 yr, n3 of 7	11056
	f: 1st birth: 4 yr	9650
	f: 1st birth: yearling	657
	f: 1st birth: 3-4 yr	4643
	f: min 2 yr	10888

	f: 2nd yr	4643
	m: spermatogenesis: 18 mo	8477
	m: 1st sire: 2+ yr	8477
	m: yearling	658
	f/m: 18-20 mo some; mode 3-4 yr	3571
Saiga tatarica	f: 1st mate: 7-8 mo	4643
	f: 1st mate: 7-8 mo	560, 561
	f: 1st mate: 1.75 yr	12107
	f: 7-8 mo	2215
	m: 1st mate: 20 mo	560, 561
	m: 1st mate: 2nd yr	4643
Sigmoceros lichtensteini	f: 1st birth: 2 yr	10158
	f: 16-18 mo	1118
Sylvicapra grimmia	f: 1st mate: 8 mo; n1	334
	f: 1st mate: ~16 mo; n1	11789
	f: 1st conceive: 8-9 mo	10158
Syncerus caffer	f: 50% cl present: 5 yr	4077
	f: 50% ovulate: 3.5 yr	9991, 9994
	f: 50% preg: 4.8 yr	1599
	f: 50% preg: 5 yr	9991, 9994
	f: 1st birth: 4 yr	8515
	m: testes wt, sperm: 2.5-3 yr	4077
	m: 1st mate: 7-8 yr	8515
	m: 2.5-3 yr	8515
	f/m: ~2 yr	5763
Tetracerus quadricornis	f: 1st birth: 1 yr 9 mo; n2	37
Tragelaphus angasi	f: 1st ovulate: 14-18 mo; 50% 20 mo	269, 271
	f: 1st birth: 21.75 mo	2603
	f: 1st birth: 2 yr	7774
	m: spermatogenesis: 12 mo-2 yr	269, 271
	m: 1st mate: > 5 yr	269, 271
Tragelaphus eurycerus	f: 1st conceive: 29 mo; r27-31; n2	8867
	f: ~2 yr	11022
	m: ~2.5 yr	11022
Tragelaphus imberbis	f: 1st birth: 36.66 mo	2603
Tragelaphus oryx	f: 1st estrus: 15-24 mo	10367
	f: 1st estrus: 20.2 mo; r17-37; n19	9311
	f: 1st mate: 28.7 mo; ±2.3	10023
	f: 1st conceive: 13 mo	10895
	f: 1st birth: 2 yr min; wild 4 yr	10158
	f: 1st birth: 26-37 mo	10300
	f: 1st birth: 26-46 mo	8690, 9311
	m: 15-24 mo	10367
	m: 18 mo min	10158
Tragelaphus scriptus	f: 1st conceive: 18 mo	8712
	f: 1st ovulation: 11 mo	7537
	f: 1st birth: 535.5 d; r509-562; n2	11347
	f: 1st birth: 24.75 mo	2603
	f: 14 mo	10158
	m: spermatogenesis: 11 mo	7537
	m: 10.5 mo	10158
Tragelaphus spekei	f: 1st birth: 16-23 mo	2485
	f: 1st birth: 20 mo; n1	9693
	f: 1st birth: 22-28 mo	6192
	f/m: full maturity: ~4 yr	5763
Tragelaphus strepsiceros	f: 1st mate: 17 mo	10023, 10027
	f: 1st preg: 17 mo	9986
	f: 1st birth: 2 yr	2275
	f: 1st birth: ~3 yr	11786
CYCLE LENGTH		
Addax nasomaculatus	estrus: 1.5-2 d	5115
Aepyceros melampus	estrus: 24-48 h	3147
	12-29 d	3147
Ammotragus lervia	estrus: few h	4535

Antidorcas marsupialis	16 d; mode 16; r14-17; n18 cycles, 1 f	6510
Antilocapra americana	27.4 d; sd6.6; r12-40; n11 cycles, 4 f	8644
Antilope cervicapra	estrus: ~24 h	1649
	estrus: 24 h	10628
	5-6 d; not cycle length	10628
	20 d	4860
Bison bison	estrus: 9-28 h	7143
	~3 wk	7143
Bison bonasus	estrus: lasts ~6-8 d	4643
	20.6 d; 80% 18-22; r12-28	5999
	mode 20-30 d; r10-70; interestrous intervals	5229
Bos frontalis	estrus: 1-4 d	9593
	3 wk	9545
	26.3 d; r7-37; n15 cycles, 1 f	9593
Bos javanicus	3-4 wk	11088
Bos taurus	29.3 d; mode 21-22; 60% 18-25; n500 cycles, 200 f	10905
Boselaphus tragocamelus	estrus: 4 d	9175
Bubalus bubalis	estrus: 19.6-19.8 h	467
	estrus: 21.69 h; r16-27	9445
	estrus: 2-3 d	2936
	r20-22 d; n14 f	5496
	21.39 d; r19-25; n46 cycles, 7 f	9445
	22-28 d	2936
	22 d; se0.6; 30% 31-60; 49% 11-30; r10-88; n235 cycles, 87 f	4187
	25.05 d; se0.41	8927
	32-50 d	9994, 10938
Capra hircus	18.4 d; ±5.6; 70% 18-24; r5-27; n20 cycles, 6 f	8230
	19-21 d	9223
Capra ibex	estrus: <24 h - several d	2081
	20 d; sd1; n13 cycles, 10 f	10530
Capricornis crispus	20-21 d	5178
Cephalophus rufilatus	estrus: 0.5-1 d	2605
Cephalophus zebra	estrus: 0.5-1 d; n1	9691
Damaliscus dorcas	estrus: <24 h	2296
Gazella gazella	28 d; n2 cycle, 1 f	9175
Gazella subgutturosa	estrus: 12 h	9175
Gazella thomsoni	estrus 1-3 ?d	4972
	1 ?wk	4972
Kobus kob	estrus: <24 h	1434
	6-13 d	7540, 7541
	20-27 d	1434
Madoqua kirki	estrus: ~2 d	5763
Madoqua saltiana	estrus: ~2 d	5763
Oryx dammah	luteal phase: 16-18 d	2815
	21-22 d	2815
	21.7 d; r20-23; n3 cycles, 1 f; from graph	6581
Ovibos moschatus	19.6 d; sd0.96; n19 cycles, 6 f	9331
Ovis ammon	estrus: 3 wk to 1-1.5 mo	4643
Ovis aries	estrus: ~1 d; lambs; ~1-3 d; ewes	4109
	estrus: 45.3 h; n49	12011
	15 d; r10-18 d; 1st ovulation no estrus	4109, 4111
	16.7; n43 cycles, 5 f	12011
	24.9 d; se3.3; n8 cycles, 3 f	12010
Ovis canadensis	estrus: 1-3 d	4908
	estrus: 2 d	10950
	~28 d	10950
Raphicerus campestris	estrus: ~4 d	1684
	3 mo	1684
Syncerus caffer	23 d	9994
Tragelaphus angasi	19 d	270
	24 d; r10-34; n5 cycles, 1 f	269, 271
Tragelaphus eurycerus	estrus: 1-2 d	1381
	estrus: ~3 d	8867
	21-22 d	8867

	22 d; n1 f	1381, 2720
Tragelaphus oryx	estrus: 1-2 d	10367
	estrus: 3 d	8690
	22.16 d; r22-23; se0.04	10367
	21-26 d	8690
GESTATION		
Addax nasomaculatus	260.5 d; r257-264; n2	2603
	265 d; r292-315	8421
	10-11 mo; according to natives	1352
Aepyceros melampus	5-6 mo	11697
	6 mo; field data	10041
	6 mo	3145
	195-200 d	3147, 10158
	195-204 d	8965
	~6.5 mo	5764
	196 d	3146
	196 d	10023
	196 d	1230
	204 d ea; n2	633
Alcelaphus buselaphus	238 d; n1	2603
	8 mo	4582
	~8 mo	10042
	~8 mo	3956
	~242 d	2382
	~9 mo	5764
Ammotragus lervia	12-15 wk	4535
	5 mo	1352
	154-161 d	8223
	22-23 wk	1355
	155 d; sd5	6523
	160 d	6652
	165 d	9734
Antidorcas marsupialis	167 d	1230
	168 d	10023, 11213
	~171 d	923
	177 d; r174-180; n1 f	6510
Antilocapra americana	217-252 d	11208
	7.5 mo	6387
	~246 d; field data	6217
	250 d	12019
	252 d; r249-255	4647
	~9 mo; field data	2607
Antilope cervicapra	5 mo	7616
	5 mo; field data	6541
	6 mo	1355
	6 mo	10628
Bison bison	~9 mo; field data	3559, 3560
	~9 mo	188
	9 mo	1355
	9-9.5 mo; not reliable	5229
	~285 d	4501
	~9.5 mo	7144
Bison bonasus	261-283 d	4643
	262 d	2109
	264.29 d; sd4.20; r254-272; n132	5999
	265 d; 56% 260-269; r256-283; n52	5229
	40-41 wk	3571
	9 mo	3822a
Bos frontalis	~8-9 mo	2802
	9 mo	5047
	9 mo	4582
	298.2 d; r293-301; n5	9593
Bos grunniens	8.5-9 mo	1355
	9 mo	4582

	~9 mo; field data	8740
	280 d	1969
Bos javanicus	285-300 d	11088
	297 d; se5	6524
	300 d; r329-338	8421
Boselaphus tragocamelus	8 mo	4582
	<8 mo 7 d	1355
	247 d	9175
	250 d; n1	6145
	280-300 d	1046
Bubalus bubalis	~10 mo	1355
	~10 mo	6654
	317 d	4187
	320 d; r312-334	10939, 10940
Bubalus depressicornis	9.5-10 mo	4582
Budorcas taxicolor	200-220 d; n1	459
	7-8 or 10-11 mo	7800
	8 mo; field data	11419
Capra caucasica	5 mo? (no data provided)	8458
	150-160 d	4643
Capra falconeri	155 d	9547
	162-170 d	9175
	~6 mo	4643
Capra hircus	<5 mo	11389
	5 mo	9223
	5 mo; field data	9668
	150-155 d	9175
	<156 d	9357
	165-170 d; field data	2080
Capra ibex	22-23 wk	3571
	155-170 d	9175
	165-170 d	2081
	167 d; sd3; n63	10530
	170-180 d	4643
	~6 mo; field data	6652
Capricornis crispus	200-230 d	11996
	210-220 d	5947
	212 d; r211-213	5178
Cephalophus dorsalis	7.5-8 mo	2605
	256 d; n1	2603
Cephalophus maxwelli	120 d	74
Cephalophus monticola	167 d	1230
	205 d	7204
	207 d; se6; mode 206-210; r196-220; n16	1047
Cephalophus niger	126 d; n1	3175
Cephalophus ogilbyi	4 mo	5277
Cephalophus rufilatus	223 d daughters, 241 d sons	2605
	236.3 d; r223-245; n3	2603
Cephalophus sylvicultor	4 mo	5277
	151 d; n1	3175
Cephalophus zebra	222.8 d; r190-241; n5	9691
Connochaetes spp	8-8.5 mo	1355
Connochaetes gnou	8.5 mo	4582
	8.5 mo	10042
	250-260 d	7204
	9 mo; field data	10041
Connochaetes taurinus	239-243 d	2449
	~8 mo; field data	441
	8-8.5 mo	3099
	251 d; n1	10349
	252 d; r249-255; n2	10351
	8.5 mo	11697
	~255 d; r8-9 mo	9993
	~9 mo	10637

Damaliscus dorcas	223 d; n1	2603
	225 d	10023, 10027
	7.5 mo; field data	5702
	238-254 d	10158
	8 mo	9328
	~8 mo; field data	6657
	~8 mo; field data	6204
	~8 mo; field data; 237/8, 254 d; n2; 1st mating to birth	2295, 2296
	288-308 d	1230
Damaliscus hunteri	279 d; n1	5249
Damaliscus lunatus	7-8 mo; field data	1790, 1792
	235-241 d	5764
	8 mo; field data	11697
	8 mo; field data	7470
Gazella cuvieri	~160 d	8100
	165-180 d	3567
Gazella dama	200 d	3567
	~7 mo	345
Gazella dorcas	3 mo	8223
	3 mo	254
	3 mo	5575
	169-174 d; n4; 1 stillborn 181 d	10059
	172.5 d; r171-174; n4	2600, 2603
	6 mo	493
Gazella gazella	6 mo	493
Gazella granti	~6 mo	5764
	195 d	3247
	198.7 d; r198-199; n3	2600, 2603, 2606
Gazella leptoceros	162.5 d; r156-169; n2	2600, 2603
Gazella rufifrons	5 mo	5277
	exactly 6 mo	345
	187.5 d; r186-189; n2	2600, 2603
Gazella soemmerringi	6-7 mo	5764
Gazella spekei	178.6 d; r169-190; n15 lt, 8 f	8991
	~7 mo	4462
Gazella subgutturosa	5-5.5 mo	9175
	5.5-6 mo	4643
Gazella thomsoni	implant: 60 d	4972
	115 d; r114-116; n2; mating to birth	4972
	3.5-3.75 mo; n2; presumed mating to birth	4972
	150 d	3247
	est 150-170 d	5122
	5-6 mo	5764
	165 d	2449
	168 d; r152-186; from fetal wt	1196
	191 d; r185-195; n8	9180
	199 d; r189-214; n19 f	4972
	223 d; r222-224; n2	9180
Hemitragus hylocrius	180-190 d; postpartum estrus to birth	11750
Hemitragus jemlahicus	6 mo	9540
	~6 mo	1355
	~6 mo; field data	6652
Hippotragus equinus	274 d; r268-280; n2	2603
	275 d	11758
	~275 d	2449
	~275 d	5764
	276-287 d	10158
	10 mo; field data	254
	10 mo; field data	5575
Hippotragus niger	240-248 d	11758
	265 d; r261-271; n3	2603
	266 d; r259-272; n4; 1st mount to birth	4085, 4086
	9 mo	4582
	9 mo	11697

	272 d	6801
	275, 266 d, n2	5972
	~10 mo; field data	3104
Kobus ellipsiprymnus	4+ mo	5052
	est 8 mo	4326
	243 d	2439
	9 mo	4582
	272-287 d	2449
	280 d	10158
Kobus kob	~6 mo	345
	235-240 d	7540
	236-252 d	1432, 1434
	261-271, 266-267 d, 8 mo 26 d	11037
	~9 mo	5763
	10 mo	7591
Kobus leche	7 mo	11697
	215-248 d	2449
	7-8 mo	9181
	8 mo; field data	333
Kobus vardoni	~8 mo	10158
	~9 mo	5763
Litocranius walleri	6.5-7 mo	6410
	203 d; n1	9636
Madoqua kirki	170-174 d	5763
	170-174 d; n1	5625
	171 d; r169-174; n3	2598, 2603
	172 d	4977
	25 wk	4622
Nemorhaedus goral	~6 mo	6654
	6-8 mo	9175
	~7 mo; field data	3034
	~8-9.5 mo; n4	2652
	250-260 d	4643
Neotragus batesi	~6 mo	5763
Neotragus moschatus	180 d	5191
Oreamnos americanus	147 d	5299
	6 mo; field data	1237
	185.8 d; sd1.34; n9	5112
	~191 d; field data	4933
Oreotragus oreotragus	5 mo	10158
	7-7.5 mo	2170
	214 d	7531
Oryx dammah	242-256 d	5896
	246 d; n1	2603
	9 mo; field data	3781
Oryx gazella	8.5-10 mo	4582
	264 d; n1	2603
	~264 d	1230
	265 d	8421
	265-300 d	5764
	~9 mo	11212
Oryx leucoryx	240 d	7531
Ourebia ourebi	192 d	6315
	210 d	7531
	210-211 d	10158
	7 mo	11697
	7 mo	5763
	217-220 d	2608
Ovibos moschatus	~8-8.5 mo; field data and 1 captive birth	10738, 10740
	244-252 d; n4	6381
	250 d	9330
	250-275 d	8798
	8.5-9 mo	5300
	9 mo	2364

Ovis ammon	~5 mo	6654
	~5 mo	4643
	21-22 wk	3571
	5.5 mo	9175
	~6 mo; field data	6652
	~7 mo; field data	8740
Ovis aries	134 d	9175
	150 d	9547
	150 d	7735
	151 d; se1.34; n162	1181, 4111
Ovis canadensis	174.2 d; sd1.70; r171-178; n20	9816
	175 d	3667
	176 d; n1	1021
	179 d; sd6	10950
	180 d (maybe slightly more); field data	3535
	~180 d	10257
	6 mo; field data	10105
	~6 mo	1355
	~6-7 mo; field data	6652
Ovis dalli	5.5-6 mo; field data	7625
	171 d; se2.21; field data	7883, 7884
	174 d; r172-176; n2	3667
Ovis musimon	148-159 d	8467
	153 d; n2	9318
	~5.5 mo	10946
Ovis vignei	~6 mo	1355
	~6 mo	6654
	~6 mo; field data	6652
Pantholops hodgsoni	~7 mo; field data	8972
	~8 mo; field data	8740
Pelea capreolus	~261 d	1230
Procapra gutturosa	~6 mo	4643
	186, 189-192 d; n2	7395
Pseudois nayaur	est 4-5 mo; few field data	11419
	est 159-160 d; field data	11778
	160 d	9547
	~6 mo; field data	9541
Raphicerus campestris	168 d; min interbirth interval	922
	210 d	7531
	7 mo	11697
Raphicerus sharpei	7 mo	10158
Redunca arundinum	7-7.5 mo	5763
	~215 d	2449
	7.75 mo	11697
Redunca fulvorufula	~6 mo; field data	10025
	223 d; n1	2603
	236-251 d; n1	5165
Redunca redunca	~7.5 mo	5763
Rupicapra rupicapra	160-170 d	2079, 2080
	165-185 d	8477
	25-27 wk	3571
Saiga tatarica	5 mo	4643
	~5 mo	4645
	~5 mo; r139-152 d	560, 561
	153 d; n2	3377
Sigmoceros lichtensteini	214 d; ~8 mo	2439
	237 d	2449
	240 d	2710
Sylvicapra grimmia	3 mo	10158
	est 4.5-7 mo	7204
	123 d	2449
	170 d; n1	9971
Syncerus caffer	300 d	5575
	300 d	254

	11 mo	11697
	11 mo	3145
	~11 mo	5763
	~335 d	1230
	340 d	9994
	343-346 d	2449
	345 d; r343-~346; n2	11292
Tetracerus quadricornis	~6 mo	6654
	~180 d	7382
	235.5 d; r228-243; n2	9957
	240-255 d	37
Tragelaphus angasi	~6 mo	1230
	220 d; 7 mo	269, 270, 271
	252 d	7774
Tragelaphus eurycerus	282 d; n1; [reported as 294-296 d]	8867, 11972
	285 d; n1	2720
	298 d; embryo transferred 8 d after ovulation & insemination	2719
Tragelaphus imberbis	~7 mo	5763
	222 d; n1 stillborn	2603
	8 mo 10 d; r8 mo 1 d-8 mo 20 d; n3	6189
	8-8.5 mo; r8-11	6411
Tragelaphus oryx	~4.5 mo; field data	5052
	255 d	10894
	8.5 mo	8690
	8.5 mo	11697
	8.5-9 mo	4582
	260 d	2439
	260-284 d	5763
	9 mo exactly	1355
	9 mo	5277
	9 mo; field data	6058
	271 d; se2.9	10023
	273 d; r264-278; n12	10300
	276 d; ±4; n40 lt 1st calves	10895
	280.4 d; n5	9311
Tragelaphus scriptus	~5.5 or ~6 mo	5763
	179.7 d; r178-187; n3	2603, 2608
	6 mo	201
	~6 mo	7537
	*e*st r6-7 mo	11697
	214 d	2439
Tragelaphus spekei	~7.5 mo	5763
	7 mo 24 d; n1	9693
	239 d; n1	5688
	8 mo + few d; r8 mo 4 d-8 mo + 15 d; n7	6192
	247 d; r225-258	2485
	251-260 d	8421
Tragelaphus strepsiceros	~4 mo; field data	5052
	210 d	5671
	210-214 d	10023, 10027
	7-8 mo	11697
	214 d	5299
	~8 mo	11786
	261 d	8421
	~9 mo	5763
LACTATION		
Aepyceros melampus	solid food: 2.4 d; sd0.6; r2-3	7503a
	wean: 4-7 mo	5259
Alcelaphus buselaphus	solid food: 2 wk	10158
	wean: 4 mo	3956
	wean: 7 mo	10158
	wean: 8 mo	10042
Ammotragus lervia	wean: 4 mo	1352

Antidorcas marsupialis	solid food: 2 wk	10158
	wean: 120-122 d; n1 f	6510
Antilocapra americana	solid food: 3 wk	12019
	wean: 3.5-4 mo	5828
	wean: 12 wk	1515a
	wean: 4 mo	8009
Antilope cervicapra	solid food: 5 d nibbled on grass	9545
	wean: ~8-9 wk; nursing infrequent; max 12	785
Bison bison	solid food: 5 d	7143
	solid food: ~1 wk	4048
	can survive: 7-8 wk	7143
	wean: 5-9 mo	6342
	wean: 9-12 mo; n8 f with subsequent birth	4047
	wean: 17-21 mo; n5 f without subsequent birth	4047
Bison bonasus	wean: 5-9 mo	6342
	wean: 5 mo-2 yr	5229
	wean: 6-8 mo	3571
Bos frontalis	wean: 4.5 mo	3447
Bos grunniens	wean: 5-9 mo	6342
	wean: 8 mo	1969
Bos javanicus	wean: 10 mo	11088
Budorcas taxicolor	solid food: 2 wk	8798
	solid fodd: 2nd mo	7800
Capra caucasica	solid food: 1 mo	4643
Capra falconeri	wean: 5-6 mo; into winter	4643
Capra hircus	wean: 4-5 mo; rejection by mother	9175
	wean: 6 mo	11389
Capra ibex	solid food: < 1 mo	4643
	wean: 3 mo	6764
	wean: 1 yr	8594
	wean: ~1 yr	3571
Capricomis crispus	solid food: 1 mo	11996
Cephalophus monticola	solid food: browse at 2 wk	5763
Cephalophus sylvicultor	wean: 4-6 wk	6642
Cephalophus zebra	solid food: within 1 wk	11021
Connochaetes gnou	solid food: 4 wk	11356
	wean: 6-9 mo	11356
Connochaetes taurinus	solid food: 10 d	10637
	solid food: 10-14 d	10158
	wean: 7-8 mo	11697
	wean: 8 mo	10158
	wean: ~1 yr with new calf, 16 mo no new calf	10637, 10638
Damaliscus dorcas	solid food: 7 wk grazes 40% of the day	2296
	wean: 4 mo	2746
	wean: 6 mo	6204
Gazella dorcas	can wean: 3 wk	5997
	wean: 75-90 d	493
Gazella gazella	wean: 75-100 d	493
Gazella subgutturosa	solid food: 8-10 d	4643
	wean: 1-6 mo	4643
Gazella thomsoni	wean: ~2 mo	5764
	wean: 4.7 mo; r1.5-7	4972
Hemitragus hylocrius	solid food: 14 d; nibbled	11750
Hippotragus equinus	solid food: 40 d	5248
	wean: 6 mo	10158
Hippotragus niger	wean: 6 mo	11758
	wean: 8 mo	4086
Kobus ellipsiprymnus	wean: 6-8 mo	10269
	wean: ~6-8 mo	5763
	wean: 276 d; lact stops 180-210 d	10158
Kobus kob	wean: ~180 d	7541
	wean: 6-7 mo	1432
Kobus leche	wean: 3-4 mo	7368
	wean: ~5-6 mo	10158

Madoqua kirki	nurse: short period	5625
	wean: ~6 wk	5763
	leave parents: ~6 mo	4977
Madoqua saltiana	wean: ~6 wk	5763
Nemorhaedus goral	solid food: ~1 mo	4643
Neotragus moschatus	solid food: few d	5763
Oreamnos americanus	solid food: 1st d	1237
	solid food: 2-3 d	4933
	nurse: est 3 mo; stay with f 1 yr	9122
	wean: ~4 mo	1237
Oreotragus oreotragus	wean: 4-5 mo	10158
Oryx leucoryx	solid food: 30 d	4943
Ourebia ourebi	wean: 4-5 mo	2439
Ovibos moschatus	solid food: 2-3 wk	8798
	wean: 5-9 mo	10770
	nurse: est until next birth; n1	8033
Ovis ammon	solid food: ~2 wk	3571
	solid food: 1 mo	4643
	wean: 4-5 mo	3571
Ovis canadensis	solid food: 1 wk	3667
	solid food: 10 d	4643
	solid food: 14 d	9815
	solid food: ~2 wk?	1011
	solid food: 3 wk-1 mo	2476
	wean: 4-6 mo	3667
	wean: 5-6 mo	1011
Ovis dalli	solid food: 1 wk	3667
	wean: 3-5 mo	1166a
	wean: 4-5 mo	3667
Ovis musimon	wean: 6 mo	10946
Procapra gutturosa	solid food: 10 d	4643
	solid food: 40 d	7395
Raphicerus campestris	solid food: 2 wk	1684
	wean: 3 mo	1684
	wean: 3 mo	5763
Rupicapra rupicapra	solid food: 1 mo	4643
	solid food: 2 mo	2079
	wean: 2-3 mo	2079
Saiga tatarica	solid food: 2-2.5 mo	4643
	solid food: 3-4 d	561
	solid food: 4-8 d	4643
	solid food: 8 d	560
	solid food: 3 wk	3377
	solid food: 1 mo	4645
	wean: 6 wk	3377
	wean: 2.5-3 mo	560, 561
	wean: 2.5-4 mo	4643
	wean: 4 mo	10200
Syncerus caffer	suckle: 4-5 mo	8515
	wean: 5-6 mo	8515
	wean: 10-12 mo	1599
	wean: 12 mo	4077
Tragelaphus angasi	wean: 7 mo	269, 271
Tragelaphus eurycerus	solid food: 1-2 wk	11022
Tragelaphus oryx	solid food: min 1 d	11044
	dependent: 3 mo	4801
Tragelaphus strepsiceros	wean: 6 mo	9986
INTERLITTER INTERVAL		
Addax nasomaculatus	355 d; 33.7% 275; r227-741	2487
	1 yr	6741
Aepyceros melampus	365 d	10023
Alcelaphus buselaphus	326 d; r271-366	3955
	1 yr	3956

Ammotragus lervia	182 d; r178-184; n3	9734
	0.5-1 yr	1352
	<7 mo; n2	4020
Antidorcas marsupialis	1 yr	10038
Antilocapra americana	1 yr	12019
Antilope cervicapra	6 mo; r5-7; n2	6541
	6 mo	7616
	6-9 mo	10628
	197 d; n6	36
	1 yr	9545
Bison bison	44-76 wk	4048
	1 yr	5229
	54 wk; se5; n37	4047
	1.5 yr	3559
	1-2 yr	6585
Bison bonasus	1-2 yr	3571
	2 yr	4643
Bos frontalis	1 yr	9545
	14-15 mo; r394-906 d; n1 f	3447
	not until 3rd yr	2802
Boselaphus tragocamelus	1 yr	6654
Bubalus bubalis	1 yr	6654
Budorcas taxicolor	1 yr	8798
Boselaphus tragocamelus	436 d; n8	36
Capra caucasica	1 yr	4643
Capra hircus	7-10 mo	9357
	1 yr	4643
	2 yr	2080
Capra ibex	1 yr	10530
	1 yr	4643
	2 yr	2081
Capricornis crispus	18 mo; ovarian histology	5826
Cephalophus monticola	265 d; se21.4; r208-446; n39 lt, 9 f	11349
Cephalophus sylvicultor	399 d median	5996
Connochaetes taurinus	1 yr	10637, 10638
Damaliscus dorcas	1 yr	10023, 10027
Gazella cuvieri	0.5-1 yr	8100
Gazella dama	0.5 yr	254
	0.5 yr	5575
Gazella dorcas	310.7 d; median 271; r177-965; n95 intervals, 13 f	5997
	1 yr	8223
Gazella gazella	7-8 mo or 1 yr; 2 locations	492, 493
Gazella spekei	201-212 d	8991
Gazella thomsoni	0.5 yr	5122, 5123
	0.5 yr	1328
	6-7.5 mo	4972
Hemitragus hylocrius	7.8, 8.4 mo; r7-10; n14 intervals, 2 f	11750
Hemitragus jemlahicus	1 yr	6654
Hippotragus equinus	10-10.5 mo	11758
	317 d	10158
	1 yr	11253
Hippotragus niger	1 yr	11253
	1 yr	11758
Kobus ellipsiprymnus	1 yr	11253
Kobus vardoni	1 yr	11253
Litocranius walleri	min 223 d	6410
	8-9 mo; n5, 1 f	6410
Madoqua kirki	~6 mo	5625
	0.5 yr	5625
	1 yr	4622
Nemorhaedus goral	1 yr	6654
	1 yr	4643
Oreamnos americanus	1 yr	9121
Ourebia ourebi	0.5 yr	11253

Oreotragus oreotragus	1 yr	11253
Ovibos moschatus	1 yr	1474
	1 yr	8033
	2 yr	10770
	2 yr	10740
	2 yr	10770
	2 yr	1474
Ovis ammon	1 yr	3571
	1 yr	4643
Ovis canadensis	1 yr	4643
Ovis dalli	~1 yr	1447
Ovis musimon	1 yr	10946
Ovis vignei	1 yr	6654
Pantholops hodgsoni	1 yr	6654
Procapra gutturosa	1 yr	4643
Raphicerus campestris	r168-270 d; n5; 4 intervals 168-177, 1 was 270	922
	6 mo?	11697
	~8 mo; n3 intervals	1684
Redunca arundinum	0.5-1 yr	11253
Redunca fulvorufula	10.2 mo; r8-13; n6	5165
	12-14 mo	10158
Rupicapra rupicapra	1 yr	3571
	1 yr	2079, 2080
	1 yr	4643
Saiga tatarica	1 yr	4645
	1 yr	561
Sigmoceros lichtensteini	1 yr	11253
Sylvicapra grimmia	4 mo	11253
	259.5 d; 2se11.6; r232-298	11348
Syncerus caffer	15 mo; 95%ci 13.3-17.5	9991, 9994
	18 mo; 95%ci 15.6-21.3	9994
	18-24 mo; ovaries	5763
	18 or 24 mo; 2 locations	4077
	20-26 mo	1599
	2 yr	11253
	2 yr	10158
Tetracerus quadricornis	316 d; r285-347; n2	37
Tragelaphus angasi	231-303 d	270
	r257-293 d; n6 birth, 1 f	7774
	297 d	269, 271
Tragelaphus imberbis	min 8 mo; n300	6408
	8-10 mo	6411
Tragelaphus oryx	~10 mo	5763
	331.7 d; r277-553; n11	8690
	337 d; r284-553; n51	9311
	354.5 d; se19.9	10023
	370.8 d; r289-602; n6	10300
Tragelaphus scriptus	0.5 yr	11253
	7.5 mo; n1	7537
	8 mo; n1 f	8712
	249.1 d; se8.6; n7	11347
Tragelaphus spekei	min 11.6 mo, 64% < 12 mo	2485
Tragelaphus strepsiceros	1 yr	10023, 10027
SEASONALITY		
Addax nasomaculatus	birth: Jan-mid Apr; Chad	2897
	birth: peak Sept-Oct; Circle d'Agadez, Niger	1352
	birth: yr rd; captive	2487
	birth: yr rd; n19; zoo	2603
	birth: yr rd; Chad	6741
Aepyceros melampus	testes size: peak Jan-Mar, low Oct-Dec; Zimbabwe	5697
	testes size: peak Apr, low June; S Africa	1334
	testes physiology: peak May, low Sept; Transvaal, S Africa	10022, 10041
	testes size: peak Nov-June, low July-Oct; Zululand, S Africa	267
	mate: Mar; Zululand, S Africa	267

	mate: Mar?; Angola	11697
	mate: Mar-May; Zululand, S Africa	7204
	mate: Apr-May; S Africa	1334
	mate: Apr-May; S Africa	6928
	mate: Apr-mid July; Kruger Nat Park, S Africa	3145, 3148
	mate: peak mid May; n Botswana	9157
	mate: end Apr-May; 10-20 d; nw Zimbabwe	7660
	mate: peak major May, minor Oct; Zululand, S Africa	268
	mate: May-Aug, peak May; Zambia	11463
	mate: mid May-mid June; Zimbabwe	2275
	mate: peak May-June; Transvaal, S Africa	10022, 10023, 10041
	mate: yr rd, peaks Dec, May; Rwanda	7470
	conceive: Mar-Apr; Serengeti, e Africa	5260
	conceive: peak May; S Africa	3149
	ovulation: May; Zululand, S Africa	267
	birth: Sept-Oct; Angola	11697
	birth: Sept-Dec; S Africa	331, 333
	birth: end Oct-early Nov; Botswana	9157
	birth: Nov-Dec; ne S Africa	3145, 3148
	birth: Nov-Dec, if dry Jan-Feb; Kruger Nat Park, S Africa	8514
	birth: Nov-Dec; S Africa	5778
	birth: Nov-mid Jan, peak Dec, low July-Oct; zoo	1230
	birth: Nov-Jan; Zululand, S Africa	7204
	birth: Nov-Jan; S Africa	10158
	birth: Nov-Feb, peak Nov; Matopos Nat Park, Zimbabwe	11785
	birth: peak Nov-Jan; S Africa	10041
	birth: Dec-Jan; Zimbabwe	2274, 2275
	birth: Dec-Jan, 2 wk; Transvaal, S Africa	8965
	birth: Dec (60%)-Jan (20%); n547; S Africa	10041
	birth: end Dec-Feb, peak Jan; Namibia	5451
	birth: yr rd, peaks Feb-May, Oct-Dec; Kenya	5597
	repro: yr rd; zoo	2603
	repro: yr rd; Kenya	6413
Alcelaphus buselaphus	mate: Feb-Mar; Transvaal, S Africa	10041
	mate: Apr-May; Zaire	477
	mate: end Apr-June; Uganda	5764
	mate: peak mid Oct-mid Nov; S Africa	1153
	mate: yr rd, peaks June-July, Nov; Kenya	3956
	conceive: yr rd; n41, 2 yr; Kenya	3247
	conceive: yr rd, peaks June-July, minor Nov; Kenya	3956
	conceive: peak Feb-Apr; S Africa	1153
	birth: peaks Feb-Mar, minor July; Kenya	3956
	birth: Aug-Oct, peak Sept; Cape Province, S Africa	7204
	birth: peak Sept-Oct; S Africa	2382
	birth: peak Oct; zoo	1230
	birth: Oct-Dec, 82% Oct-Nov; n278; Transvaal, S Africa	10041, 10042
	birth: mid Dec-mid Feb; Ethiopia	6416
	birth: end Dec-mid Feb; W Africa	375
	birth: yr rd, peaks before rain; Kenya	3955
	repro: after Oct; s of Botswana	7204
	repro: yr rd, peaks Feb-Mar, July-Aug; Kenya	6413
Ammodorcas clarkei	mate: yr rd, peak Mar-May; Mudugh Province, Somalia	7167
	mate: 30% Oct; Somalia	1890
	birth: peak Oct-Nov; Somalia	1890
	birth: yr rd; Mudugh Province, Somalia	7167
Ammotragus lervia	mate: mid July-mid Aug; Niger	1352
	mate: Oct-Nov; zoo, Ukraine	6523
	mate: yr rd, peak spring; TX	4021
	birth: start spr, sometimes Jan; Morocco	8223
	birth: 70% Mar-Apr; n26 1 yr; panhandle, TX	4021
	birth: peak Mar-Apr, low Jan, July; n176; zoo	5818
	birth: Mar-May; zoo, Ukraine	6523
	birth: May, June, Oct; zoo	9734
	birth: Dec-Jan; Niger	1352

	birth: yr rd, peak Mar; n162 lt	2109
	birth: yr rd, peak Mar-May; n190; zoo	12128
	birth: yr rd, peak spring; zoo	1230
Antidorcas marsupialis	testes size: peak Apr-May, low Jan; S Africa	10039, 10041
	mate: Feb-Apr, some June, Sept, Nov; S Africa	2297
	mate: June-Aug, peak end July; Namibia	923
	mate: yr rd, peak Mar-July; S Africa	10023, 10039, 10041
	mate: yr rd, peak May; n23; zoo	8339
	ovulation: end May-June; Transvaal, S Africa	10038
	birth: Mar-June, Aug-Nov, peak Oct; Cape, S Africa	923
	birth: peak Apr-Sept; s Africa	9703
	birth: Aug; Namaqualand, Namibia	9945
	birth: Aug-Sept; S Africa	2297
	birth: Sept-Dec; Botswana	1393
	birth: peak Sept-Jan, few Apr-July; n1655; S Africa	10023, 10036, 10038, 10041
	birth: Dec-Jan; W Africa	8351
	birth: yr rd, peak spring-early summer; zoo	1230
Antilocapra americana	testes large: July-Oct; WY, MT, CO	8010
	mate: Aug-Sept; Mexico	6387
	mate: Aug-Oct; w US	1474
	mate: mid Aug-Oct; zoo	2607
	mate: peak Sept; NM	6217
	mate: Sept-Oct; MT	1424
	mate: mid Sept-early Oct; WY	1871a
	mate: mid Sept-mid Oct; WY	4647
	mate: Oct; w US	5146
	conceive: Sept-Oct; MT	5828
	birth: Feb; Baja CA, Mexico	10809
	birth: Feb; s TX	11208
	birth: Apr-May; sw US	1474
	birth: May; NM	502
	birth: May-June; nw US	1474
	birth: May-June; w N America	12019
	birth: mid May-mid June; zoo	2607
	birth: end May-June; WY	4647
	birth: June; Mexico	7145
	birth: July; ID	11208
Antilope cervicapra	estrus: yr rd; zoo	785
	spermatogenesis: yr rd; TX	1649
	spermatogenesis: yr rd; captive	3109
	spermatogenesis: yr rd; captive, Italy	10628
	spermatogenesis: yr rd, peak Apr, mid Aug-mid Oct; India	9545
	mate: Feb-Mar; India	6653, 6654
	mate: Feb-Apr; Nepal	6365
	mate: May-June; Nepal	7382
	mate: yr rd; zoo, S Africa	1230
	mate: yr rd, peak May; n82; zoo	8339
	birth: Jan-Mar, May, Sept; n11 lt; zoo	44
	birth: Feb-May, Sept-Nov; few data; zoo, Italy	10628
	birth: Mar, June-July, Sept, Nov; n9; captive	785
	birth: Apr-May, Oct-Nov; captive	6541
	birth: May-July, peak June; n143; zoo	5818
	birth: Nov-Dec; Nepal	7382
	birth: yr rd; India	6653, 6654
	birth: yr rd; captive	2109
	birth: yr rd; captive, TX	7617
Bison bison	mate: mid June-Sept; WY	7068
	mate: peak Aug; US	188
	mate: July-Oct; NWT, Canada, WY	1474
	mate: July-Oct; WY	1871a
	mate: mid July-early Sept; NWT, Canada, WY	7144
	mate: peak July 21-mid Aug; SD, NE	4501
	mate: end July-mid Aug, 46% July 30-Aug 2; MT	6584
	mate: Aug; NWT, Canada	10216

	mate: Aug; Canada	3559, 3560
	birth: Mar-June; US	188
	birth: Mar-Oct; n36; zoo	5818
	birth: peak mid Apr-May; Yellowstone Park, WY	7068
	birth: Apr-Nov; n91; captive	2109
	birth: mid Apr-May; WY	7068
	birth: 80% end Apr-mid May; MT	9386
	birth: end Apr-Oct; SD, NV	4501
	birth: May; Canada	3559, 3560
	birth: May-early June; NWT, Canada	10216
	birth: peak May; WY, NWT, Canada	7144
	birth: peak May; NE, SD	4501
	birth: May-July; n Canada	3559
Bison bonasus	mate: Aug-mid Sept; USSR	4643
	mate: Aug-Sept; Germany	3571
	mate: Aug-Dec, peak Aug-Oct; Poland	5999
	mate: end summer; Poland	5229
	birth: low Jan, Feb n29; zoo	5818
	birth: Apr-June; WY	1871a
	birth: Apr-Dec, peak May-July; Poland	5999, 8846, 8954
	birth: May-early June; Poland	6652
	birth: May-mid June; USSR	4643
	birth: July; n2; zoo	2109
	birth: yr rd, peak May; n187; captive	5229
	birth: yr rd, peak May-July; Germany	3571
	birth: yr rd, peak May-July; zoo	11879
Bos frontalis	mate: Dec-Jan; India, se Asia	6652
	mate: peak Dec-Jan; India	2802
	mate: yr rd; India	9545
	birth: Jan-May; Aug, Sept; n16; zoo	12128
	birth: May-Oct; India	2802
	birth: peak June; n48; captive	2109
	birth: Aug-Sept, some Apr-June; India, se Asia	6652
	birth: peak Aug-Sept; India	2802
	birth: Sept-mid Mar, peak Dec-Jan; India	9545
	birth: yr rd; Malaysia	5047
	birth: yr rd; captive	3447
Bos grunniens	mate: Sept; Tibet	8740
	birth: June; according to natives; Tibet	8740
	birth: yr rd, low Feb-Mar; n31; zoo	5818
	birth: yr rd; n4; zoo	12128
Bos javanicus	birth: 14/25 Feb-Mar; captive; USSR	6524
	birth: rainy season; se Asia	6652
Boselaphus tragocamelus	estrus: May-July; zoo	4535
	mate: peak Nov; n23; zoo	8339
	mate: possible peak Nov-Dec; e Rajasthan, India	9545
	birth: Jan-Mar; n5; zoo	44
	birth: Apr-Oct, peak July; n126; zoo	5818
	birth: June-Oct; India	9545
	birth: yr rd; n61; zoo	12128
	birth: yr rd; zoo	1230
	birth: yr rd; India	9545
Bubalus bubalis	m fertile 8 mo/yr; Australia	10940
	mate: Sept; Nepal	7382
	mate: fall; India	6654
	mate: yr rd; zoo	1230
	birth: June-July; Nepal	7382
	birth: 90% Dec-Apr; Australia	10939, 10940
	birth: yr rd, peak July-Sept, low Dec-Apr; Pakistan	6737
Bubalus depressicornis	mate: yr rd; zoo	1230
	birth: Feb-July; n14; captive	8708
	birth: Feb-Mar, May-June, Oct; few data; zoo	2669
	birth: Mar, May, Sept-Nov; n7; zoo	12128
	birth: yr rd, Feb-July, Oct; zoo	2669

Budorcas taxicolor	mate: end July-early Sept; China	8798
	mate: end July-Sept; China	9547
	birth: May-June; Szetchwan, China	3034
Capra caucasica	mate: Jan?; zoo	8458
	mate: end Nov-early Jan; USSR	4643
	birth: mid May-3rd wk June; captive to mid July; USSR	4643
	birth: end May-June; n24; zoo	12128
	birth: June; zoo	8458
Capra falconeri	estrus: Dec; Kashmir	4535
	mate: Oct; s Pakistan	9175
	mate: Dec; n Pakistan	9175
	mate: early Nov-3rd wk Dec; USSR	4643
	mate: mid Dec-early Jan; Pakistan, Afghanistan	9547
	birth: end Apr-early May; captive to early June, peak May; USSR	4643
	birth: peak May, few Apr, June; n48; zoo	5818
Capra hircus	estrus: June-early July; Giura I, Greece	9668
	estrus: end Oct-Nov; Crete, Greece	9668
	estrus: Aug-early Sept; Antimilos, Greece	9668
	mate: Sept-Oct; Pakistan	9175
	mate: mid Sept-mid Oct; Pakistan	9547
	mate: Oct; Scotland, UK	7051
	mate: mid Nov-mid Dec; 2 wk; USSR	4643
	mate: Dec-mid Jan; Swiss Alps	2080
	mate: Dec-Feb; Taurus Mt, Turkey	11389
	mate: yr rd; New Zealand	9357
	birth: Apr-May; Taurus Mt, Turkey	11389
	birth: end Apr-mid June; n22; zoo	12128
	birth: end May-early June; USSR	4643
	birth: end May-mid June; Swiss Alps	2080
	birth: Jan; Antimilos, Greece	9668
	birth: end Feb-Mar; Samotha Lake, Greece	9668
	birth: end Jan-Mar; Pakistan	9175
	birth: end Feb-Mar 20; Pakistan	9547
	birth: Mar; Scotland, UK	7051
	birth: Mar; Tavolara I, ne of Sardinia, Italy	9668
	birth: Apr-May; Crete, Greece	9668
	birth: Nov; Giura I, Greece	9668
	birth: every 2-4 mo; Santa Catalina I, CA	1930
Capra ibex	estrus: Nov-Dec; Himalayas, India, Tibet	4535
	estrus: Nov-Jan; semiwild; Germany	10363
	mate: Nov-early Jan; USSR	4643
	mate: Nov-Mar; Switzerland	7912
	mate: Dec-Jan; Switzerland	4210
	mate: Dec-mid Jan; Alps, Europe	2081
	mate: Dec-early Feb; Switzerland	10530
	mate: mid Dec-early Jan; Germany	3571
	mate: Jan; Swiss Alps	6652
	mate: yr rd; Ethiopia	7912
	birth: Apr-May, later higher elevations; n Tian Shan, China	3189
	birth: end Apr-3rd wk June, peak May; USSR	4643
	birth: May; Pakistan	9175
	birth: May-June; central Asia	6652
	birth: May-June; Switzerland	7911
	birth: May-July; zoo	8595
	birth: May-Aug; Switzerland	10530
	birth: (end May) June-early July; Germany	3571
	birth: peak early-mid June; Alps	2081
	birth: end June-early July; Swiss Alps	6652
	birth: spring; Israel	6764
Capra pyrenaica	birth: mid Apr-early May; Pyrenees, France/Spain	6652
	birth: May; Spain	112

Capricornis crispus	mate: Oct-Dec; n2 yr; captive	5178
	mate: Oct-Nov; Japan	10817
	birth: end May-early June; Kasabori, Japan	104
Cephalophus adersi	mate: yr rd; captive	5763
Cephalophus dorsalis	m fertile: yr rd; zoo	2605
Cephalophus maxwelli	birth: Jan-Mar, Aug-Sept; captive	74
	birth: peak Jan-Mar, Aug-Sept; Nigeria	4354a
Cephalophus melanorheus	lact f: Apr; n1; Tanzania	186
Cephalophus monticola	birth: Jan-Feb; Uganda, w Kenya	5763
	birth: Feb; n2; e Africa	185
	birth: peak Sept-Nov, low June; n49; zoo	11346
	birth: yr rd; captive	1230
	birth: yr rd; n38; captive	1047
	birth: yr rd; Cape Province, S Africa	5777
	repro: yr rd; S Africa	10158
Cephalophus natalensis	birth: Oct-Nov; S Africa	5781
Connochaetes gnou	testes size: peak Mar-May; S Africa	10041, 10042
	mate: Jan, May-June; n26; zoo	8339
	mate: mid Mar-mid May; Transvaal, S Africa	10041, 10042
	birth: Nov-Mar, slight peak Dec; captive	1230
	birth: mid-end Nov; Pretoria Game Reserve, S Africa	11355
	birth: Dec (54%)-Jan (38%); n408; S Africa	10041, 10042
	birth: mid-end Dec; Lombard Nature Reserve, S Africa	11355
Connochaetes taurinus	mate: end Mar-Aug, peak Apr-mid June; e Africa	10638
	mate: mid Apr-mid May; Tanzania	9993
	mate: June-?; Angola	11697
	conceive: Mar-Apr, peak Apr, none Sept-Oct; Zululand, S Africa	440, 441
	conceive: peak Mar-mid June; Tanzania	9994
	birth: Jan-Feb, 80% 3 wk period; Tanzania	3102, 3105
	birth: prob Jan-Feb; Zimbabwe	2275
	birth: end May-early July; zoo	6522
	birth: Aug-Nov; w Zambia	331
	birth: Sept; Angola	11697
	birth: Sept-early Nov; S Africa	1392
	birth: Oct-Jan; Namibia	9945
	birth: Oct-Nov; Tanzania	3099
	birth: peak Nov-Dec; Luangwa Valley, Zambia	331
	birth: Nov-Dec, peak Dec; Zululand, S Africa	441
	birth: Nov-Jan, peak Dec; Matopos Nat Park, Zimbabwe	11785
	birth: end Nov-Jan; Kruger Nat Park, S Africa	8514
	birth: mid Nov-Dec; ne S Africa	3145
	birth: mid Nov-mid May, peak Jan-Mar; Kenya	10637, 10638
Damaliscus dorcas	testosterone: peak Feb; S Africa	10029
	testes size: peak Mar-Apr, low Oct; Transvaal, S Africa	10032, 10041
	mate: peak Jan; Cape, S Africa	6204
	mate: peak Jan-mid Mar; S Africa	2295, 2296
	mate: Mar-Apr; Transvaal, S Africa	6657
	mate: Mar-Apr; Transvaal, S Africa	2746
	mate: peak Mar-May; S Africa	10032, 10041
	mate: Apr-June, peak mid Apr; S Africa	5702
	mate: Nov-mid Feb; zoo	1230
	conceive: Mar-May; Transvaal, S Africa	7976
	birth: June, Aug; n2; zoo	208
	birth: Sept-Oct; S Africa	1391
	birth: peak Sept-Oct; Cape Province, S Africa	6204
	birth: Sept-Nov; S Africa	2295, 2296
	birth: Nov, sometimes Feb; S Africa	5702
	birth: Nov-Dec; S Africa	9328
	birth: 88% Nov-Dec; n364; S Africa	10032, 10041
	birth: mid Nov-Dec; Transvaal, S Africa	2746
	birth: mid Nov-Dec; Transvaal, S Africa	6657

Damaliscus lunatus	mate: Dec-Feb; Rwanda	7470
	mate: peak mid Dec-Jan; S Africa	5456
	mate: Feb-Mar; Uganda	5308
	mate: Mar-Apr; Botswana	1792
	mate: May-July; Angola	11697
	birth: June-Sept, peak July-Aug; Zambia	331
	birth: Aug-Nov; Rwanda	7470, 7476
	birth: Sept-Nov; Kruger Nat Park, S Africa	3145
	birth: Sept-Dec; Angola	11697
	birth: Sept-Dec, peak Oct; S Africa	5456
	birth: Oct; S Africa	3618
	birth: Nov; Botswana	1792
Gazella cuvieri	birth: peak spring, smaller peak fall; captive	8100
Gazella dama	birth: Sept-Oct; Circle d'Agadez, Niger	1352
Gazella dorcas	mate: Sept; Israel	493
	birth: Mar; Israel	493
	birth: Mar-May, July-Aug, Oct; n7; zoo	12128
	birth: Aug-Oct; Circle d'Agadez, Niger	1352
	repro: yr rd; Morocco	8223
Gazella gazella	mate: peak Dec-Jan; Upper Galilee, Israel	492
	conceive: Dec; Upper Galilee, Israel	493
	preg f: Jan; n1; nw Iran	6293
	birth: peak May-July; Upper Galilee, Israel	492
	birth: June; Upper Galilee; yr rd; lower Galilee, Israel	491, 493
	repro: end monsoon-early Oct, Mar-Apr; Pakistan	9175
Gazella granti	conceive: peak Dec; Tanzania	10271
	conceive: yr rd; n44 2 yr; Kenya	3247
	birth: Jan-Feb, Apr, Sept; e Africa	7856
	birth: Jan-Feb, June-Aug, Nov-Dec; e Africa	5764
	birth: peak June; Tanzania	10271
	birth: yr rd; n24; zoo	2603
	repro: yr rd; Kenya	6413
Gazella soemmerringi	mate: Oct; Somalia	5764
	birth: Apr; Somalia	5764
	birth: est Apr; Somalia	2978
Gazella subgutturosa	mate: Nov-Dec; USSR	4643
	mate: Dec; Pakistan	9175
	birth: Apr-May; USSR	4643
	birth: Apr-mid July, Sept-Oct, peak Apr-May; n22; zoo	12128
	birth: peak end Apr-early June; captive	9175
	wean: Oct; USSR	4643
Gazella thomsoni	sperm present: yr rd; s Kenya, n Tanzania	5122
	mate: dry season, Jan-Feb, July-Aug; captive; Kenya	4972
	mate: yr rd, peak Aug-Oct; Kenya	6922
	mate: yr rd; Tanzania	1328
	mate: 1 mo after birth: e Africa	5764
	conceive: yr rd; n44, 2 yr; Kenya	3247
	preg f; yr rd; Tanzania	1196
	birth: Jan-Feb, July; e Africa	5764
	birth: peaks Nov-Jan, July-Aug; captive, Kenya	4972
	birth: yr rd, peaks Jan-Mar, June-July; Tanzania	1328
	birth: yr rd, peak Apr, Nov; s Kenya, n Tanzania	5122
	birth: yr rd, peak May, Nov; n Tanzania	9180
	birth: yr rd; n59; zoo	5818
Hemitragus hylocrius	birth: yr rd; Asia	6654
Hemitragus jemlahicus	estrus: Dec; Kashmir, Himalayas	4535
	estrus: winter; zoo	4487
	mate: Oct-mid Jan; Nepal	9547
	mate: mid Nov-Dec; Tibet	9540
	birth: May-Sept, peak June; n181; captive	2109
	birth: May-early Oct, peak May-July; n115; zoo	12128
	birth: June; zoo	4487
	birth: June-July; Himalayas	6652, 6654

Hippotragus equinus	mate: Jan; Zaire	254
	mate: Jan; Zaire	5575
	mate: peak May-Aug; Zaire	11253
	mate: Oct-Dec (calculated); Angola	11697
	mate: yr rd; e Africa	5764
	birth: Jan-Mar, May, Nov; Zimbabwe	1794
	birth: peak Feb-mid May; Zaire	11253
	birth: Apr-Aug?; few data; Kruger Nat Park, S Africa	3145
	birth: Aug-?; Angola	11697
	birth: Aug-Oct; s, s central Africa	9752
	birth: Sept-Apr; s, e Africa	5764
	birth: Nov; Zaire	254
	birth: Nov; Zaire	5575
	birth: yr rd, peak Mar-July; Percy Fyfe Nature Reserve, S Africa	10027
	birth: yr rd; few data; Zimbabwe	1794
	repro: yr rd; S Africa	10158
Hippotragus niger	mate: Mar-May; Angola	11697
	mate: Aug-Sept; Angola	3104
	mate: yr rd; zoo	1230
	conceive: May-July; Transvaal, S Africa, Zimbabwe	11758
	birth: Dec-Feb; Angola	11697
	birth: Dec-June, peak early Feb; Matopos Nat Park, Zimbabwe	4085
	birth: Jan-early Feb; ne Botswana; S Africa	10158
	birth: Jan-Mar, peak Feb; n Transvaal; S Africa	10158
	birth: Jan-Mar, peak Feb; Transvaal, S Africa, Zimbabwe	11758
	birth: end Jan-mid Mar; Kruger Nat Park, S Africa	8514
	birth: Feb-Mar; S Africa	3145
	birth: Feb-Apr, peak Mar; Matopos Nat Park, Zimbabwe	11785
	birth: Mar, Aug, Oct-Nov; Zimbabwe	1794
	birth: Mar-Aug, peak Feb-Mar; Percy Fyfe Nature Reserve, S Africa	10027
	birth: Apr-May, Oct-Nov; Zaire	11253
	birth: Apr-June; Kenya	9742
	birth: Apr-Aug; zoo	1230
	birth: May-July; Angola	3103, 3104
	birth: yr rd, July, Nov-Dec; few data; Zimbabwe	4084, 4086
	birth: Sept-Oct; s, s central Africa	9751
Kobus ellipsiprymnus	estrus: May-July; 3 wk; zoo	4535
	mate: June-Aug; Zambia	333
	mate: yr rd; zoo	1230
	preg f: Mar; n1; Tanzania	186
	birth: Jan-Feb, July-Aug; n27; zoo	5818
	birth: peaks Jan-Mar, Sept-Oct; Kruger Nat Park, S Africa	3145
	birth: Feb-Nov; Zambia	333
	birth: end Feb-Apr; Zimbabwe	2275
	birth: Aug; W Africa	8350
	birth: Oct-May, peaks Oct, Feb-Mar; Kruger Nat Park, S Africa	8514
	birth: peak Oct-Nov; Zaire	11253
	birth: peak Dec-Mar, low Nov; zoo	1230
	birth: Dec-July, peak Feb-Mar; n Natal, S Africa	7197
	birth: mid Dec-Feb; British E Africa	5206
	birth: end Dec; Niger	374
	birth: yr rd; Zambia	331
	birth: yr rd, poss peak dry season; Zaire	11271
	birth: yr rd, peak Jan-Feb; Transvaal, S Africa	4655, 4656
	repro: yr rd; Kenya	6413
Kobus kob	mate: Mar-Apr; Pibor, Akobo Valleys, Uganda	5763
	mate: May-Aug; Ivory Coast	7591
	mate: yr rd; w Rift Valley, Uganda	1434, 1436
	mate: yr rd; Uganda	6407
	conceive; Jan-Apr; n16/17; Uganda, Sudan	3543
	birth: est peak Jan-Feb; Zaire	11271
	birth: ~Nov-Dec; Senegal, s Sudan	5763

	birth: Dec-Mar; Ivory Coast	7591
	birth: yr rd; Ivory Coast	3659
	birth: yr rd; Uganda	5763
	birth: yr rd, slight peak Jan-Feb; Uganda	1432
Kobus leche	mate: Oct; Angola	11697
	mate: peak end Oct-Dec; Zambia	333
	mate: end Oct-Dec; Zambia	7368
	mate: mid Nov-mid Feb; Zambia	9676
	mate: yr rd, peak May; nw Botswana	6378
	mate: yr rd, peak mid Nov-Jan; Zambia	9181
	preg f: Oct; n1; Namibia	9945
	birth: Jan, Mar-May, July-Aug, Dec; n17; zoo	12128
	birth: Apr-Sept, peak mid July-mid Aug; Zambia	331, 333
	birth: May-Sept, some Oct, Nov; Zambia	194
	birth: May-Dec, peak mid July-mid Aug; Zambia	7368
	birth: Aug-Sept; Angola	11697
	birth: Oct-Nov; upper Zambesi, Zambia	9750
	birth: peak Oct-Dec; Botswana	10158
	birth: yr rd, peak mid Aug; Zambia	9181
Kobus vardoni	birth: Jan-Feb, June-July; Zaire	11253
	birth: peak Jan-Apr; Luangwa Valley, Zambia	
	birth: peak May-Sept, minor July-Aug; Zambia	331, 333
	birth: Apr-Dec; S Africa	2413
	birth: probably Nov-Dec; Zambia	9749
	birth: yr rd, poss peak during 1st heavy rains; central Africa	5763
Litocranius walleri	birth: during rains; ne Africa	5764
	birth: prob yr rd; n12; Kenya	6410
	repro: yr rd; Kenya	6413
Madoqua kirki	testes size: peak May; Tanzania	5625
	spermatogenesis: Nov, low Jan-Feb, July-Aug; Tanzania	5625
	mate: yr rd: peak June-July, Nov-Dec; Tanzania	5625
	birth: Dec-Apr; S Africa	10158
Madoqua saltiana	preg f: yr rd; Ethiopia, Horn of Africa	5763
Nemorhaedus goral	mate: Sept-Nov; Korea	1310a
	mate: Sept-Jan; captive	2652
	mate: end Sept-Nov; USSR	4643
	mate: early-mid Nov; China	3034
	birth: Apr-May; hunters report; Pakistan	9175
	birth: May; China	3034
	birth: May-June: India	6654
	birth: end May-mid June, few July-Aug; USSR	4643
	lact f: May, Aug; n2; Nepal	7382
Neotragus batesi	mate: yr rd, peaks between wet-dry seasons; Gabon	5763
	birth: yr rd, peaks end each rainy season; Gabon	3191, 3192
Neotragus moschatus	mate: yr rd; e Africa	5763
	birth: Aug-Feb; S Africa	10158
	birth: peak Nov-Feb; e Africa	5763
	birth: mid Nov-mid Dec; Zululand, S Africa	5772
	birth: prob yr rd; captive	5191
Neotragus pygmaeus	birth: Sept; n1; Liberia	184
Oreamnos americanus	mate: Oct-Dec; Rocky Mt, w N America	1474
	mate: end Oct-early Dec; ID	1237
	mate: Nov; BC, Canada	4933
	mate: Nov-Dec; ID, MT	1237
	mate: Nov-mid Dec; MT, SD	9121
	mate: mid Nov-early Dec; WY	1871a
	birth: May-June; Rocky Mt, N America	1474
	birth: mid May-mid June; ID, MT	1237
	birth: mid May-mid June; WY	1871a
	birth: end May-early June; MT, SD	9121
	birth: end May-early June; BC, Canada	4933
	birth: end May-mid June; MT	6381a
	wean: Sept; MT, SD	9121

Oreotragus oreotragus	preg f: Feb-Nov; Zambia	11788
	preg f: Sept; n1; Namibia	9945
	preg f: yr rd; S Africa	10158
	birth: July-Aug; Zaire	11253
	birth: Nov; Ethiopia	2798
	birth: Nov-Dec?; Kruger Nat Park, S Africa	8514
Oryx dammah	mate: Oct-Nov; Chad	3781
	birth: Jan, Mar-June, Sept-Dec; n12; zoo	12128
	birth: July-early Aug; Chad	3781
	birth: yr rd; n41; zoo	5818
Oryx gazella	mate: peak Nov-Dec; S Africa	10041
	birth: Jan-Mar; e Africa	5207
	birth: prob yr rd, peak Aug-Sept; n22; S Africa	10041, 11212
	birth: yr rd, peak Sept; S Africa	2985
	birth: yr rd; zoo	1230
	birth: yr rd; Circle d'Agadez, Niger	1352
	birth: yr rd; e Africa, Namibia	5764
	birth: yr rd; S Africa	10158
	repro: yr rd, peak June-Aug; Kenya	6413
Oryx leucoryx	birth: May-Sept; Saudi Arabia	10395
Ourebia ourebi	birth: July; Angola	11697
	birth: Sept-Nov; Kruger Nat Park, S Africa	8514
	birth: Oct-Dec; Zimbabwe	10803
	birth: Dec-Jan, May-June; Zaire	11253
Ovibos moschatus	estrus: Aug; e Greenland	5300
	mate: July-Aug; n N America	1474
	mate: July-Aug; Norway	8593
	mate: Aug; NWT, Canada	10738, 10740
	mate: Aug-Sept; Jameson Land, ne Greenland	10770
	mate: Aug-early Oct; arctic	8798
	mate: Aug-Oct; Arctic, Canada	9330
	birth: June-Feb; captive, ALB, Canada	8033
	lact: May-June; captive, ALB, Canada	8033
	birth: Apr-early May; e Greenland	5300
	birth: Apr-May; NWT, Canada	6241
	birth: Apr-May; Arctic, Canada	9330
	birth: Apr-mid June; Nunivak I, AK	6381
	birth: Apr-July, peak May; n23; zoo	5818
	birth: mid Apr-mid June; NWT, Canada	10738, 10740
	birth: end Apr-mid May, peak early May; Jameson Land, ne Greenland	10770
	birth: May-early June; Arctic	8798
Ovis ammon	mate: Aug; Mongolia	8739, 8740
	mate: Oct; Nepal	7382
	mate: Oct-Nov; according to natives; Tibet	3034
	mate: Oct-Dec, rare Aug-Jan; Germany	3571
	mate: mid Oct-mid Jan; USSR	4643
	mate: Nov?; Chinese Turkestan, Pakistan	9175
	mate: Dec-Jan; Tibet	6652, 6654
	birth: end Feb-June, peak end Mar-early May; Germany	3571
	birth: Mar, June; Mongolia	8740
	birth: Apr-June; USSR	4643
	birth: May-June; Tibet	3034
	birth: May-June; Tibet	6652, 6654
	birth: May-June; Nepal	7382
Ovis aries	estrus: yr rd, peak June-Nov; captive, Niger	12010
	estrus: mid Oct-Nov; Outer Hebrides, UK	4109, 4111
	estrus: July-Jan; captive, Niger	12011
	mate: Oct-mid Dec; Scotland, UK	1181
	ovulation: yr rd, peak June-Nov; captive, Niger	12010
	birth: Mar-Apr; Scotland, UK	1181, 4111, 5307
	birth: Apr-June; India	9547
	birth: Apr-May; Corsica, France, Sardinia, Italy	6652
	birth: mid Apr-early May; Pakistan	9175

	birth: Nov-June; captive, Niger	12011
Ovis canadensis	estrus: mid Nov-mid Dec; n21; Rocky Mt, N America	3667
	mate: end June-Nov, peak Sept-Oct, some Jan-Mar; CA	11578
	mate: July-Sept; AZ	10950
	mate: Nov-Dec; ID	10950
	mate: July-mid Oct; NV	6392
	mate: end July-early Dec; NV	144
	mate: peak Aug; AZ	9384
	mate: Oct-Nov; Rocky Mt, N America	6652
	mate: mid Oct-early Mar; WY	7346
	mate: Nov-Dec; Rocky Mt, N America	1474
	mate: Nov-Dec; ALB, Canada	3667
	mate: mid-end Nov; BC, Canada	1011
	mate: mid Nov-mid Dec; USSR	4643
	mate: mid Nov-mid Dec; CO	10257
	mate: mid Nov-Dec; ID	10105
	mate: mid Nov-Dec; w US	5146
	mate: end Nov-mid Jan, peak Dec; WY	7016
	birth: Jan; CA	11578
	birth: Jan-Mar; AZ	10950
	birth: May-June; ID	10950
	birth: Jan-Mar, peak Feb-Mar; AZ	9384
	birth: Jan-mid Apr; NV	6392
	birth: Feb-July; NV	144
	birth: Mar-Apr; TX	2351a
	birth: end Apr-mid June, peak end May; BC, Canada	1011
	birth: May; Rocky Mt, N America	6652
	birth: May-June; Rocky Mt, N America	1474
	birth: May-July, Oct; n15; zoo	5818
	birth: mid May-?, peak June 1; ID	10105
	birth: mid May-mid June; CO	10257
	birth: peak end May; ALB, Canada	3243
	birth: end May-mid June; USSR	4643
	birth: end May-mid July, peak mid June; WY	4950
	birth: end June-early July; MT	10411
	birth: yr rd; NV	4333
Ovis dalli	estrus: end Nov-mid Dec; n16; BC, Canada	3667
	mate: Oct-Nov; AK	7625
	mate: Nov-Dec; AK	7883
	mate: mid Nov-early Dec; n N America	7884
	birth: May; YUK, Canada	1448
	birth: May; BC, Canada	3667
	birth: May; AK	7883
	birth: May; AK	8844
	birth: May-June, peak May; AK	7625
	birth: end May-early June; n N America	7884
Ovis musimon	spermatogenesis: peak Oct-Dec; Czechoslovakia	7579
	estrus: mid Mar-early May; Czechoslovakia	10945
	mate: mid Oct-Dec; Corsica, France	8467
	mate: Oct-Dec; Germany	10946
	mate: Nov-Jan; zoo	207
	mate: Nov-Dec; Corsica, France	9318
	birth: Jan-Oct, peak Mar-May; n115; zoo	12128
	birth: Feb-Dec, peak Mar-Apr; n>100; zoo	5818
	birth: mid Mar-early May; Czechoslovakia	10945
	birth: Mar-May; Germany	10946
	birth: Apr, Oct; n2; E Germany	1048
	birth: mid-end Apr; Corsica, France	8467
	birth: mid Apr-early May; zoo	207
	birth: spring-early summer; zoo	1230
Ovis nivicola	mate: Dec; Koryak Mt, Siberia, Russia	10722
	birth: mid-end June; Koryak Mt, Siberia, Russia	10722

Ovis vignei	mate: Sept; India	6652, 6654
	birth: Mar-Apr; India	6654
	birth: Mar-July, peak May-June; n14; zoo	12128
	birth: June; Astor	6652
Pantholops hodgsoni	mate: Nov-Dec; Russia	9547
	mate: mid Nov-Dec; Tibet	3034
	mate: end Nov; Tibet	8972
	mate: end Nov-Dec; Tibet	8740
	mate: winter; according to natives; Tibet	6653, 6654
	birth: end June-early July; Tibet	8972
	birth: July; Tibet	8740
	birth: summer; according to natives; Tibet	6653, 6654
Pelea capreolus	mate: Apr; S Africa	10158
	birth: Nov-Dec; e Transvaal highveld, S Africa	3095
	birth: mid Nov-Dec; S Africa	5783
	birth: yr rd, peak Sept-Dec; zoo	1230
Procapra gutturosa	mate: end Nov-early Jan; USSR	4643
	mate: end fall; Mongolia	182
	birth: May; Mongolia	8739
Procapra picticaudata	mate: Oct-Jan, peak Dec-Jan; Tibet	3034
	mate: end Dec-Jan; Tibet	8740
	birth: June-Aug; Tibet	3034
Pseudois nayaur	estrus: Nov; Tibet	9541
	estrus: Nov-Jan; Nepal	11449
	mate: mid Dec-end Jan, peak early Jan; Nepal	11778
	birth: mid May-mid June; Tibet	9541
	birth: end May-June; Nepal	11778
	birth: end May-early Sept; n43; zoo	12128
	birth; June; zoo	9175
	birth: June-Aug; Pakistan	9175
Raphicerus campestris	mate: spring; S Africa	5782
	mate: yr rd; s Africa	5763
	preg f: yr rd; n32; Botswana	10154
	birth: Apr-May, Sept-Dec; Namibia	9945
	birth: Apr-June, Oct, Jan; Kruger Nat Park, S Africa	8514
	birth: May, Sept; Angola	11697
	birth: Dec; S Africa	5782
	birth: yr rd; S Africa	10158
Raphicerus melanotis	conceive: peak Mar-Apr; S Africa	7977
	conceive: peaks Apr-May, Sept-Nov; S Africa	269
	birth: peak Oct-Dec; n18; S Africa	7977
Raphicerus sharpei	testes size; no change; Zimbabwe	5699
	birth: peak early summer; Nyassaland to ne Transvaal, S Africa	10392
	repro: est yr rd; S Africa	10158
Redunca arundinum	mate: yr rd, peak May-June; S Africa	5469
	preg f: Mar; n1; Tanzania	186
	birth: May, Dec; Angola	11697
	birth: Dec-Mar; S Africa	5779
	birth: yr rd; zoo	1230
	birth: yr rd; Zambia	331
	birth: yr rd, peak Dec, May; Zululand, se Africa	5763
	birth: yr rd, peak Dec-May; S Africa	5468, 5469
Redunca fulvorufula	mate: Mar-May; Transvaal, S Africa	5164
	mate: peak Apr-May; S Africa	10025
	preg f: May; n1; Kenya	4926
	birth: Sept-Feb; Kruger Nat Park, S Africa	8514
	birth: peak Oct-Jan; S Africa	10158
	birth: Dec-Feb, wet season, some Nov; Transvaal, S Africa	5164
	birth: peak summer; zoo	1230
	birth: yr rd, peak 78% Oct-Dec; S Africa	10025
Redunca redunca	mate: end June-early July; e Africa	5203

Rupicapra rupicapra	mate: Oct-early Dec; French Alps	8477
	mate: Oct-Dec; Europe	2079, 2080
	mate: end Oct-early Dec; USSR	4643
	mate: end Oct-mid Dec; Germany	3571
	mate: mid-end Nov; Austria	10888
	mate: Dec-Feb; Austria	7177
	conceive: Nov 12%, Dec 70%; Alps, Europe	11056
	ovulation: mid Nov; Switzerland	3663
	birth: end Apr-mid May; USSR	4643
	birth: end Apr-June; Germany	3571
	birth: May; French Alps	8477
	birth: mid May-mid June; Europe	2079, 2080
	birth: est June-July; Austria	7177
Saiga tatarica	mate: Dec-Jan; USSR	560, 561
	mate: Dec-Jan; Europe, w USSR	4645
	mate: end Nov-Dec; Europe	560
	mate: last 1/3 Dec; zoo	8641
	birth: Apr; zoo	12107
	birth: Apr; Europe	560
	birth: Apr-May, peak early May; USSR	2215
	birth: Apr-May; zoo	2668
	birth: end Apr-May, peak early May; USSR	560, 561
	birth: May; Russia	3162
	birth: May; USSR	4643
	birth: May-early June; Germany	8641
	birth: May-June; Europe, w USSR	4645
Sigmoceros lichtensteini	mate: mid Dec; Zambia	2710
	birth: May-Sept, peak July: nw plateau, Zambia	331, 333
	birth: June-Oct, peak July-Aug; Zambia	7369
	birth: peak July-Sept; Zaire	11253
	birth: Sept; e Africa	5052
	birth: Sept; S Africa	10158
	birth: Oct-Nov; S Africa	9030
	birth: Oct-Nov: Luangwa Valley, Zambia	331, 333
Sylvicapra grimmia	preg f: yr rd; n39; Botswana	10154
	birth: end May-July; Angola	11697
	birth: yr rd; Zimbabwe	11789
	birth: yr rd; Zimbabwe	331, 333
	birth: yr rd; S Africa	10158
	birth: yr rd, peaks Jan-Feb, June-Aug; Zimbabwe	9135
	birth: yr rd, peak summer; zoo	1230
	lact f: Jan; n1; Tanzania	186
	repro: yr rd, peak dry season; Zimbabwe	2275
Syncerus caffer	estrus: Jan?; Angola	11697
	mate: Mar-May; Kruger Nat Park, S Africa	3145
	mate: Mar-May; S Africa	8515
	mate: peak June-July, low Aug-Sept, some Oct-May; Tanzania	9994
	conceive: peak Apr-July; Tanzania	9994
	birth: Jan-Mar; S Africa	9748
	birth: Jan-Apr, peak Jan-Feb; S Africa	8515
	birth: Jan-Apr, peak Jan-Feb; Kruger Nat Park, S Africa	3145
	birth: perhaps Jan-Apr; Zambia	2488
	birth: peak Mar-May; Tanzania	9994
	birth: Apr, Oct, Dec; Zaire	11271
	birth: Apr-June; Zaire	11253
	birth: Oct-Jan; Angola	11697
	birth: Nov-Apr, peak Jan-Feb; Botswana	1599
	birth: Nov-July, peak May-Apr, low Aug-Oct; e Africa	9991
	birth: Dec-Oct, peak July; Tanzania	9994
	birth: Dec-Feb; British E Africa	5205
	birth: yr rd; Zambia	331
	birth: yr rd; Kruger Nat Park, S Africa	8514
	birth: yr rd; n8; zoo	1230
	birth: yr rd, peak Apr-June; Zaire	11253

	birth: yr rd, peaks May, Dec; Uganda	4077
	repro: yr rd; Zimbabwe	2275
Tetracerus quadricornis	mate: July-Sept; Nepal	7382
	mate: during rainy season; India	37
	birth: Jan-Feb; Nepal	7382
	birth: Jan-Feb; India	6654
Tragelaphus angasi	spermatogenesis: yr rd; ne Natal S Africa	269
	conceive: yr rd, peaks Feb-May, Oct; ne Natal, S Africa	269, 270
	mate: peak Apr-May, Oct; ne Natal, S Africa	269
	mate: July, Oct; but not sampled all yr; Mozambique, Malagasy	6525
	preg f; June-Nov; se Zimbabwe	2357
	birth: Feb, Apr, Sept, Oct; few data; Mozambique, Malagasy	6525
	birth: Apr-May, July-Nov peak Aug-Oct; Kruger Nat Park, S Africa	8514
	birth: July-Aug; Zululand, S Africa	7857
	birth: Aug-Nov; few data; Kruger Nat Park, S Africa	3145
	birth: Oct-Jan; ne Natal, S Africa	269
	birth: Nov; se Zimbabwe	2357
	birth: yr rd, bimodal; S Africa	10158
	birth: yr rd; zoo	1230
Tragelaphus eurycerus	mate: peak Apr, Nov; e Africa	5763
	birth: peak July-Sept; e Africa	5763
Tragelaphus imberbis	mate: probably yr rd; e Africa	5763, 6413
	birth: yr rd; n40; Kenya	6411
Tragelaphus oryx	testes size: peak Mar; S Africa	10041
	estrus: May-July; 3 wk; zoo	4535
	estrus: Dec-Jan; Angola	11697
	estrus: yr rd, peak May-July; captive	10367
	mate: May-June; zoo	6058
	mate: Dec-Jan; Angola	11697
	mate: yr rd, peak probably Feb; S Africa	10023, 10041
	mate: yr rd, peak May; n16; zoo	8339
	birth: Feb, May, Sept-Dec; few data; captive	10300
	birth: Feb-Mar; captive	6058
	birth: Feb-Mar; zoo	6058
	birth: Mar; e Africa	5052
	birth: peak Apr-Sept; Zambia	2488
	birth: May-Nov, peak July-Sept; e Africa	5763
	birth: June-July; Zaire	11253
	birth: June-Aug, peak Aug; se Africa	9753
	birth: July-Aug; Zimbabwe	11784
	birth: Aug-Sept calculated; Angola	11697
	birth: Aug-Sept; Kruger Nat Park, S Africa	3145
	birth: Aug-Oct; captive	8690
	birth: Aug-Nov; Matopos Nat Park, Zimbabwe	11785
	birth: Dec-Aug, 71% Feb-Apr; zoo	10895
	birth: yr rd; zoo	1230
	birth: yr rd; n103; zoo	12128
	birth: yr rd; n408; zoo	10894
	birth: yr rd; e Africa	4801
	birth: yr rd, peak Aug-Sept; Kruger Nat Park, S Africa	8514
	birth: yr rd, peak Aug-Oct; Zimbabwe	9311
	birth: yr rd, 30% Nov; S Africa	10023, 10041
Tragelaphus scriptus	conceive: Apr-May; Zambia	7537
	birth: Feb-Mar; e Africa	5204
	birth: bimodal Apr-May, Oct-Nov; S Africa	10158
	birth: June; Angola	11697
	birth: July-Nov, peak Oct; few data; Kruger Nat Park, S Africa	3145
	birth: Oct-Nov; Zambia	7537
	birth: mid Oct-mid Dec, sometimes Feb; Cape Province, S Africa	5773
	birth: peak rainy season; Zaire	11253
	birth: prob yr rd; n24; Kenya	201

	birth: yr rd; zoo	1230
	birth: yr rd; n7; captive	11347
	birth: yr rd; n20; zoo	2603
Tragelaphus spekei	mate: Mar; n1; zoo, Netherlands	5688
	mate: yr rd, peak May; n21; zoo	8339
	birth: peak Apr; Uganda	5763
	birth: Nov; n1; zoo, Netherlands	5688
	birth: probably yr rd; Kenya	8165
	birth: probably yr rd; Zambia	331
	birth: yr rd; zoo	6192
	birth: yr rd; n61; zoo	5818
	birth: yr rd; S Africa	10158
	repro: yr rd; captive	2485
Tragelaphus strepsiceros	testes size: peak June, low Nov; Transvaal, S Africa	10031
	mate: May-July calculated; Angola	11697
	mate: June; Transvaal, S Africa	10023, 10027, 10031
	mate: June-July; S Africa	11043
	mate: July; Zimbabwe	9986
	mate: peak Sept; Tanzania	5763
	mate: yr rd, peak May; n21; zoo	8339
	preg f: July-Feb; few data; Zambia	11781
	birth: Jan-Feb; Kruger Nat Park, S Africa	8514
	birth: peak Jan-Apr; Zambia	11781
	birth: Jan-Apr, peak Feb; Zimbabwe	11786
	birth: mid Jan-Feb; ne S Africa	3145
	birth: Feb-Mar; Zimbabwe	9986
	birth: Feb-Mar; Zimbabwe	2275
	birth: peak Mar; S Africa	11043
	birth: July-Jan; n21; zoo	5818
	birth: Oct; Angola	11697
	birth: during rains; Tanzania	5763
	birth: end summer; S Africa	10023, 10027
	birth: yr rd; Zambia	331
	birth: yr rd, peak Jan-Mar; zoo	1230